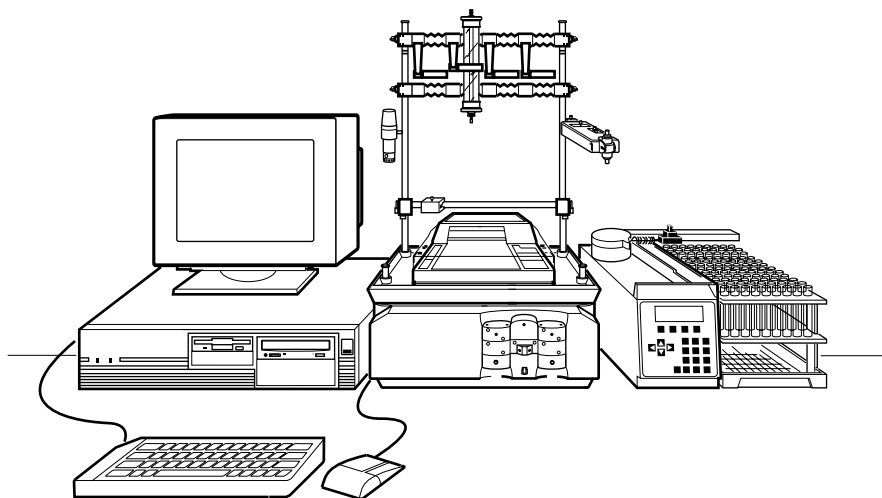




BIOLOGIC HR CHROMATOGRAPHY SYSTEM

INSTRUCTION MANUAL

BIO-RAD

SAFETY



Caution/Warning

Disconnect power to any BioLogic HR system component before servicing. No user-serviceable parts are inside any component. Refer servicing to Bio-Rad service personnel.

This instrument is intended for laboratory use only.

The BioLogic HR system is certified to meet the I.E.C. 1010* standard for safety of laboratory equipment. Certified products are safe to use when operated in accordance with the instruction manual. This safety certification does not extend to other chromatography equipment or accessories not I.E.C. 1010 certified, even when connected to this BioLogic HR system.

The BioLogic HR system conforms to the class A standards for Electromagnetic Emissions, intended for laboratory equipment applications. It is possible that emissions from this product may interfere with some sensitive appliances when placed nearby or on the same circuit as those appliances. The user should be aware of this potential and take appropriate measures to avoid interference.

This instrument should not be modified or altered in any way. Alteration of this instrument will void the manufacturer's warranty, void the I.E.C. 1010 certification, and create a potential safety hazard for the user.

Bio-Rad is not responsible for any injury or damage caused by the use of this instrument for purposes other than for which it is intended or by modifications of the instrument not performed by Bio-Rad or an authorized agent.

*I.E.C. 1010 is an internationally accepted electrical safety standard for laboratory instruments.

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1.0 INTRODUCTION

1.1 OVERVIEW

The BioLogic HR is a full-featured, gradient chromatography system specifically designed for the high resolution purification of proteins, peptides and other biomolecules where recovery of biological activity is of primary concern.

The BioLogic HR system is microprocessor controlled, with an easy-to-use graphic interface and menu-driven software for manual operation, system setup, method editing and run operations.

The flexible control architecture allows the seamless integration of a wide variety of configurations with other Bio-Rad and non-Bio-Rad components to meet your purification requirements.

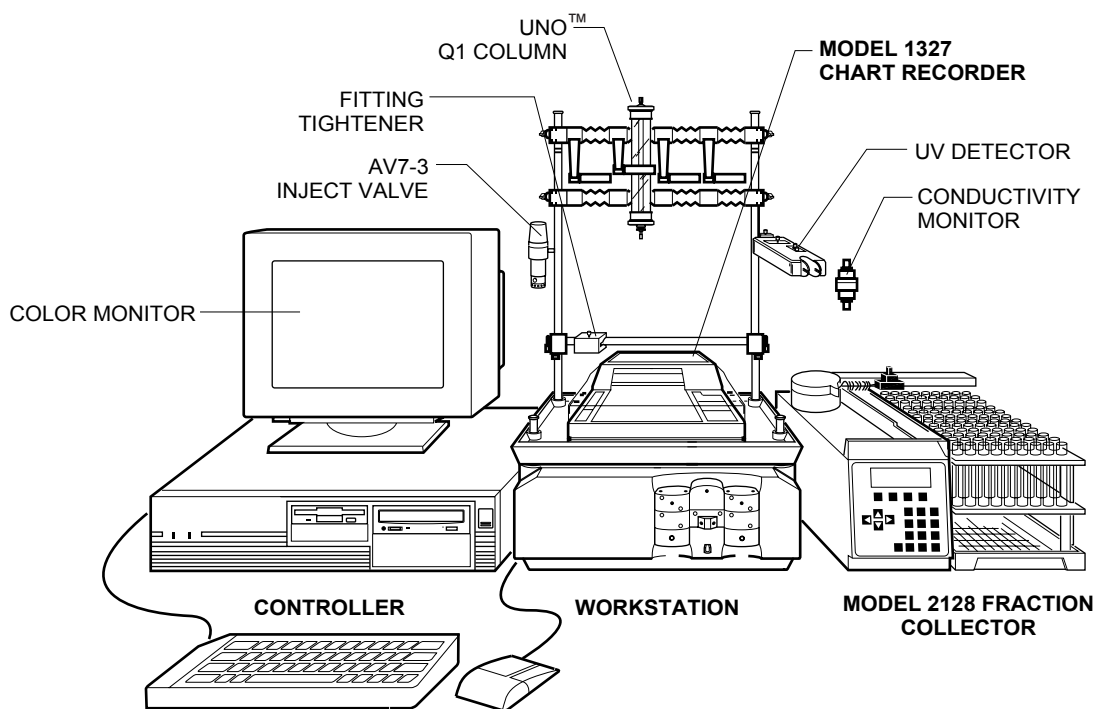


Figure 1. BioLogic HR System

1.2 FEATURES

The BioLogic HR system provides the following features:

- A space-saving modular and stackable design which minimizes the System footprint on the bench. The BioLogic HR Controller, Workstation and optional peripherals are linked by a single “instrument bus” cable in any configuration, making electrical connections simple and fast.
- Control software specifically designed for the bio-chromatographer. The pre-loaded, dedicated software is based on the easy-to-use Windows® 95 platform and is designed for multiple users in the laboratory. Programming separation methods is menu-driven, easy and intuitive using either Time-based or Volume-based steps. Method storage capacity is virtually unlimited.
- On-screen help function.
- Single-point control of chart recorders, fraction collectors, signal import modules and auxiliary pumps (e.g., Bio-Rad’s Econo Pump or a peristaltic pump) for sample loading.
- Binary gradient formation by two dual piston, low-pulsation pump heads which operate up to a maximum pressure of 1000 psi at flow-rates up to 10 ml/min. This permits the use of a wide range of both preparative and analytical columns.
- Dynamic mixing for accurate and reproducible gradient formation.
- The use of biocompatible materials throughout the flowpath to ensure maximum recovery of biological activity.
- Standard high resolution optics unit (mercury lamp) for fixed wavelength (280 nm and 254 nm) detection and a choice of preparative or analytical flow-cells. A zinc lamp and filter combination for 214 nm detection is optional. Other filters are available from Bio-Rad.
- Standard conductivity cell to monitor gradient formation.
- Simple-to-sophisticated Fraction Collection schemes are standard. Collection is volume-based with Collect All, Collection Windows, Threshold and Threshold + Collection Window modes available with a range of collector types.
- 3 ports to accommodate a range of low pressure solenoid divert and buffer select valves.
- 3 ports to accommodate a range of high pressure, automated stream select and sample injection valves.
- Sufficient tray space on the basic System Rack to hold a Model 2110 Fraction Collector or a Model 1327 Chart Recorder. Additional rack components are available to make a three-tray configuration.
- Standard column clamping arrangement to eliminate rack clutter.
- Available manual inject valve with position-sensing switch to satisfy Hold conditions programmed in a Method.
- Water-resistant integrated mini-keyboard designed for the laboratory.

1.3 UNPACKING

When you receive the BioLogic HR system, carefully inspect the shipping containers for any damage which may have occurred in shipping. Severe damage to a container may indicate damage to its contents. If you suspect damage to the contents may have occurred, immediately file a claim with the carrier in accordance with their instructions before contacting Bio-Rad Laboratories.

**Caution**

Lift items from the bottom as you remove them from their containers!

Open each of the shipping cartons and lift the contents out of its packing. Check the contents of each box against the supplied packing list. Remove the plastic bag from each unit and inspect the unit for external damage. If any part is missing or damaged, contact Bio-Rad Laboratories immediately.

1.4 DESCRIPTION OF SYSTEM COMPONENTS

The following sections identify the key features of the BioLogic HR system and available options.

Table 1.
Description of System Components

Component	Function
Controller	This is the “brain” of the system. It provides the graphic user-interface from which the system must be set up and run. From the controller, you can set up and run methods, perform simple data analysis, and store method and run data. Control of the Workstation (which includes the Gradient Pump, Valves, UV/Conductivity Detectors) and peripheral instruments (Model 2128 fraction collector, the Model EP-1 Econo Pump, and Signal Import Modules) is via the instrument bus connection.

Table 1. (continued)
Description of System Components

Component	Function
Workstation	This consists of the following: • Gradient Pump. This consists of two bio-compatible dual-piston pumpheads, labeled A and B. It is controlled from the Controller. Purge and Pause buttons (Purge A, Purge B, and Pause) are present on the front of the Workstation. The Gradient Pump has a maximum delivery of 10 ml/min at a maximum pressure of 1000 psi. The pump system includes a built-in, integrated pressure transducer and dynamic mixing chamber. • Control circuitry and connectors for the UV and Conductivity Detector signals. Signal export connectors for UV and Conductivity data at 1 V and chart recorder control (pen up/down, start/stop). • Control circuitry and connectors for the system valves (low pressure solenoid and automated high pressure inject and select valves). • An Auxiliary (AUX) connector for the Model 2110 fraction collector, generic fraction collectors, the position-sensing cable from the manual inject valve, and/or the Model EP-1 Econo Pump for sample loading. • An output power connector to the UV lamp. • Two instrument bus (phone-type) connectors.
System Rack	This is the system organizer for columns and cartridges, buffer containers, sample inject and stream select valves, UV and Conductivity Detectors, and other devices used with the system. The optional BioLogic HR Rack Expansion kit extends the rack to 3 trays.
UV Detector	The UV Detector with a mercury lamp and filters for fixed wavelength UV detection at 280 nm and 254 nm is provided with all BioLogic HR systems. An optional zinc lamp with a 214 nm filter is available, as are filters for other wavelengths. Two flow cells are available: • An analytical "Z" cell for high resolution protein chromatography applications is provided with all BioLogic HR systems. It has a path length of 5 mm for maximum sensitivity and a swept volume of only 8 µl for maximum sensitivity at flow rates from 0.1 to 10 ml/min. • A cuvette cell is available as an option. It is for preparative work where high sensitivity is not an absolute requirement. It has a path length of 2 mm and a swept volume of 30 µl.
Conductivity Detector	A Conductivity flow cell, which rests in a specially designed holder on the front of the UV Optics unit, provides instantaneous, on-line readings of conductance for monitoring a salt gradient and optimization of column chromatographic conditions. It has a nominal swept volume of 2 µl.
pH Monitor	The pH Monitor, which is available as an option, provides real-time monitoring of buffer pH. It is designed for flow-through pH readings.
Valves	The following valves are available as options to the BioLogic HR system. • High pressure (to 1000 psi) valves include: AV7-3 Automated Inject Valve, V7-3 Manual Inject Valve, AV9-8 Automated Stream Select Valve, V9-8 Manual Stream Select Valve. • Automated, low pressure (to 30 psi) solenoid valves for applications such as buffer selection and flow diversion include: SV5-4 Select Valve and SV3-2 Diverter Valve.

Table 1. (continued)
Description of System Components

Component	Function
Sample Loops	The following sample loops are available for use with the inject valves: 25 μ l, 50 μ l, 100 μ l, 250 μ l and 500 μ l, 1.0 ml, 2.0 ml and 5.0 ml. The 1.0 ml loop is provided with the System. In addition, the DynaLoop 25 and DynaLoop 90 Sample Loops are available for automated introduction of large sample volumes.
Columns	Bio-Rad makes a number of different columns. A description of each type of column is located in Chapter 2, section 2.6, Columns and Column Fittings.
Auxiliary Load Pump	The Model EP-1 Econo Pump may be used as an auxiliary pump for loading large sample volumes onto low pressure columns. This is available as a system option. For loading large volumes onto high pressure columns, load directly through the Gradient Pump via a select valve, or use the Bio-Rad DynaLoop sliding-piston sample loop.
Fraction Collector	Bio-Rad offers two fraction collectors as system options. • The Model 2128 provides X-Y motion drop dispensing over any of 5 types of racks. The racks accommodate a wide range of tube diameters and lengths, microtiter plates, microtubes, and "bottle size" fractions. The Model 2128 is suited for both analytical and preparative applications. It is controlled by the BioLogic HR Controller via the instrument bus. • The Model 2110 uses a stationary drop-dispensing head to collect up to 80 fractions in a carousel. To divert non-peak material to waste, use the SV3-2 Diverter valve. The Model 2110 is controlled by the BioLogic HR Controller via the AUX connector on the Workstation. • Other fraction collectors may be used with the BioLogic HR system as long as tube advances can be initiated by an active low pulse (>100 ms). (Refer to the fraction collector's documentation for discussion of its active low pulse.) The SV3-2 Diverter valve is required for full operation. Fraction advance marks (event marks) for a generic collector are embedded in the UV signal sent from the Workstation to a chart recorder. Generic collectors are controlled by the BioLogic HR Controller via the AUX connector on the Workstation. Regardless of which fraction collector is chosen, the fraction volume and collection parameters such as Collect All, Collection Windows, Threshold, and Threshold + Collection Windows can be controlled from the BioLogic HR Controller.
Chart Recorder	Bio-Rad offers the Model 1327 Chart Recorder as a system option. This is a dual pen chart recorder for Conductivity and UV Detector readings. Stop/Start and Pen Up/Down functions are controlled by the BioLogic HR Controller via the BioLogic HR Workstation. Chart speed is set at the recorder itself (the BioLogic HR Controller does not control this function). Event marks including fraction advance marks are embedded in the UV signal sent from the Workstation to the chart recorder. The BioLogic HR system may be used with any generic single or dual-pen chart recorder, provided Stop/Start and Pen Up/Down pin-out information is available.

Table 1. (continued)
Description of System Components

Component	Function
Signal Import Module	Up to two Signal Import Modules (SIMs-HR) may be connected to the BioLogic HR using the instrument bus. This module allows you to connect the pH Detector to the BioLogic HR. There is also a 3-pin connector which may be used to connect other detectors (such as variable wavelength, UV, fluorescence, and RI detectors) which transmit an analog signal of up to 2.5 V. The SIM-HR converts the detector's analog signal to a digital signal and communicates the information to the BioLogic HR system.
Tubing	The following tubing is supplied with the unit: System Tubing: 1/16" OD, 0.020" ID Tefzel® tubing. The approximate volume of 1 cm of tubing is 2 µl. Pump Inlet Tubing: 1/8" OD, 0.062 " ID PTFE tubing. The approximate volume of 1 cm of tubing is 20 µl.
Fittings	Flangeless 1/4-28 flat bottom fittings.
Auxiliary Pump	Bio-Rad's Model EP-1 Econo Pump is available as a system option. It is a two-channel bidirectional variable speed peristaltic pump for low-pressure chromatography. It offers a full range of features to facilitate ease of use as a stand-alone pump or as an accessory to the Biologic HR system to load large volumes, especially onto low-pressure columns.

Tefzel is a registered trademark of E.I. du Pont de Nemours & Co.

2.0 DESCRIPTION OF THE BIOLOGIC HR SYSTEM COMPONENTS

2.1 CONTROLLER

The Controller is the “brain” of the BioLogic HR system. It consists of the Color Display Monitor, the Controller unit, the keyboard and mouse devices, and a floppy disk drive. **Note:** The sound capabilities of the CD-ROM drive and the audio and mic connectors on the back of the controller are not for use with the BioLogic HR system.

Table 2.
Controller's Front Panel Features

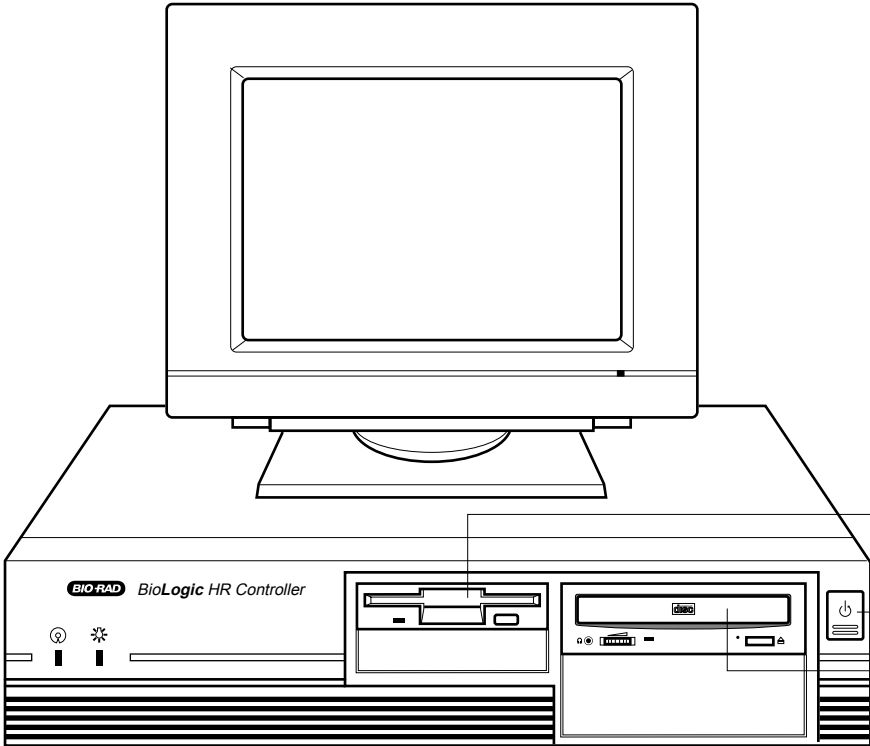
	
Feature	Description
CD ROM Drive	Use this drive to load updates of the BioLogic HR operating software. To open the drive, press the button on the front of the drive.
Keyboard & Mouse	These two interface devices (not shown) are used to control the system. The keyboard can withstand the minor spills associated with use in a laboratory.
Floppy Disk Drive	Use it to backup and restore methods onto floppy disk. Press the button next to the drive slot to manually eject diskettes from the drive.
Power Switch	Turns on/off the Controller and Monitor.

Table 3.
Controller's Rear Panel Connectors

Connector	Description
	Instrument bus (phone jack connector): The Instrument Bus allows all components in the BioLogic HR System to communicate with the BioLogic HR Controller. The components may be cabled together in any order via their respective Instrument Bus connectors. This is termed “daisy-chaining.” The presence of a component on the Instrument Bus is recognized when it is switched on. Even when one component is switched off, another component “daisy-chained” to it can be controlled by the Controller. Note that the Controller has only one phone jack connector.
	Video connector: Use this connector to connect the Color Monitor to the Controller.
	Keyboard connector: Use this connector to connect the Keyboard to the Controller. It supports any standard PC compatible keyboard.
	Mouse connector: Use this connector to connect the mouse to the Controller. It supports any standard PC compatible mouse or trackball.
	Parallel connector: Use this connector when connecting any parallel printer. Laser and Dot Matrix printers are examples of parallel printers. Refer to your Windows® 95 and/or your printer documentation for instructions on how to set up the printer for use with the BioLogic HR System. Some printer drivers are pre-installed on the Controller.
	Power Connector: This is the power connector for the Controller.

2.2 Workstation

The Workstation is the “heart” of the BioLogic HR system. It contains the following:

- Gradient Pump. This consists of two bio-compatible dual-piston pumpheads, labeled A and B. The Gradient Pump is rated to 1,000 psi (maximum), with a flow rate range of 0.1 to 10 ml/min. This module also contains an integrated pressure transducer and dynamic mixer.
- The control circuitry for the Gradient Pump, the UV and Conductivity Detectors, and the Valves (solenoid and automated high pressure valves). Connectors for all valves and Detector inputs are on the rear panel along with output connectors for UV and Conductivity signals to a chart recorder.
- An AUX combicon connector used for (a) controlling fraction advances of the Model 2110 and generic fraction collectors, (b) receiving a signal from the position sensing switch of the V7-3 Manual Inject Valve, and (c) starting and stopping the Model EP-1 Econo Pump for sample loading.
- Power supply for the Gradient Pump, the UV optics unit, the Conductivity cell, and the Workstation electronics.

Table 4.
Workstation's Front Panel Controls

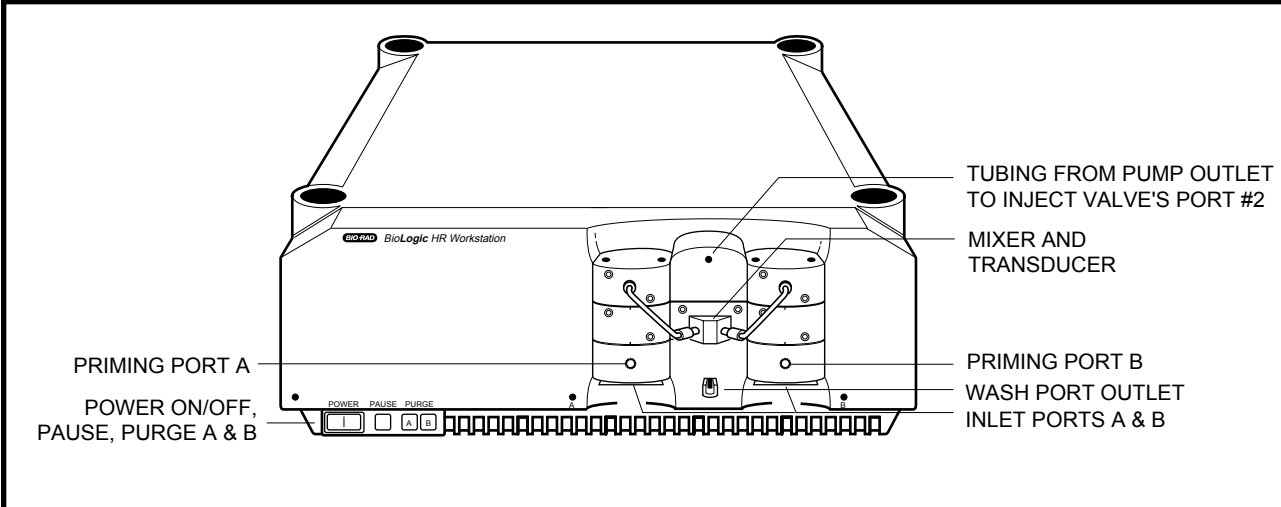
	
Feature	Description
Power Button	Press this switch to turn on power to the Workstation.
Pause Button	Stops the Gradient Pump and pauses a running Method. When the Pause button is pressed, the status LEDs for pumps A and B change from green to flashing red. This button is useful in the event of a leak in the plumbing (the fittings or tubing) or if there are not enough tubes in the Fraction Collector (Panic stop!). To resume the run, • Press the button again, or • Press the Continue button in the Run screen, or • Press the START button in the Manual screen.

Table 4. (continued)
Workstation Front Panel Controls

Feature	Description
Purge A & B Buttons	These buttons are used to purge the tubing lines during setup of the system. When the purge buttons are pressed, the pump heads run at their (default) maximum flow of 7.0 ml/min. (To alter the purge flow rate, select Edit User Preferences from the Options pull-down menu.) When the buttons are pressed, the green LEDs for pumps A and B flash. To stop the flow, press the purge buttons again and note the LEDs go off. These buttons do not operate when the system is running a Method. Note: It is important to move any in-line inject valve to its purge position to avoid exposing a column to high flow rates when the purge function is used.
Alert Light	This LED serves two functions: Constant red: Indicates that the pump has shut down due to a mechanical or electrical problem. This includes a shutdown due to the pressure limit being exceeded. Flashing red: Indicates that a pre-programmed ALARM step has been reached in a running Method (e.g., to indicate that a manual valve position needs changing). This LED will flash simultaneously with an audible tone from the Controller.
Plumbing Connections	Ports on the front of the Workstation: Pump Head Inlet ports: Buffer inlet lines attach to the ports at the bottom of each pump head. These ports accept standard 1/4-28 flat-bottom fittings. The pump head inlet tubing is 1/8" OD, 0.062" ID PTFE tubing with flat bottom fittings which are supplied in the fittings kit. Pump Head Output and Pressure Transducer Mixer Input ports: Output from the two pump heads is to the integral Pressure Transducer/Mixer, located between the pump heads. These ports are pre-plumbed using standard 1/4-28 flat-bottom fittings and 1/16" OD, 0.020" ID Tefzel tubing. Mixer Output port: The output from the mixer accepts standard 1/4-28 flat-bottom fittings. Use the 1/16" OD, 0.020" ID Tefzel tubing with Super Flangeless flat bottom fittings supplied in the fittings kit. Pump Head Priming port: There is a port on each pump head which is used when priming the pump. The port accepts any size syringe with a luer (not luer lock) fitting. This draws buffer along the inlet line and to the pump head. Twist the port counter-clockwise to open it. Pump head washout port: The angled port at the top of each pump head is used for rinsing the piston behind the piston seals to remove crystallized salts. It accepts any syringe with a luer fitting. The rinse output is at the lower front of the pump housing.

Table 5.
Workstation's Rear Panel Connectors

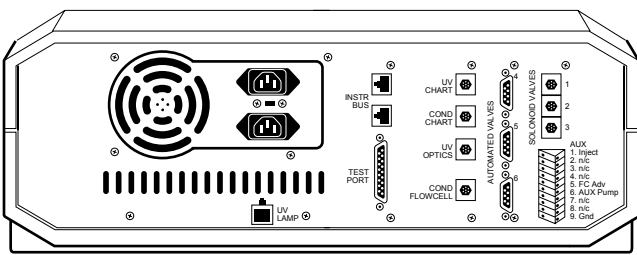
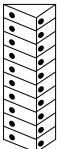
																					
Connector	Description																				
 AUX 1. Inject 2. n/c 3. n/c 4. n/c 5. FC Adv 6. AUX Pump 7. n/c 8. n/c 9. Gnd	Aux connector: This 9-pin connector is used to connect a variety of peripheral modules which do not communicate with the BioLogic HR Controller over the Instrument Bus. <table border="1"> <thead> <tr> <th>Pin #</th><th>Description</th></tr> </thead> <tbody> <tr> <td>1</td><td>Inject. The V7-3 manual inject valve reports its position (either LOAD or INJECT). This information may be used to satisfy a Hold command which has been programmed in a Method Protocol.</td></tr> <tr> <td>2</td><td>n/c. No connection</td></tr> <tr> <td>3</td><td>n/c. No connection</td></tr> <tr> <td>4</td><td>n/c. No connection</td></tr> <tr> <td>5</td><td>FC Adv. Model 2110 and generic Fraction Collector Advance output.</td></tr> <tr> <td>6</td><td>AUX Pump. A Stop-Start command is sent to a pump (e.g. Bio-Rad EP-1).</td></tr> <tr> <td>7</td><td>n/c. No connection</td></tr> <tr> <td>8</td><td>n/c. No connection</td></tr> <tr> <td>9</td><td>GND. Ground</td></tr> </tbody> </table>	Pin #	Description	1	Inject. The V7-3 manual inject valve reports its position (either LOAD or INJECT). This information may be used to satisfy a Hold command which has been programmed in a Method Protocol.	2	n/c. No connection	3	n/c. No connection	4	n/c. No connection	5	FC Adv. Model 2110 and generic Fraction Collector Advance output.	6	AUX Pump. A Stop-Start command is sent to a pump (e.g. Bio-Rad EP-1).	7	n/c. No connection	8	n/c. No connection	9	GND. Ground
Pin #	Description																				
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8	n/c. No connection																				
9	GND. Ground																				
	Solenoid Valves: For connecting Bio-Rad's low pressure solenoid valves (SV5-4 and SV3-2) to the system.																				
	Automated Valves: For connecting Bio-Rad's high pressure automated valves (AV7-3 Inject and AV9-8 Stream Select) to the system.																				
	Cond Flowcell: For connecting the Conductivity flow cell to the system.																				
	UV Lamp: This specialized 6-pin square connector provides electrical power to the mercury or zinc lamp in the UV Optics module's lamp housing.																				
	UV Optics: For connecting the UV Optics module to the system.																				

Table 5. (continued)
Workstation Rear Panel Connectors

Key	Description
	Cond Chart: For conductivity signal output to a single or dual pen chart recorder. An 8-pin mini-DIN to banana plug cable (System Cable 4) for connection to the Model 1327 Chart Recorder is available. Connect the red line to the positive (+) terminal and the black line to the negative (–) or ground terminal of channel 2 (CH2). The chart recorder should be set to 1 V full scale. If you are using the Model 1327 Chart Recorder, move all buttons to the “green” settings.
	UV Chart: For UV signal output to a single or dual pen chart recorder. In addition, when the Bio-Rad Model 1327 is used, chart recorder Pen Up/Down, Stop/Start commands, and event marks are sent from this connector. The Bio-Rad Model 1327 dual pen recorder needs an 8-pin mini-DIN to standard DIN cable (System Cable 2) available from Bio-Rad. Generic chart recorders require an 8-pin mini-DIN to breakout cable (System Cable 7) available from Bio-Rad. When a Signal Import Module signal replaces the standard BioLogic HR UV Detector, use System Cable 20 to control a Bio-Rad Model 1327 dual pen recorder. The chart recorder should be set to 1V.
	Power Cord Connector: This grounded 3-prong connector is for input power to the Workstation and output power to any unit connected to the Workstation. The Workstation’s input power cord should be plugged into a 3-prong grounded power outlet.
	Instr Bus: These RJ-45 modular phone connectors and their cables connect the Workstation to the other components in the system. The Instrument Bus handles all communications between the Controller and each of the components in the system. For example, the Instrument Bus may connect the Workstation to the Model 2128 Fraction Collector. Components can be connected in any order in the system.
	Test connector: This connector is reserved for internal Bio-Rad use.

2.3 UV AND CONDUCTIVITY DETECTORS

The UV detector consists of the Optics bench, a filter tray, a flow cell, and a lamp. It is designed to hold the Conductivity Monitor flow cell if required.

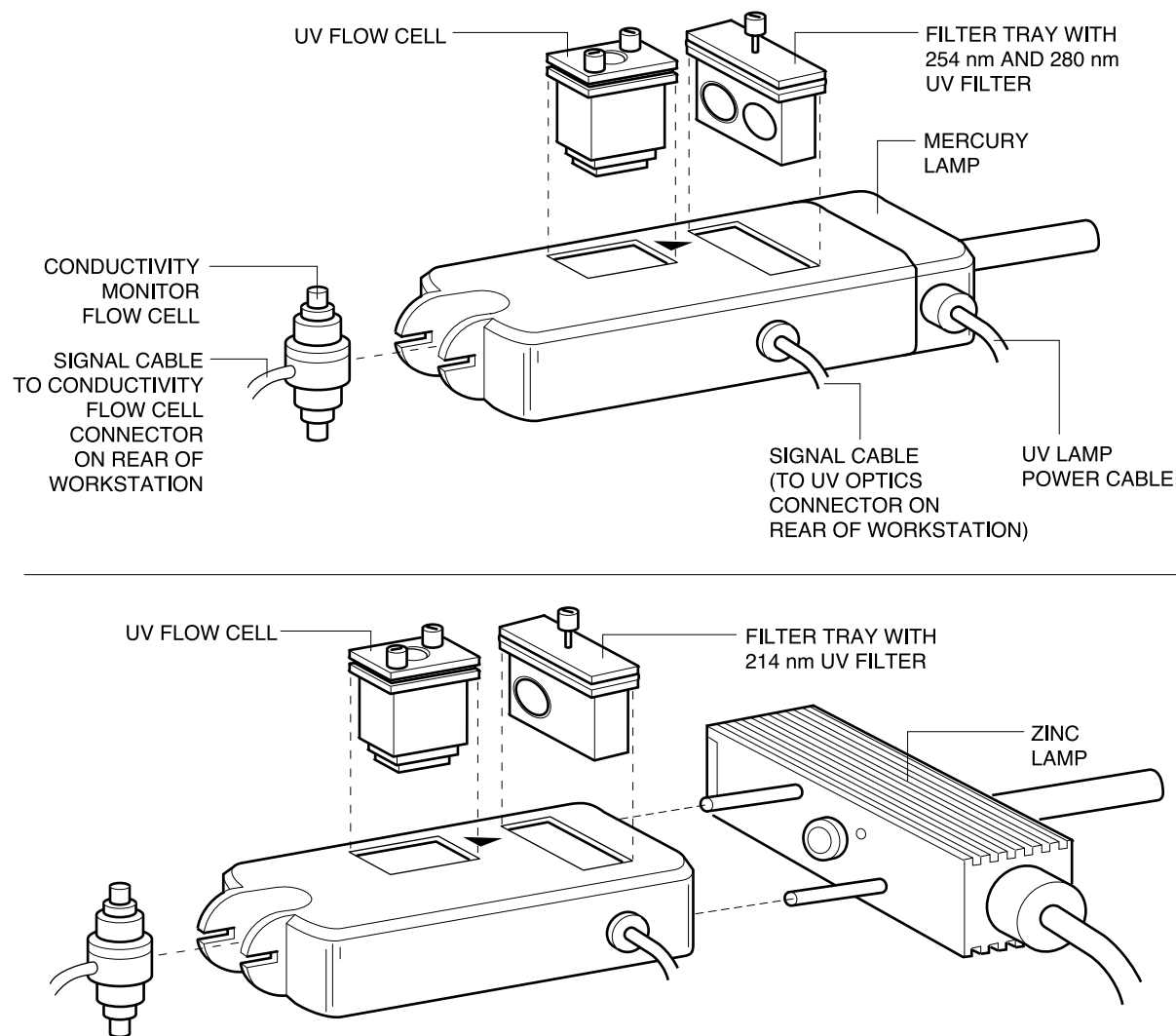


Figure 2. Model OM-II UV Optics Module and Conductivity Flow Cell

2.3.1 UV Detector

The UV Detector is a single beam, fixed wavelength UV absorbance detector specifically designed for high resolution protein chromatography. The portability of this detector enables it to be positioned close to a column outlet, minimizing system dead volume and remixing of peaks.

The following configurations are available:

- **Mercury Lamp and 280/254 nm filters.** The mercury lamp and 280 nm and 254 nm filters are standard. The filters are both held by a single tray. To switch filters, rotate the filter holder.
- **Zinc Lamp and 214 nm filter.** The Zinc lamp and 214 nm filter are optional. The zinc lamp attaches directly to the optics module in place of the mercury lamp.
- **Mercury Lamp and other filters.** Filters for different wavelengths are also available from Bio-Rad. These include 313 nm, 365 nm, 405 nm, 436 nm, and 546 nm filters, which are supported by the mercury lamp.

The UV detector offers two types of flow cells, both using 1/4-28 flat-bottom fittings:

- For high resolution detection at flow rates between 0.1 and 10 ml/min, use the analytical “Z” Flow Cell, which is standard with the system. The cell has a path length of 5 mm, a swept volume of 8 μ l, and is rated to 750 psi.
- For applications demanding less sensitivity, a 2 mm path length “preparative” Cuvette Flow Cell is available as an option. It has a swept volume of 40 μ l and is rated to 750 psi. This cell is best suited to applications at flow rates greater than 0.5 ml/min.

The UV Optics module receives power via the Workstation, to which it is connected by the UV Lamp cable. The UV detector communicates with the system via the UV Optics cable, which plugs into the UV Optics connector on the back of the Workstation.

The UV detector sensitivity range is specified in AUFS (Absorbance Units Full Scale). This is set at the Controller in either the Manual or Run screens. The ranges available are: AUFS 2.0, 1.0, 0.5, 0.2, 0.1, 0.05, 0.02, 0.01, 0.005, 0.002, 0.001. A Zero Baseline button is available in the Manual and Run screens and a programmable Zero Baseline command is available as part of a Method Protocol. **Note:** Refer to the discussion of the chart recorder for setting the chart recorder range.

The UV Optics module is made up of two parts, the optics bench (which holds the flow cell and filter tray) and the lamp housing. The two sections are held together by mounting screws, two guide pins, and a gold grounding pin.

Figure 2 shows how the flow cell and the filters can be changed. **Note:** Be sure to consult the documentation supplied with the flow cell and filter, as specific installation instructions may vary, based on the Optics module's model number and serial number.

To replace a burnt-out lamp, refer to Chapter 11, Maintenance, or to the instruction sheet for the replacement lamp.

2.3.2 Conductivity Monitor

The Conductivity Monitor consists of a small portable flow cell and the control circuitry which resides in the BioLogic HR Workstation. This monitor provides a means to check equilibration conditions and the formation of a salt gradient, for example, in ion exchange chromatography and hydrophobic interaction chromatography. This data helps in optimizing purification protocols and column cleaning procedures.

The Conductivity Monitor consists of the following:

- Flow cell with Inlet/Outlet ports: For connection of the tubing, use 1/4-28 flat bottom fittings. The flow cell allows flow in either direction.
- Signal Cable: For connection to the back of the Workstation. Electrical power for the Conductivity Detector is drawn off the signal cable.

The flow cell can be plumbed immediately after the UV flow cell or at any other point in the flowpath. A holder for the Conductivity Monitor is built into the UV Detector. The swept volume in the cell is a nominal 2 μ l.

Conductivity is measured in mS/cm. The BioLogic HR system allows the selection of conductivity chart recorder ranges of 500, 250, 100, 50, 25, 10, 5 mS/cm in the Manual and Run screens. When setting the range, choose one that will accommodate the maximum conductivity expected during the chromatography run. Keep in mind that although the chart recorder range can be manipulated during a chromatography run, the data will be stored unattenuated, allowing for re-scaling at a later time.

The Conductivity Monitor can be calibrated by either entering the cell constant or by calibrating the cell against a known standard. The "Conductivity Flow Cell Constant Calibration" feature is found under the Utilities pull-down menu. The cell constant is supplied with every conductivity flow cell and can be found on the tag attached to the cable.

2.4 BIO-RAD SYSTEM OPTIONS

Bio-Rad offers a number of system options for use with your BioLogic HR system. The following instrument options are described in this section.

- Signal Import Module-HR (SIM-HR)
- pH Monitor
- DynaLoop Large Sample Volume Loop
- Model EP-1 Econo Pump
- Models 2110 and 2128 Fraction Collectors
- Model 1327 Chart Recorder
- BioLogic HR Enviro-Case
- Uninterruptable Power Supply (UPS)
- AS-100 Autosampler

2.4.1 Signal Import Module-HR (SIM-HR)

For certain chromatographic applications, it may be desirable to observe and record additional detector information. This is accomplished using the Signal Import Module-HR (SIM-HR). The SIM-HR allows for connection of both a pH monitor and another detector which outputs an analog signal of from -2.5 Volts to +2.5 Volts. Instruments that may be connected in this way could include a variable wavelength UV detector, a refractive index detector, or a fluorescence detector. The SIM-HR digitizes the analog signal and transmits it to the BioLogic HR Controller. Up to two SIMs-HR may be connected to the BioLogic HR system.

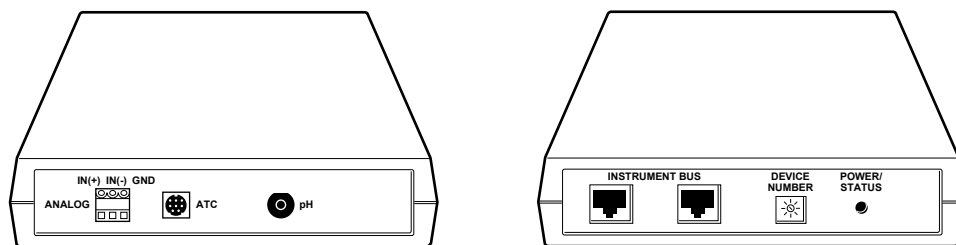
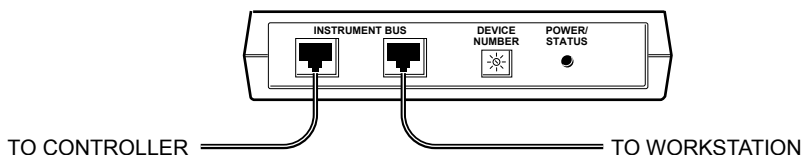


Figure 3. Signal Import Module-HR (front and rear views)

The output from the pH probe is connected to the pH connector (a BNC connector) on the SIM-HR; the output from any other detector is connected at the 3-pin connector (+, -, gnd) of the SIM-HR. The Automatic Temperature Compensation (ATC) connector is reserved for future upgrades and enhancements.

Cable 17 (the instrument bus cable) connects the SIM-HR to the Controller and the Workstation. (Refer to Figure 4.) If a second SIM-HR is to be used, one of the Instrument Bus cables is used to connect the two SIMs-HR. The SIM-HR is described in detail in its separate documentation.

CONNECTING A SINGLE SIM-HR



CONNECTING TWO SIMs-HR

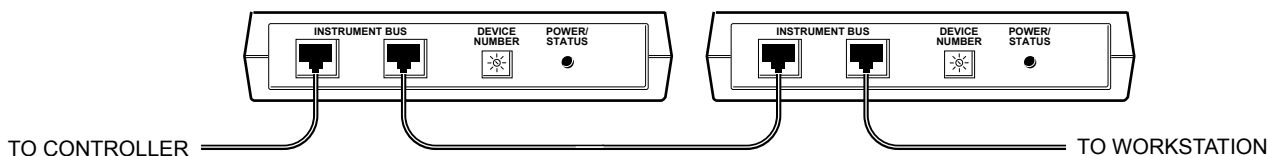


Figure 4. Cable Connections to the Signal Import Module-HR

2.4.2 pH Monitor

The pH Monitor provides real-time monitoring of buffer pH and is designed for flow-through pH recordings. The control circuitry for the pH Monitor resides in the BioLogic HR. The pH Monitor consists of the following:

- Flow cell with Inlet/Outlet ports: The flow cell is plumbed immediately after any backpressure device using 1/16" OD tubing and 1/4-28 flat bottom fittings. Either port may serve as the inlet port, leaving the remaining port as the outlet port. The swept volume in the flow cell is a nominal 50 μ l. A holder for the pH Monitor is built into the flow cell. The flow cell should be positioned so that the flow path is angled upward to promote bubble clearance.
- pH electrode: The pH electrode is a Ag/AgCl double junction combination electrode consisting of both a pH electrode and a reference electrode built into a single body. The sealed reference design eliminates the need to add filling solutions and minimizes reference dryout. The double junction design includes a chamber filled with 3.5M KCl between the Ag/AgCl electrode and the flow stream. Consequently, this pH electrode is fully compatible with buffers (such as Tris) that would otherwise interact aberrantly with Ag/AgCl.
- Signal Cable: For connection to the SIM-HR. Electrical power for is drawn off the signal cable.

The pH electrode and flow cell are described in detail in their separate documentation.

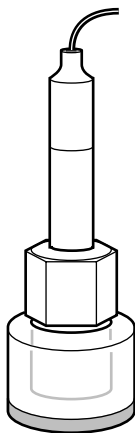


Figure 5. pH Electrode and Flow Cell

2.4.3 DynaLoop Large Sample Volume Loop

The DynaLoop is a sliding-piston (dynamic) loop which allows injection of a large sample volume, or allows repeat injections of a single sample. The DynaLoop is available in both a 25 ml and 90 ml size. It is plumbed identically to other sample loops, in ports 3 and 6 of the AV7-3 or V7-3 inject valve. For instructions on how to use the high pressure inject valve with the DynaLoop, refer to Section 4, Advanced System Applications. The DynaLoop is described in detail in its separate documentation.

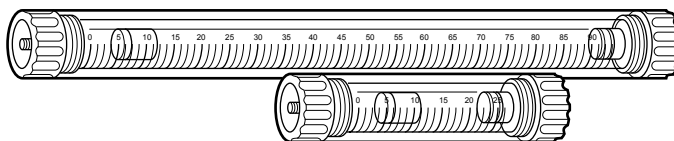


Figure 6. DynaLoop

2.4.4 Model EP-1 Econo Pump

The Model EP-1 Econo Pump is a two-channel, bi-directional, variable speed peristaltic pump for low-pressure chromatography. It offers a full range of features to facilitate ease of use as a stand-alone pump or as an accessory to the BioLogic HR System to load large volumes, especially onto low pressure columns. Refer to Chapter 8, Sample Loading, for instructions on using the EP-1 pump with the BioLogic HR System.

This instrument is described in detail in its separate documentation.

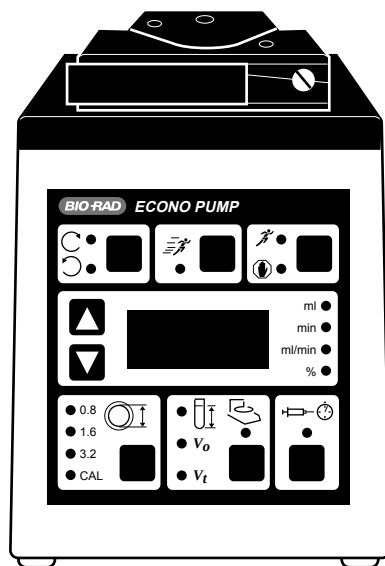


Figure 7. Model EP-1 Econo Pump

2.4.5 Model 2110 Fraction Collector

The Model 2110 uses a stationary, drop-dispensing head and collects up to 80 fractions in a motor-driven carousel. It uses standard 13 x 100 test tubes. An optional adapter is available for use with 1.5 ml microcentrifuge test tubes.

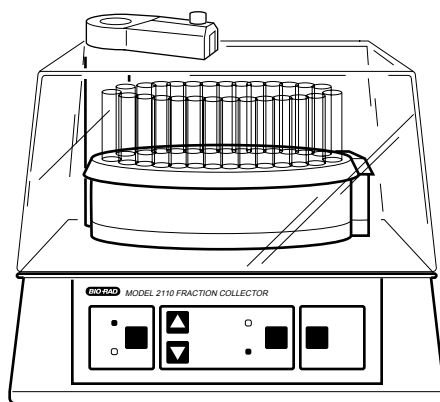


Figure 8. Model 2110 Fraction Collector

When controlled by the BioLogic HR system, fraction collection is specified by volume. The fraction collection scheme is programmed in the Method Editor software where a Delay Volume feature is standard. **Note:** Collection by time or drop count is not available with the BioLogic HR system.

The SV3-2 diverter valve is required by the Model 2110. When this valve is configured as the *fraction collector diverter valve* in the Setup screen, collection parameters such as Collect All, Threshold, Collection Windows, and Threshold + Collection Windows become available in the Method Editor.

The Model 2110 is connected to the BioLogic HR Workstation via the Workstation's AUX connector (black wire to pin 5 and white wire to pin 9). **System Cable 5** is included for this purpose.

The Model 2110 is plumbed to the BioLogic HR system using 1/16" OD inlet tubing which connects directly to the fraction collector's drop former, without the need for additional fittings.

This instrument is described in detail in its separate documentation.

2.4.6 Model 2128 Fraction Collector

The Model 2128 provides X/Y motion drop-dispensing across 5 available racks which accommodate a wide range of tube diameters and lengths, microtiter plates, microtubes, and “bottle size” fractions. It is suited for both analytical and preparative applications.

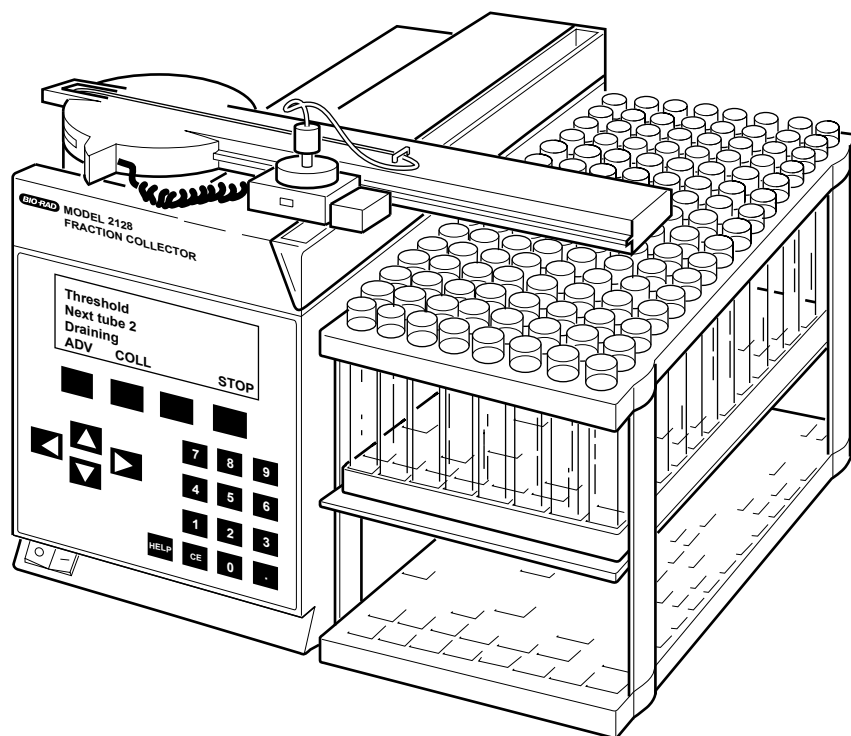


Figure 9. Model 2128 Fraction Collector

The following racks are available.

Rack name	Software ID number	Tube Description	Capacity
Rack #1 (Standard)	1	12 and 13 mm diameter; 180 mm height (max.)	128 (16 X 8) tubes
Rack #2	2	16 to 18 mm diameter; 180 mm height (max.)	78 (13 X 6) tubes
Micro-Adapter	3	Microplate (96 wells in 8 X 12 format)	3 plates (288 wells)
Micro-Adapter	4	Microtubes (capless tubes required)	128 microtubes
Prep-Adapter	5	Bottles of any size	10 bottles

The Model 2128 is controlled by the BioLogic HR Controller via the Instrument Bus. Look for an available Instrument Bus Connector on the rear of the collector. When the Model 2128 is configured in the Setup screen, collection schemes such as Collect All, Threshold, Collection Windows, and Threshold + Collection Windows (all with Delay volume, if required), automatically become available in the Method Editor. The starting and ending tube numbers also may be specified. The Model 2128's optional diverter valve mounts on the collector's arm and minimizes liquid spills during fraction advances. If this valve is used, it is automatically sensed by the system and hence is **not** configured by the user in the Setup screen.

Note: Model 2128 Fraction Collectors manufactured prior to 1995 may require the BioLogic HR Interface Kit, catalog number 750-0310.

The Model 2128 is plumbed to the BioLogic HR system using 1/16" OD inlet tubing and either of the following fittings:

- To connect directly to the dispenser head, use 1/4-28 flat bottom fittings.
- To connect to the optional on-arm diverter valve, use the 10-32 nut and ferrule fittings supplied with the valve.

This instrument is described in detail in its separate documentation.

2.4.7 Model 1327 Chart Recorder

The Controller outputs the UV analog data signal, pen up/down, and Stop/Start commands to the Model 1327 chart recorder via the Workstation's UV Chart connector (cabled to the chart recorder using **System Cable 2** mini DIN to standard DIN). The Conductivity analog data signal is sent from the Workstation's Conductivity Chart (**Cond Chart**) connector to channel 2 of the chart recorder using **System Cable 4** (mini-DIN to banana plugs). The chart recorder should be set to all green settings (1V). The chart recorder is described in detail in its separate documentation.

Note: The chart speed is set at the recorder. The BioLogic HR system does not control this function.

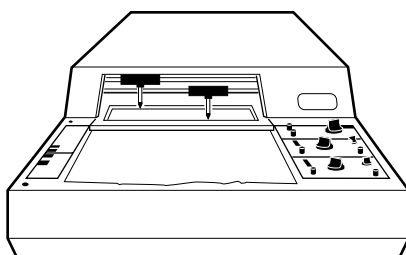


Figure 10. Model 1327 Chart Recorder

Event marks are recorded on the Chart Recorder for the following:

- Fraction Collector Advance: When the Fraction Collector advances to the next tube.
- A manual event mark from the Run screen. Pressing the **Event Mark** button on the Run screen allows you to "mark" events as they happen.

Non-Bio-Rad UV detectors: If the Model 1327 chart recorder is used in conjunction with a SIM-HR and a third-party detector to replace the standard BioLogic HR UV Detector, **System Cable 20** should be used to control the recorder. Channel 1 signals should be sent directly from the third-party detector using a bare wires-to-banana plug cable. In this case, the appropriate input voltage range must be selected on the recorder. Refer to the documentation for your non-Bio-Rad UV detector.

2.4.8 BioLogic HR Enviro-Case

The Enviro-Case is an enclosure for the BioLogic HR Controller and color display monitor which allows the system to be set up in a condensing, high humidity environment. It shields the moisture-sensitive components of the system from water contamination. For questions about the BioLogic HR Enviro-Case, consult with your local Bio-Rad representative.

2.4.9 Uninterruptible Power Supply (UPS)

In certain laboratory environments, a UPS may be required. A UPS is available through Bio-Rad in both 110 V and 220 V configurations. For questions about this UPS, consult with your local Bio-Rad representative.

2.4.10 AS-100 Autosampler

Bio-Rad's Model AS-100 Autosampler is a refrigerated, automatic sampler that provides automatic and unattended injections of up to 100 individual samples.

By creating a protocol to schedule the injection, you can increase throughput to the system. The installation and use of the AS-100 with the BioLogic HR is discussed in greater detail in Chapter 8, Sample Loading.

This instrument is described in detail in its separate documentation.

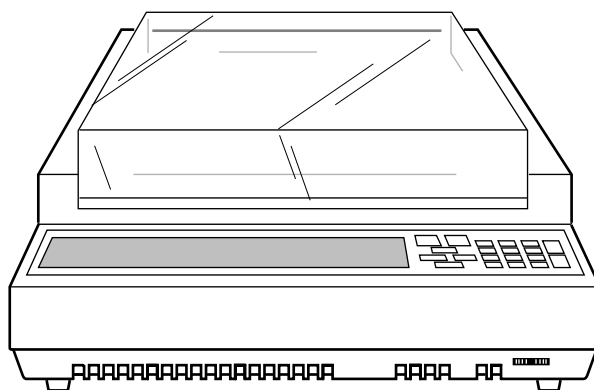


Figure 11. AS-100 Autosampler

2.5 NON-BIO-RAD SYSTEM OPTIONS

Non-Bio-Rad fraction collectors and chart recorders may be used with the BioLogic HR system, as discussed below:

2.5.1 Generic Fraction Collectors

A non-Bio-Rad collector may be used as an integral part of the BioLogic HR system providing its tube advance function can be initiated by an active low TTL signal (>100 ms). Electrical connection from the collector is to pin 5 (Frac Advance) and pin 9 (ground) of the Workstation's AUX connector. See the pin-out information in your collector manual.

With an optional SV3-2 diverter valve configured as a Fraction Collector Diverter Valve in the Setup screen, the BioLogic HR Controller will control all collection parameters including advanced features such as collection by Threshold and/or Collection Windows. If the optional valve is not used, then only Collect All is available.

Fraction advance marks to a Bio-Rad chart recorder are embedded on the UV signal from the Workstation.

2.5.2 Generic Chart Recorders

A non-Bio-Rad chart recorder may be used as an integral part of the BioLogic HR system. If the BioLogic HR system's UV Detector is used, a mini-DIN to breakout cable (**System Cable 7**) must connect the Workstation's UV Chart connector to the chart recorder. Pen Up/Down and Start/Stop functions are available providing the control polarity of the generic recorder is compatible. See the pin-out information in your chart recorder manual.

Conductivity signals require a Bio-Rad mini-DIN to banana plug cable (**System Cable 4**).

The input signal voltage should be set to 1 V for both BioLogic HR UV and Conductivity detector signals. Chart speed is set at the recorder itself, it is not controlled by the BioLogic HR system.

2.5.3 Printers

Any Windows-compatible printer is compatible with the BioLogic HR system. The printer must include a connecting cable and a printer driver (unless the printer driver is included in the Windows disks supplied with the BioLogic HR system).

To install a printer driver, exit the BioLogic HR software. If the printer driver disk is available, follow the instructions that are included with that disk. If the printer driver disk is not available, click on the Start button and from Settings, select Control Panel.

In the Control Panel, double-click on the Printers icon to open the Printers window. Then double-click on the Add Printer icon and follow the on-line instructions. (**Note:** If your printer does not appear in the list of printers available, contact the printer manufacturer for a recommendation on which driver to use.)

2.6 COLUMNS AND COLUMN FITTINGS

Bio-Rad offers a variety of column chemistries and formats. The following pages provide a summary of the different column types and column fittings for use with the BioLogic HR system. Details on each of these columns is provided in Appendix E, Column Information. Bio-Rad columns for the BioLogic HR use the following matrices:

- Bio-Scale chemistries are based on 10 µm Macro-Prep® supports. They are offered in various sizes and allow for scale-up separation and purification. These medium-pressure columns are ideally suited for use on the BioLogic HR system.
- The Econo-Pac cartridge chemistries are based on 50 µm Macro-Prep® supports and are ideal for first step purification.
- UNO columns use the new Continuous Bed matrix which contains an advanced polymer matrix that is completely homogeneous. These columns are superior to beaded supports in resolution, binding capacity, speed, and value.

2.6.1 Anion Exchange: Q Strong Anion Exchange

The Q-strong anion exchanger chemistry is available in the following formats:

- **Bio-Scale Q Prepacked Medium Pressure Columns:** These columns are designed for high resolution separations of proteins, peptides, and polynucleotides in analytical to semipreparative medium pressure applications. They are available in four column sizes. Methods developed on the Bio-Scale columns can be transferred to production scale using the Macro-Prep® supports. Refer to bulletins 1880, 1946, and 2079.
- **UNO Q Biochromatography Columns:** These columns are designed to handle separations at high flow rates with low back pressure. Instead of a traditional bed of packed beads or particles, each column contains an advanced polymer matrix, called the Continuous Bed matrix, which is nonporous and homogeneous. The matrix is designed to maximize resolution, binding capacity, and speed. Refer to bulletins 2116 and 1946.
- **Econo-Pac Q and High Q Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. They are based on 50 µm Macro-Prep® supports. Refer to bulletin 1946.
- **Macro-Prep® Q and High Q Support:** This is a strong anion exchanger containing quaternary amine functional groups. It is ideal for rapid purification of acidic and neutral proteins and peptides. Refer to bulletins 1840 (A-100), 1917, and 1985.

2.6.2 Anion Exchange: S Strong Cation Exchange

The S strong cation exchanger chemistry is available in the following formats:

- **Bio-Scale S Pre-packed Medium Pressure Columns:** These columns are designed for high resolution separations of proteins, peptides, and polynucleotides in analytical to semipreparative medium pressure applications. They are available in four column sizes. Methods developed on the Bio-Scale columns can be transferred to production scale using the Macro-Prep® 50 µm supports. Refer to bulletins 1881, 1946, 2026, and 2079.
- **UNO S Biochromatography Columns:** These columns are designed to handle separations at high flow rates with low back pressure. Instead of a traditional bed of packed beads or particles, each column contains an advanced polymer matrix, called the Continuous Bed matrix, which is nonporous and homogeneous. The matrix is designed to maximize resolution, binding capacity, and speed. Refer to bulletins 2116 and 1946.

- **Econo-Pac S Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. They are based on 50 μ m Macro-Prep® supports. Refer to bulletin 1985.
- **Macro-Prep® High S Support:** This is a strong cation exchanger containing sulfonic acid functional groups. It is ideal for purification of basic and neutral proteins and peptides. Refer to bulletins 1840 (A-200), 1917, and 2079.

2.6.3 Anion Exchange: DEAE Weak Anion Exchange

The DEAE weak anion exchanger chemistry is available in the following formats:

- **Bio-Scale DEAE Prepacked Medium Pressure Columns:** These columns are designed for high resolution separations of proteins, peptides, and polynucleotides in analytical to semipreparative medium pressure applications. They are available in four column sizes. Methods developed on the Bio-Scale columns can be transferred to production scale using the Macro-Prep® 50 μ m supports. Refer to bulletins 1930, 1946, 2026, and 2079.
- **Econo-Pac DEAE Blue Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. They are based on 50 μ m Macro-Prep® supports. Refer to bulletins 1946 and 2026.
- **Macro-Prep® DEAE Support:** This is a weak anion exchanger containing diethylaminoethyl functional groups. It is ideal for purification of acidic and neutral proteins and peptides. Refer to bulletins 1840 (A-400) and 1942.

2.6.4 Anion Exchange: Carboxy Methyl (CM) Weak Cation Exchange

The CM weak cation exchanger chemistry is available in the following formats:

- **Econo-Pac CM Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. They are based on 50 μ m Macro-Prep® supports.
- **Macro-Prep® High CM Support:** This is a weak cation exchanger containing carboxylate functional groups. It is ideal for purification of basic and neutral proteins and peptides. Refer to bulletins 1840 (A-300) and 1946.

2.6.5 Ceramic Hydroxyapatite (CHT)

The CHT chemistry is available in the following formats:

- **Bio-Scale CHT Type I Prepacked Medium Pressure Columns:** These columns are designed for high resolution separations of proteins, peptides, and polynucleotides in analytical to semipreparative medium pressure applications. They are available in four column sizes. Methods developed on the Bio-Scale columns can be transferred to production scale using the Macro-Prep® 50 μ m supports. Refer to bulletins 1929, 1946, 2026, 2079, and 2156.
- **Econo-Pac CHT-II Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. They are based on 50 μ m Macro-Prep® supports. Refer to bulletin 1946.

- **Macro-Prep® CHT Support:** Macro-Prep® ceramic hydroxyapatite (CHT) overcomes the physical and chemical instability of crystalline hydroxyapatite and is available in two types. Type I has a high protein binding capacity and better binding capacity for acidic proteins. Type II is better suited for proteins that elute early and for nucleic acids. Refer to bulletins 1842 (C-100), 1927, 1971, 1986, 2156, and CHTGEN.

2.6.6 Size Exclusion Chromatography (SEC)

The SEC supports are available in the following formats:

- **HPLC SEC Columns:** Bio-Sil and Bio-Select silica based columns separate compounds by the mechanism of size exclusion. The technique is based on diffusion in and around highly porous spherical silica beads. The columns are recommended for the separation of peptides, proteins, and nucleic acids, for desalting or buffer exchange, and for molecular weight or molecular constant determination. Columns are available in both stainless steel (Bio-Sil columns) and biocompatible PEEK plastic hardware (Bio-Select columns). Each column is shipped with a free vial of Bio-Rad's protein standards. Refer to bulletins 1737 and 1946.
Note: For columns packed with particles less than 10 µm, reductions in flow rate may be necessary in order to stay within the 1,000 psi limit of the BioLogic HR system.
- **Econo-Pac P6 Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. Refer to bulletin 1946.
- **Macro-Prep® SE Support:** These gels are used for applications ranging from the separation of protein aggregates (dimers, trimers, etc.) from purified antibody solutions to the separation of complex mixtures of bio-molecules directly from cell culture feedstreams. SEC is commonly used as a final polishing step in the preparation of many bio-pharmaceuticals and has numerous applications as a quality control tool for assessing product purity and concentrations. Refer to bulletin 2190.

2.6.7 High Pressure Reversed Phase Columns

The following type of column is available:

- **Hi-Pore reversed phase columns:** These columns are commonly used for the purification and analysis of small proteins (<50 kd), peptides, oligonucleotides, and amino acids. Refer to bulletin 1946.
Note: For columns packed with particles less than 10 µm, reductions in flow rate may be necessary in order to stay within the 1,000 psi limit of the BioLogic HR system.

2.6.8 Hydrophobic Interaction Chromatography (HIC)

The HIC chemistry is available in the following formats:

- **Econo-Pac t-Butyl HIC Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. They are based on 50 µm Macro-Prep® supports. Refer to bulletins 1946 and 2079.
- **Econo-Pac Methyl HIC Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. Refer to bulletin 1946.

- **Macro-Prep® HIC Support:** HIC is offered in both methyl and t-butyl chemistries. The Macro-Prep® methyl HIC support is ideal for purification of proteins with strongly hydrophobic regions. The Macro-Prep® t-butyl HIC support is ideal for purification of proteins with few or weakly hydrophobic regions. Refer to bulletin 1841.

2.6.9 Affinity Chromatography

Affinity chromatography is available in the following formats:

- **Econo-Pac Protein A Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. Refer to bulletin 1946.
- **Econo-Pac Heparin Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. Refer to bulletin 1946.
- **Econo-Pac Blue and DEAE Blue Low Pressure Chromatography Cartridges:** These cartridges are available in 1 ml and 5 ml formats to accommodate most sample loads. They are recommended for method scouting and for first step purification of crude samples. Refer to bulletin 1946.

2.6.10 Empty Columns

The following columns are available:

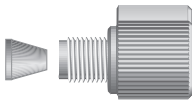
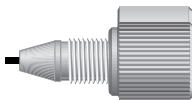
- **Bio-Scale MT empty columns:** These columns allow for easy packing of a chromatographic media, bed height adjustment, sample application, and equilibration. The availability of four column sizes (2, 5, 10, and 20 ml) allows easy scale-up of separation and purification protocols. Request bulletin 1970.
- **Glass Econo-Column columns:** Econo-Column chromatography columns are the standard for high quality, affordable low pressure chromatography columns. They accept both Econo-Column funnels and flow adapters.

2.6.11 Column Fittings

Table 6.
Columns and Column Fittings

Column Type	Description	Fittings Required
Econo-Pac® Cartridge	The Econo-Pac cartridge has one male and one female luer-lock fitting. To connect it to the BioLogic System use 1/4-28 to male and 1/4-28 to female luer adapters, provided in the kit shown to the right. One set is included in the BioLogic System Fittings Kit.	Catalog # 732-0113 Econo-Pac® Cartridge to BioLogic System Fittings Kit 
Econo-Column®/ Low Pressure Columns and flow adapters	Econo-Columns have two male luer-lock fittings. To connect to the BioLogic HR system, use female luer to 1/4-28 adapters. If a flow adapter with flexible tubing is to be used, attach the barbed-to-male luer fitting (included in this kit), then connect the female luer to 1/4-28 adapter. If the column adapter uses PTFE tubing, attach the small length of 1/16" ID Tygon® tubing and insert the barbed male luer fitting into open end. Five cm of Tygon tubing is included in this kit. Connect as above.	Catalog # 750-0565 Econo-Column® to BioLogic System Fittings Kit  This kit includes all the parts needed to connect Econo-Columns and flow adapters to the BioLogic HR system.

Table 6. (continued)
Columns and Column Fittings

Column Type	Description	Fittings Required
FPLC® Column	Pharmacia's FPLC columns use M-6 fittings. To connect to the BioLogic System, use 1/4-28 to M6 adapter union. Two of these are supplied with the BioLogic Fittings Kit.	Catalog # 750-0561 Union, 1/4-28 to M6 
HPLC Column, or Pharmacia's RESOURCE™ Column	These columns accept 10/32 nuts. To connect this type of column to the BioLogic System, use two 1/4-28 to 10-32 adapters.	Catalog # 750-0564 HPLC Column to BioLogic System Adapters 
UNO™ Column to FPLC system	This kit includes two nuts and four ferrules to connect an UNO column to an FPLC system.	Catalog #750-0567 M6 Fittings Kit 
UNO™ Column to HPLC system	This kit includes two nuts and four ferrules to connect an UNO column to an HPLC system.	Catalog #750-0568 10-32 Fittings Kit 

3.0 SYSTEM CONFIGURATIONS

3.1 SYSTEM CONFIGURATIONS

The BioLogic HR system is modular, so that you can configure your system in an arrangement that is most comfortable to you and best meets your application requirements. When planning a configuration, consider the amount of bench space available, the instruments to be used with the system, and the devices to be used with the System Rack. The BioLogic HR system can be used with Bio-Rad as well as non-Bio-Rad components.

The BioLogic HR system can be set up wherever there is available bench space and sufficient power outlets, including in 4° cold rooms or cold boxes. For cold box or cold room use where space is limited, the compact design of the Workstation and rack and use of the single instrument bus cable to the Controller allows the Controller to be placed in a more convenient position outside the cold box (see Fig. 7 below).

If the BioLogic HR Controller and its Color Monitor **must** be used in a condensing environment, they should be placed in the system's optional protective environmental enclosure, the BioLogic HR Enviro-Case.

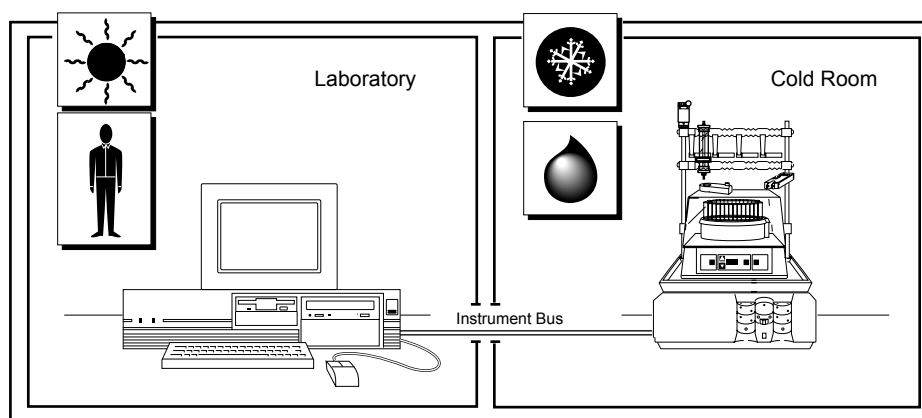


Figure 12. BioLogic HR System as it might be used with a Condensing Cold Room

All BioLogic HR systems will include as a minimum the following:

- BioLogic HR system Controller and Color Monitor, keyboard, and mouse
- BioLogic HR system Workstation
- BioLogic HR system Rack (single tray configuration)
- UV and Conductivity Detectors
- Fittings kit
- Starter kit

A variety of instruments and devices may be arranged in many different configurations. To simplify discussion, while still addressing the needs of most users, we have identified two types of configurations: The Manual BioLogic HR system and the Automated BioLogic HR system. Both systems are described as equipped with the Bio-Rad Model 1327 Dual Pen Chart Recorder. For the purpose of this manual, the two systems are distinguished by the following:

Start by looking over the discussion for each type of system, and then begin setting up your system by following the discussion which most closely describes your system's configuration. For the first time user, set-up and plumbing of the BioLogic HR system requires approximately 2 hours.

Note: Your BioLogic HR system may include components or instruments not listed in the manual or automated systems described above. Refer to the separate supporting documentation for each of the following types of devices:

- additional valves
- additional detectors
- instruments not purchased from Bio-Rad, such as a generic fraction collector or chart recorder

3.2 EXAMPLE 1 OF A SYSTEM CONFIGURATION: MANUAL BIOLOGIC HR SYSTEM

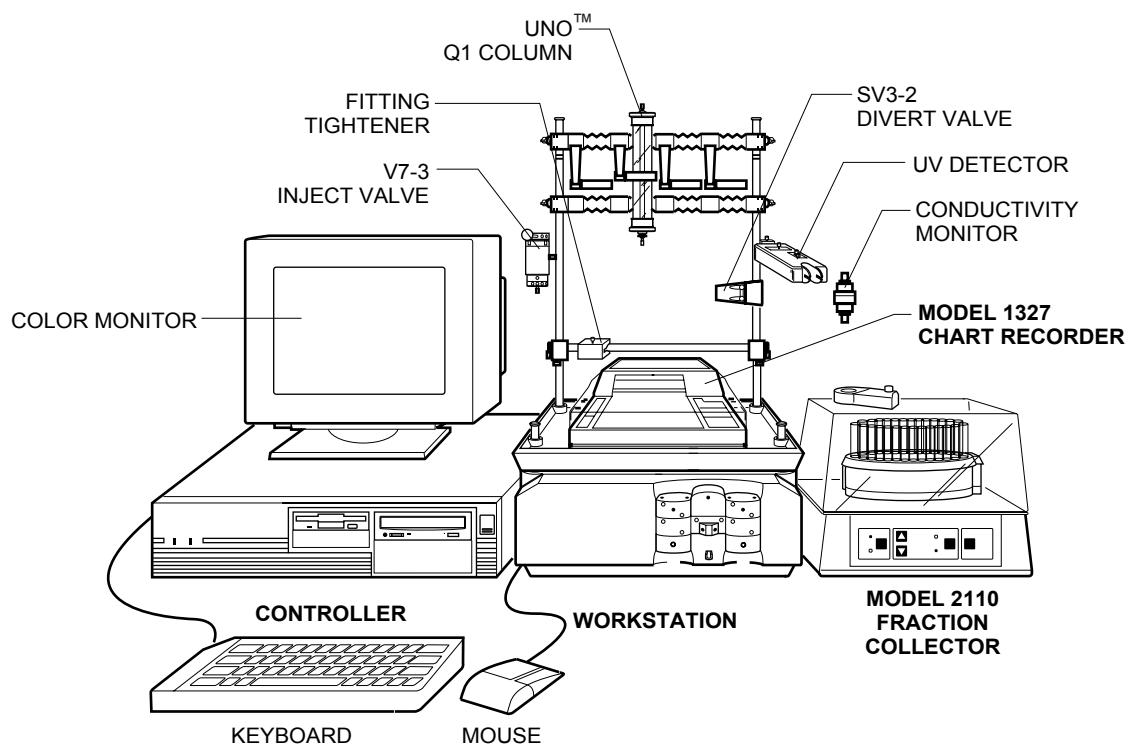


Figure 13. Manual BioLogic HR System

To set up the Manual BioLogic HR system:

1. BioLogic HR Controller.

- a. Place the BioLogic HR Controller on the bench with the color monitor on top, and turn both units so that the back side is facing you.
- b. Connect the monitor's cable connector to the connector marked "**Video.**"
- c. Connect the keyboard's cable to the connector marked "**Keyboard.**"
- d. Connect the mouse cable to the connector marked "**Mouse.**"
- e. Connect the bus cable (**System Cable 17, 18, or 19**) to the connector marked "**Inst Bus.**"
- f. Connect the power cables to the Controller and monitor. **Do not turn on these devices yet.**

2. BioLogic HR Workstation.

- a. Place the BioLogic HR Workstation next to the Controller so that the back side of the Workstation is facing you.
- b. Note there are two instrument bus cable connectors marked "**Bus.**" Either of the bus cable connectors may receive the instrument bus cable from the Controller. The other bus connector is for connection to other devices such as a fraction collector Model 2128 or a Signal Import Module.
- c. Confirm the voltage setting on the power supply. A red switch on the back of the Workstation allows you to switch between 100-110 V AC and 220-240 V AC.
- d. Connect the power cable to the Workstation. **Do not turn on this device yet.**
- e. You should now add the rack to the Workstation.

3. System Rack.

Assemble the Rack and place it on the Workstation, as described below.

- a. Fit the tapered collars into the ringed grooves on the rods. The collars are tapered so that when they are attached to the rods and the rods are fitted into the holes at each end of the tray, the rods and collars serve as the legs of the tray. With this in mind, note the following guidelines:
 - The long rods have several grooves. Attach the collars to the groove closest to the end of the rod. It doesn't matter which groove you use, as long as it is closest to the end of the rod. Attach the collars so that the taper is toward the end of the rod.
 - The short rods each have only one groove. Fit the collars onto the grooves so that the collars flare out toward the farthest end of the rod. Refer to Figure 14.
- b. Insert the two long rods into the holes at the side of the tray containing the drain.
- c. Insert the two short rods into the remaining holes.

Note: The rods are inserted from *underneath* the tray such that they produce a firm fit in the holes.
- d. Mount the 2-piece column clamping arrangement across the two long rods using the attached thumbscrews.
- e. The horizontal bar may be positioned between any of the upright bars using the rod clamps (tightened by the Allen wrench) supplied with the system.
- f. Place the rack into the four corner holes of the Workstation. Since the back of the Workstation is facing you, you will want the back of the rack (the side with the two long rods) also facing you. If a hole is covered by a green cap, remove the cap.
- g. Install the green metal Fitting Tightener in a convenient and accessible location on the rack.

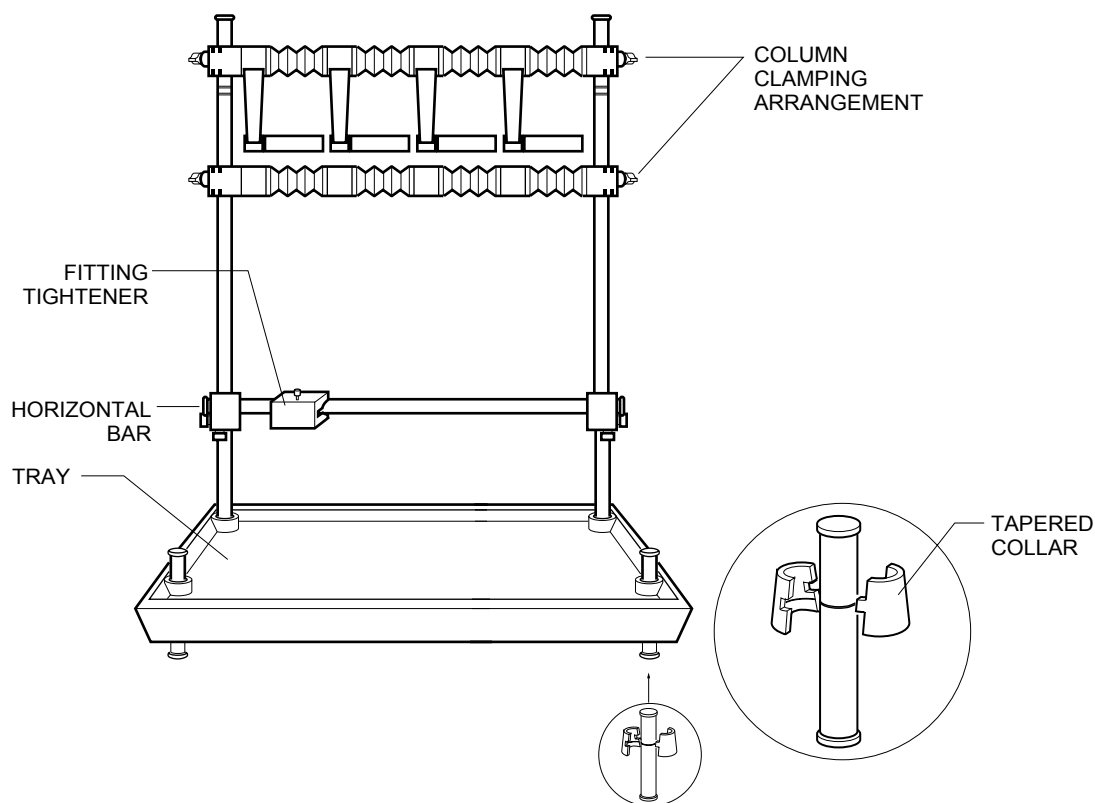


Figure 14. Rack Assembly

4. UV and Conductivity Detectors.

The procedure for connecting the UV and Conductivity Detectors provided with your system is described below. To connect a non-Bio-Rad UV Detector to the system, refer to the procedure for connecting a SIM-HR.

- Attach the UV Detector to a suitable position on the vertical or horizontal bars using the rod clamps.
- Connect the power cable (square connector) from the UV Detector lamp into the connector marked **"UV Lamp"** on the rear of the Workstation. Refer to Figure 15.
- Connect the UV Detector signal cable (mini-DIN connector) to the connector marked **"UV Optics"** on the rear of the Workstation. Refer to Figure 15.
- Connect the Conductivity Detector's combined power and signal cable (mini-DIN connector) to the connector marked **"Cond. Flowcell"** on the rear of the Workstation. Refer to Figure 15.
- The Conductivity Detector is designed to be held in the circular notch of the optics bench, but may be placed anywhere in the fluid path as desired by the user.

5. Chart Recorder Model 1327.

The chart recorder Model 1327 may be positioned on the rack shelf or on the bench.

- a. Connect control cable **System Cable 2** between the Workstation and the recorder as follows:

Note: To connect a third-part UV detector, refer to the discussion for connecting the SIM-HR.

- The mini-DIN connector is connected to the connector marked “**UV Chart**” on the rear of the Workstation. Refer to Figure 15.
- The DIN connector is connected to the single DIN connector on the side of the chart recorder.

This cable provides system control of pen up/down, event marks and paper advance, but chart speed **MUST** be set on the recorder faceplate itself.

- b. Conductivity signals are recorded on channel 2 of the recorder using **System Cable 4** as follows:

- The mini-DIN connector is connected to the connector marked “**Cond. Chart**” on the rear of the Workstation. Refer to Figure 15.
- The banana plugs are connected to the connectors marked **CH 2** on the side of the recorder (red wire to +, black wire to ground $\perp$).

- c. Set both channel inputs on the recorder to 1V. Set all other switches to their position marked in green. Connect the power adapter to the chart recorder.

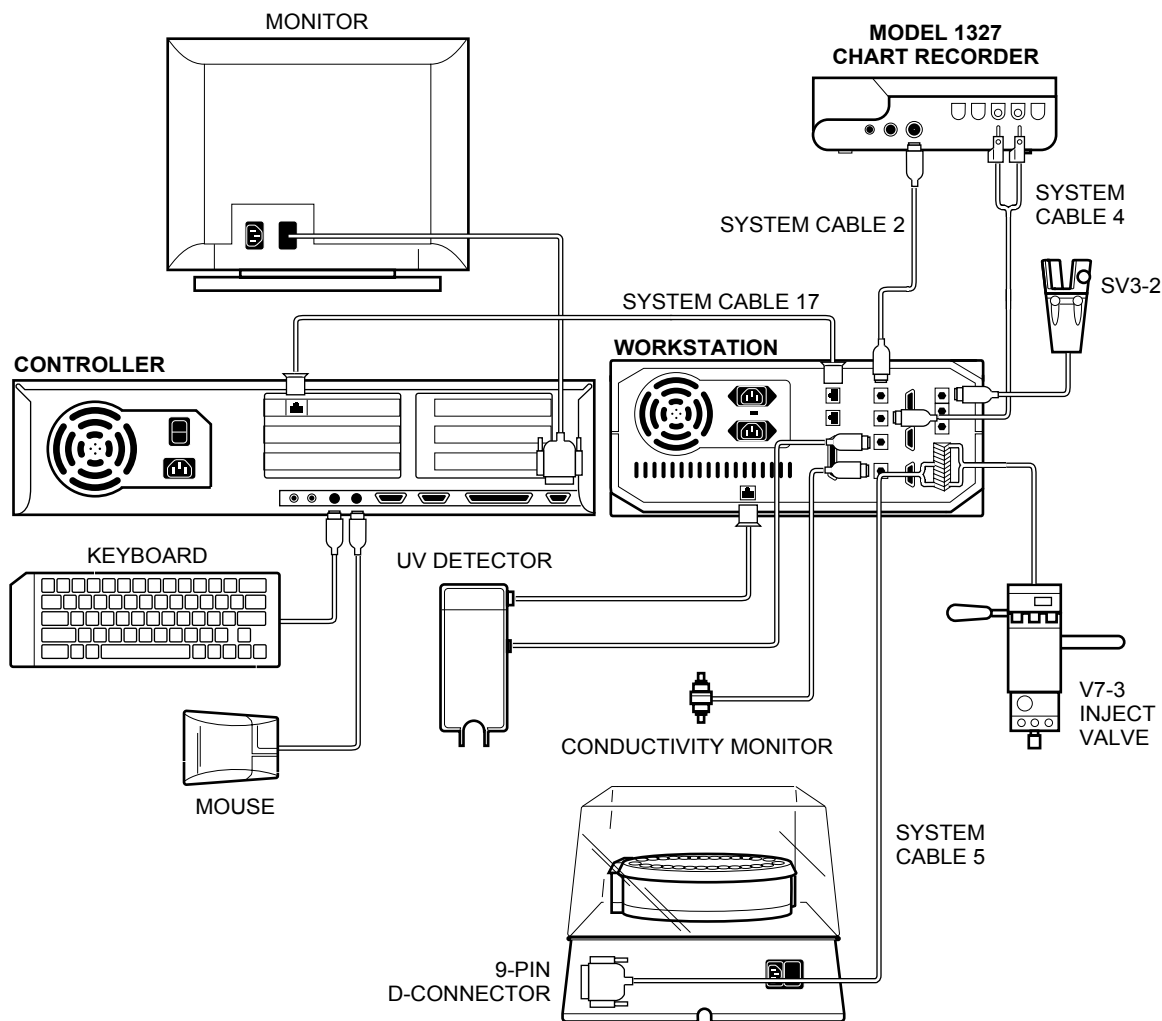


Figure 15. Manual System Cable Connections

6. Manual Injection Valve V7-3.

- Attach the manual injection valve V7-3 to a suitable position on the vertical or horizontal bars using the rod clamps. This valve contains a position-sensing device (for LOAD and INJECT positions).
- Connect one wire to pin #1 (labeled “**Inject**”) of the AUX connector strip on the rear of the Workstation. Refer to Figure 15.
- Connect the remaining wire to pin #9 (labeled “**GND**”, ground) of the AUX connector strip. There is no polarity to the wires, so they can be connected to either pin.

7. Fraction Collector Model 2110.

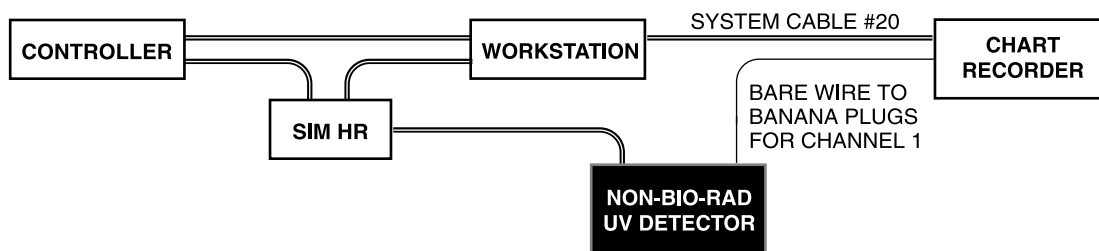
- The fraction collector Model 2110 may be positioned on the rack shelf or on the bench.
- Connect the control cable (**System Cable 5**) between the Workstation AUX connector and the fraction collector as shown in Figure 15 and described below:
 - The 9-pin “D” connector is connected into the single “D” connector on the rear of the Model 2110.
 - At the Workstation’s AUX connector strip, connect the black wire (Advance) to AUX pin #5 (labeled “**FC ADV**”, fraction collector advance) and the white wire to AUX pin #9 (labeled “**GND**”, ground). Refer to the separate Instruction Sheet for the wiring diagram and color code for System Cable 5.
- Connect the power cable.

8. Fraction Collector Diverter Valve SV3-2.

- Position the SV3-2 valve on the rack using the integral clamp.
- Connect the valve cable mini-DIN connector to any connector marked “**Solenoid Valves**” on the rear of the Workstation.

9. SIM-HR.

To connect the SIM-HR for use with optional or non-standard instruments, refer to the separate documentation provided with the SIM-HR and for the instrument to be used. If you will be connecting a non-Bio-Rad UV detector, refer to the illustration below, which shows how channel 1 signals are sent to the chart recorder by connecting the UV detector to the chart recorder using bare wires to banana plugs. **System Cable 20** transmits start/stop, pen up/down, and event mark signals.



When all the connections have been made, turn the units around and connect the other end of the instrument bus cable (**System Cable 17**) to the Controller. Plug all the power cables into an approved surge-protected power-strip and turn on all devices. The system can now be plumbed.

When all the connections have been made, turn the units around and connect the instrument bus cables (**System Cable 17**) to the Controller and the Workstation. Plug all the power cables into an approved surge-protected power-strip. In certain laboratory environments, a UPS may be required. These are available through Bio-Rad in both 110 V and 220 V configurations. For questions about this UPS, consult with your local Bio-Rad representative. The system can now be plumbed and all devices may be turned on.

3.3 EXAMPLE 2 OF A SYSTEM CONFIGURATION: AUTOMATED BIOLOGIC HR SYSTEM

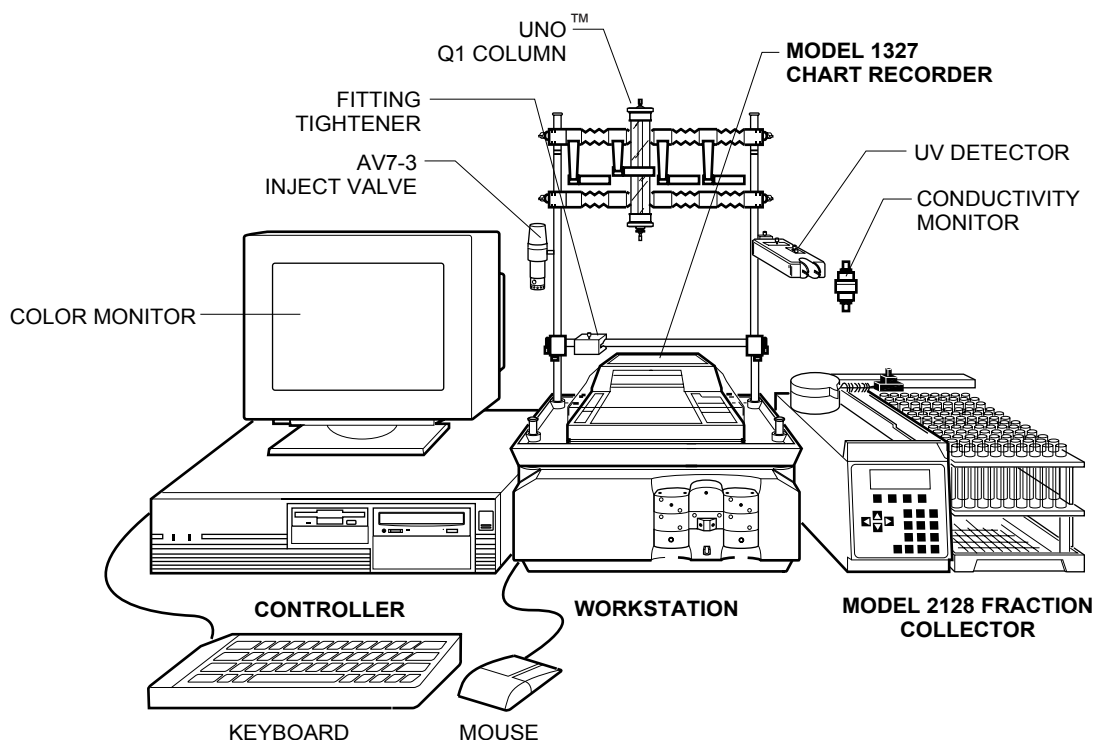


Figure 16. Automated BioLogic HR System

To set up the Automated BioLogic HR system:

1. Follow steps 1 to 5 above as for the Manual BioLogic HR system. Refer to Figures 15 and 16 when connecting the Automated BioLogic HR system.
2. **Automated Inject Valve AV7-3.**
 - a. Attach the automated injection valve AV7-3 to a suitable position on the vertical or horizontal bars using the rod clamps.
 - b. Connect the valve cable 15-pin D connector to any connector marked “**Automated Valves**” on the rear of the Workstation.
3. **Model 2128 Fraction Collector.**

Connect an instrument bus cable (**System Cable 17**) to the connector marked “Bus” on the rear of the Collector.

Note: For early versions of the Model 2128 Fraction Collector, a new interface board containing two instrument Bus connectors must first be installed in the rear of the Collector. Contact your local Bio-Rad representative for information about receiving this upgrade (Model 2128 BioLogic HR Interface Kit, Catalog Number 750-0310).

When all the connections have been made, turn the units around and connect the instrument bus cables (**System Cable 17**) to the Controller and the Workstation. Plug all the power cables into an approved surge-protected power-strip. In certain laboratory environments, a UPS may be required. These are available through Bio-Rad in both 110 V and 220 V configurations. For questions about this UPS, consult with your local Bio-Rad representative. The system can now be plumbed and all devices may be turned on.

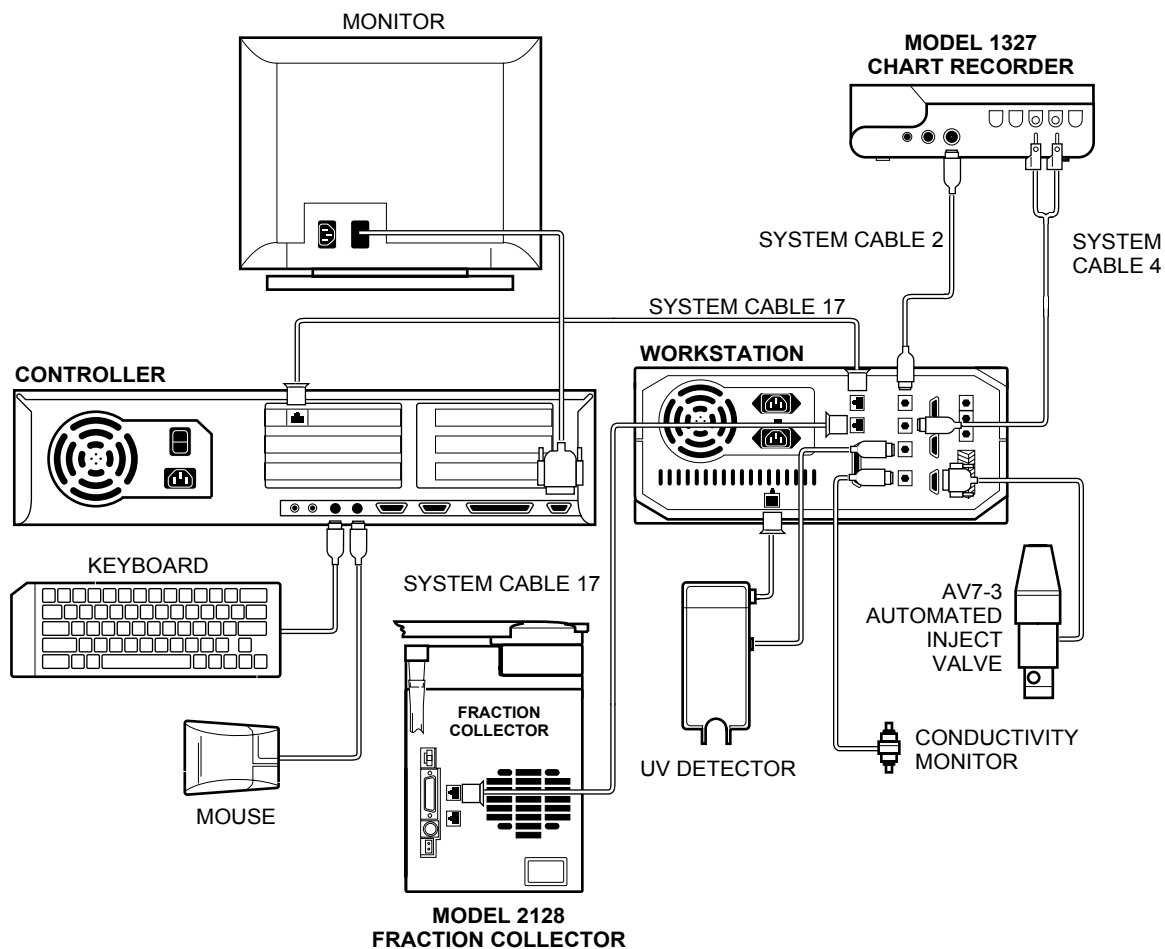


Figure 17. Automated BioLogic HR System Cable Connections

4.0 SYSTEM PLUMBING

The devices you choose and the plumbing arrangements you make may change for each application. For this reason, this chapter begins with discussion of recommended plumbing practices and general guidelines, followed by a general procedure for plumbing the major system components. Refer to Section 4, Advanced System Applications, for plumbing these valves for specific applications.

4.1 GENERAL GUIDELINES FOR PLUMBING THE SYSTEM

The BioLogic HR system uses two sizes of tubing:

- For inlets to the Workstation Gradient Pump, it is essential to use the wide-bore tubing supplied with your system, which is 1/8" OD (0.062" ID) PTFE. The approximate volume of 1 cm of this tubing is 20 µl. Using smaller bore tubing will not provide sufficient flow of liquid to the Gradient Pump. This will lead to entrapment of air bubbles in the pump heads which in turn cause an erratic delivery of liquid and may stall the pump.
- The remainder of the system is plumbed with 1/16" OD (0.020" ID) Tefzel tubing. The approximate volume of 1 cm of this tubing is 2 µl.

When plumbing the system, be sure to keep tubing lengths to a minimum. Also, all fittings should be finger-tight. Do not over-tighten, or you risk damaging the connection.

Ferrule Installation (1/8" and 1/16" tubing)

1. Slide the nut, stainless steel lock ring, and the brown PEEK ferrule, in that order, onto the tubing as shown above. The flattened end of the lock ring should face towards the nut with the tapered end facing the tapered end of the brown ferrule.

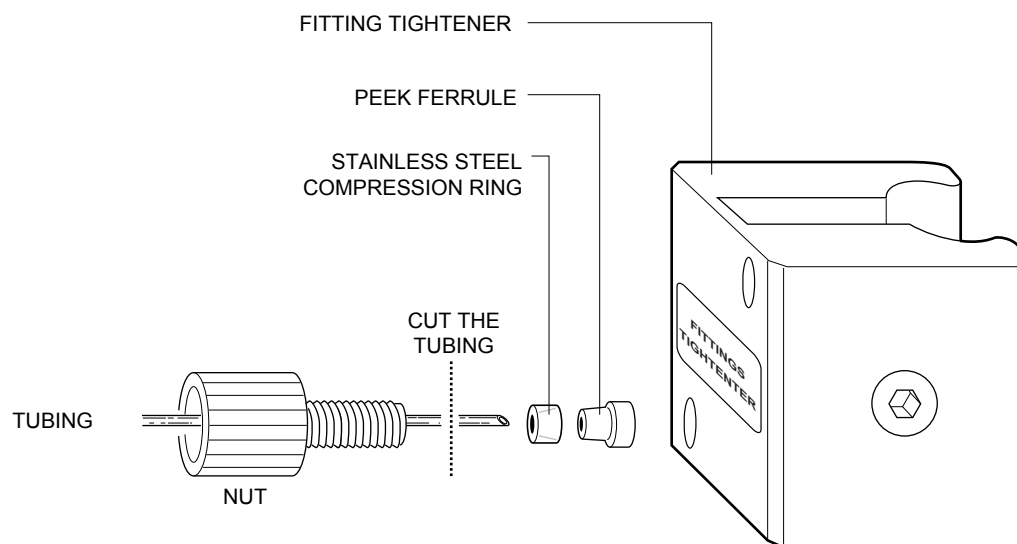


Figure 18. Making 1/4-28 flat bottom fittings

2. Allow tubing to extend slightly beyond the end of the ferrule.
3. To seat the ferrule, use the green fitting tightener which is attached to the BioLogic HR Rack. Alternatively, use the black 1/4-28 cap which is included in the fittings kit.

Note: The stainless steel lock ring is *not* in the fluid path, so biocompatibility is maintained.

4.2 EXAMPLE OF A PLUMBING ARRANGEMENT

1. **Plumbing the Workstation Gradient Pump Inlets to the Buffer Reservoirs (Figure 19).**
 - a. Cut two suitable lengths of 1/8" OD PTFE using the tubing cutter from the fittings kit.
 - b. Attach a 1/4-28 flat bottom fitting and PEEK ferrule and lock ring to one end of each piece of tubing.
 - c. Screw the tubing into the inlet connectors on the bottom of the Workstation Gradient Pump housing. **Ensure a firm connection but do not over-tighten.**
 - d. Immerse the inlet lines in a container of high quality, filtered water (e.g., HPLC grade) or buffer.

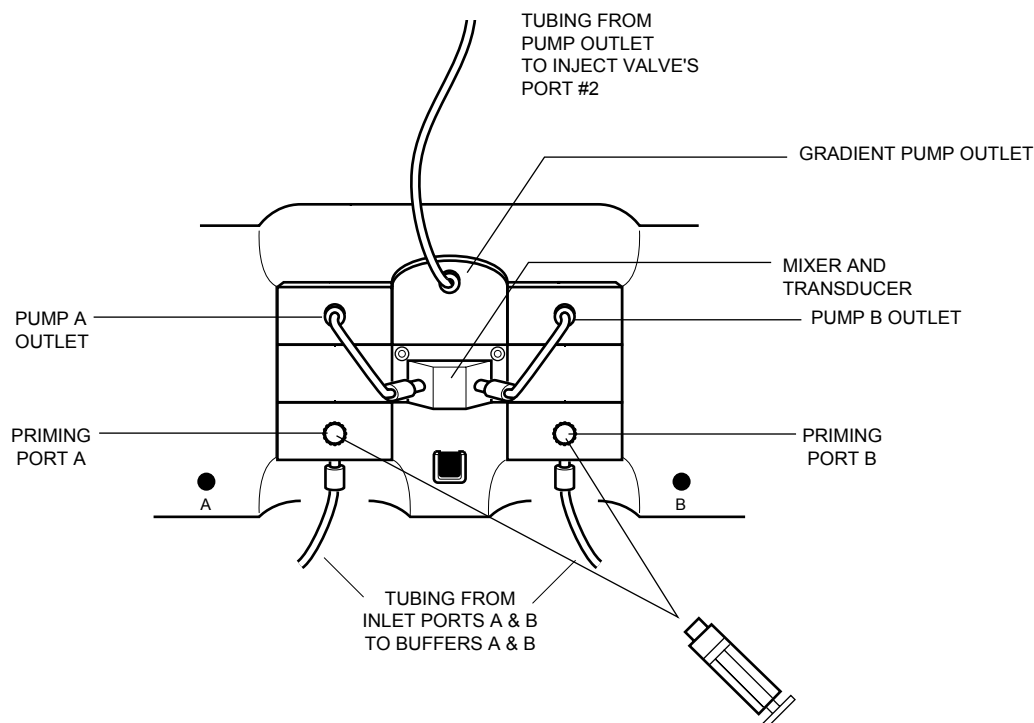


Figure 19. Plumbing Connections to the Gradient Pump

2. Plumbing from the Gradient Pump to the Injection Valve.

- Cut a suitable length of 1/16" OD Tefzel tubing that will reach from the Gradient Pump outlet to port #2 of the inject valve. Refer to Figure 20.
- Attach a 1/4-28 flat bottom fitting and ferrule to each end, as described earlier.
- Connect the tubing to the threaded port on the pump outlet and to port #2 on the inject valve and secure firmly using a twisting action. **Ensure a firm connection but do not over-tighten.**

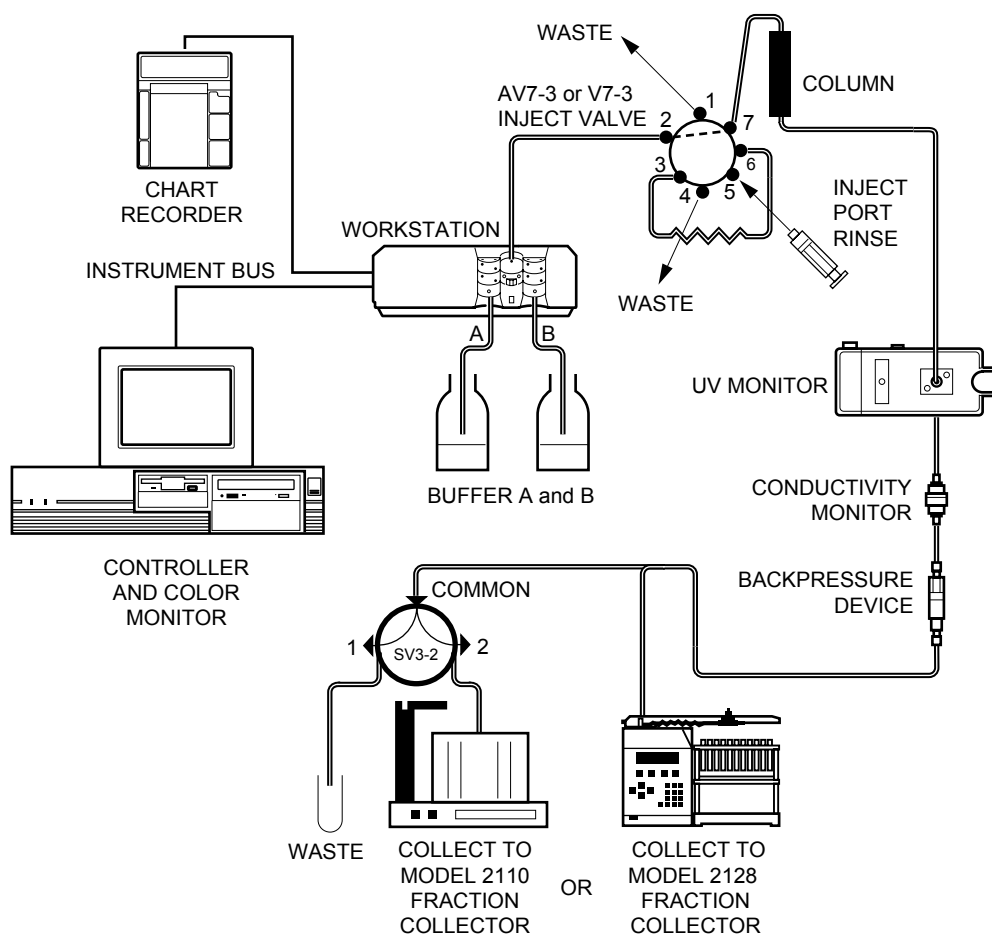


Figure 20. System Plumbing

3. Plumbing the Inject Valve V7-3 or AV7-3.

Both manual and automatic valves are plumbed in an identical manner.

- First locate the inject port assembly and the inject needle. Gently place the needle into the inject port assembly until resistance is felt. Screw this assembly into port #5 of the valve until it is secure and then withdraw the needle.
- Plumb the inject valve according to Figure 21. Additional information is located in section 4.4.

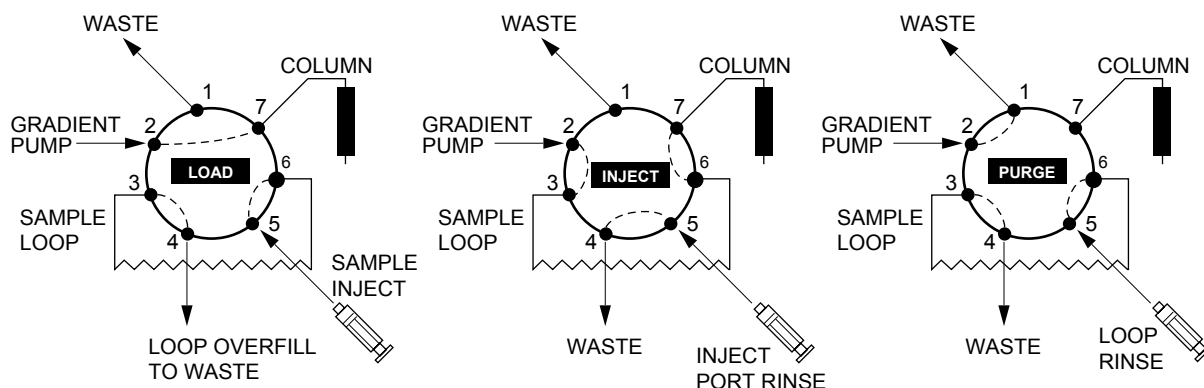


Figure 21. Inject Valve Plumbing for V7-3 or AV7-3

4. Plumbing the UV and Conductivity Detectors.

- Connect a piece of 1/16" OD Tefzel tubing between port #7 of the inject valve and the top of the UV Detector flow-cell using the 1/4-28 fittings. **Note:** The column will be positioned between port #7 and the UV detector **after** the system is purged of air.
- Cut approximately 20 cm of 1/16" OD Tefzel tubing and attach 1/4-28 fittings and ferrules to each end.

Screw one end into the bottom of the UV Detector flow-cell and the other end into the Conductivity Detector flow-cell (note that this cell is bi-directional).

- Place the Conductivity flow cell into the notch of the optics bench. This is a gentle push-fit. Note that there is a tag (with a number) attached to the conductivity cable. This number is the flow cell constant and should be entered in the software before beginning a run.
- As you pump buffer through a column, the column exerts a backpressure which serves to keep any air bubbles in solution. However, when the buffer exits the column, it returns to atmospheric pressure and air bubbles may "out-gas." As they pass through the flow cell, these bubbles may cause artifacts such as an unstable baseline. This "outgassing" is minimized by thoroughly degassing the buffer and by placing a 40 psi backpressure device **after** the UV Detector flow cell. This backpressure device puts a small backpressure on the flow cell, keeping the bubbles in solution.

Note: If using low pressure columns such as an Econo-Pac® cartridge, plumb the backpressure device between the Gradient Pump outlet and the injector valve to aid in seating the check valves to prevent permanent damage to the cartridge or column.

- Be sure to plumb the backpressure device following the direction of the arrow.

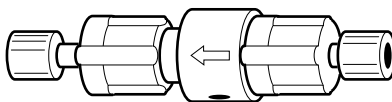


Figure 22. Backpressure Device

5. Connection to a SV3-2 Fraction Collector Diverter valve and a Model 2110 Fraction Collector.

- a. Connect a piece of 1/16" OD Tefzel tubing between the backpressure device and the SV3-2 valve common port (labeled "**Common**") using the 1/4-28 fittings.
- b. Connect a piece of 1/16" OD Tefzel tubing from port 1 of the SV3-2 valve (see figure 20) for use as a waste line.
- c. Connect a piece of 1/16" OD Tefzel tubing from port 2 of the SV3-2 valve (see figure 20) to the Model 2110 fraction collector drop-head. This tubing sits inside the drop-head, and no fittings are required.

6. Connection to a Model 2128 Fraction Collector.

Connect a piece of 1/16" OD Tefzel tubing between the backpressure device and the drop-head of the Model 2128 or the 2128 diverter valve. (For complete instructions, refer to the documentation for the Model 2128 Fraction Collector.)

Note: The tubing's length must allow unrestricted movement of the fraction collector arm.

4.3 PLUMBING BIO-RAD'S LOW PRESSURE VALVES

The BioLogic HR system may be used to control and monitor the following Bio-Rad low pressure solenoid valves:

- SV5-4 Buffer Select Valve
- SV3-2 Diverter Valve

4.3.1 Plumbing the SV5-4 Buffer Select Valve

The SV5-4 Buffer Select Valve is a low pressure, 5-port, 4-position valve primarily used for automatic buffer selection when placed before the Gradient Pump. It may also be used for buffer and/or sample selection when placed before an Auxiliary Load Pump, or for fraction collection when placed after the column. This valve is described in detail in its separate documentation. An example of how this valve is used is described in Chapter 8, section 8.1.

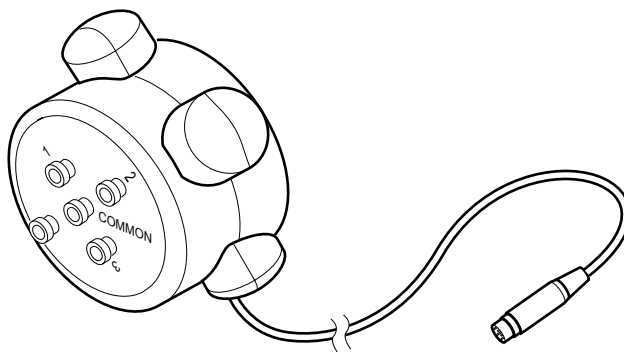


Figure 23. SV5-4 Buffer Select Valve

To install and plumb the valve:

1. Mount the SV5-4 valve to the System Rack. Select a location which will keep tubing lengths to a minimum.
2. Connect the SV5-4 valve's combined power and signal cable to any of the available Solenoid Valve connectors on the back of the Workstation (ports 1,2 or 3).
3. Use 1/8" OD PTFE tubing when plumbing the SV5-4 valve as a Gradient Pump inlet valve.

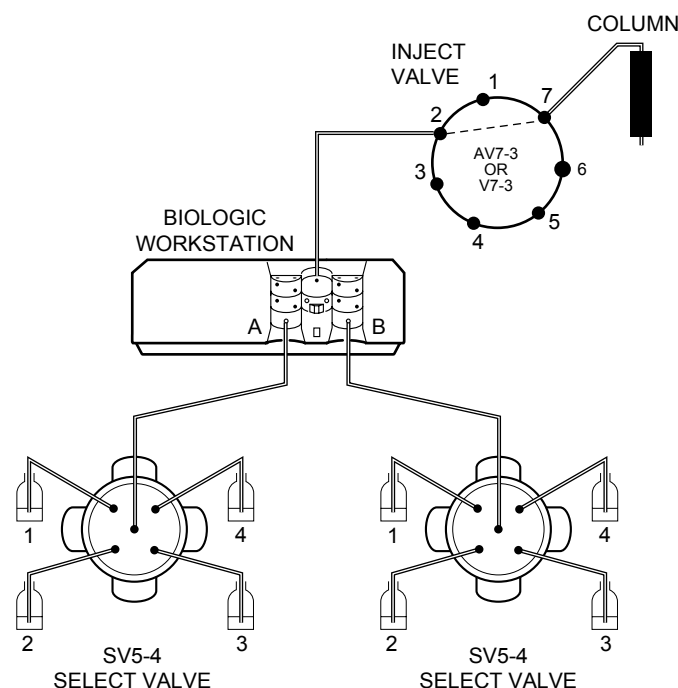


Figure 24. Configuration of the SV5-4 as a Gradient Pump Inlet Buffer Select Valve

The software's Manual mode lets you manually operate each inlet port on the SV5-4. Manual valve control is useful for the following:

- When priming the tubing and the valves with buffer and or sample prior to starting a method.
- For purging all tubing lines for cleaning purposes.

When programming a Method, the Setup screen lets you define each of the four valve positions by name. This is known as "aliasing," and it allows you to identify the valve by the name or composition of the buffer. The name that you apply in the Setup screen will appear in the Protocol screen when you program an Isocratic Flow step, a Change Valve step, or a Linear Gradient step.

4.3.2 Plumbing the SV3-2 Diverter Valve

The SV3-2 Diverter Valve is primarily used as a means to divert the eluant stream from the detectors to a fraction collector or a waste container. It may also be used as a buffer/sample select valve when placed before the Gradient Pump or an Auxiliary Load Pump. It mounts conveniently to the System Rack. This valve is described in detail in its separate documentation.

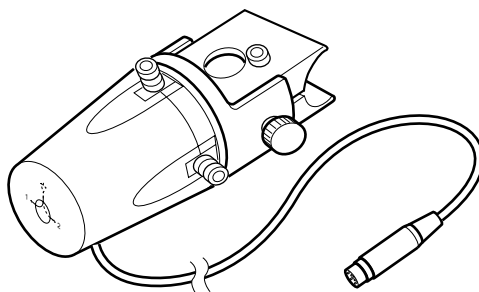


Figure 25. SV3-2 Diverter Valve

To install and plumb the valve:

1. Mount the SV3-2 valve to the System Rack.
2. Connect the SV3-2 valve's signal cable to any of the available Solenoid Valve connectors on the back of the Workstation (ports 1, 2 or 3).
3. Use 1/16" OD Tefzel tubing and 1/4-28 fittings when plumbing the SV3-2 valve. The only exception is when it is used as a Gradient Pump inlet valve (then use 1/8" OD tubing).

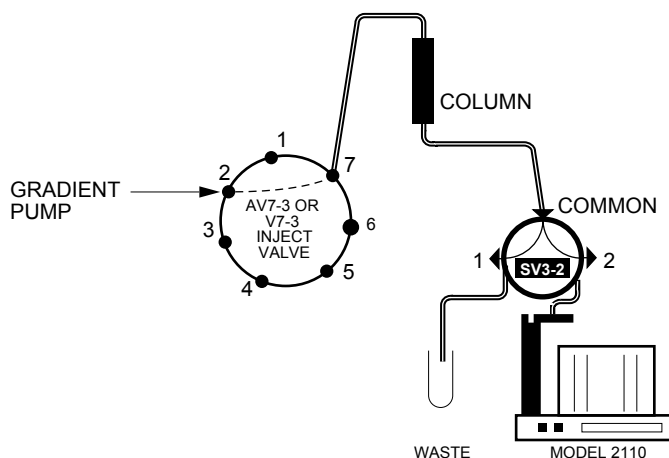


Figure 26. Configuration of the SV3-2 as a Fraction Collector Diverter Valve

The software's Manual mode lets you manually operate each inlet port on the SV3-2. Manual valve control is useful for the following:

- When priming the tubing and the valves with buffer prior to starting a method.
- For purging all tubing lines during cleaning process.

The Setup screen lets you define each of the two valve positions by name. This is known as “aliasing,” and it will make programming more convenient. For example, when the SV3-2 is placed before the Gradient Pump, you can identify the valve by the name or composition of the buffer. The name that you apply in the Setup screen will appear in the Protocol screen when you program an Isocratic Flow step, a Change Valve step, or a Linear Gradient step.

When used as a diverter valve with a Model 2110 or generic fraction collector, this valve should be aliased as such from the Setup screen. Position 1 of this valve is Waste; position 2, Collect.

4.4 PLUMBING BIO-RAD'S HIGH PRESSURE VALVES

Bio-Rad offers high pressure valves for controlling the flow of sample material. These include the following:

- AV7-3 automated and V7-3 manual inject valves
- AV9-8 automated and V9-8 manual stream select valves

These valves are discussed in detail in the following sections.

4.4.1 High Pressure Inject Valves (AV7-3 and V7-3)

The BioLogic HR system offers two types of high pressure injection valves for sample injection. There is an automated valve (the AV7-3) and a manual valve (the V7-3). These are 7-port, 3-position valves, the 3 positions being Load (position 1), Inject (position 2), and Purge (position 3). The Load Position on these valves is the default position when the system is powered up or at the end of a method run unless configured differently from the Edit User Preferences window, available from the Options menu. Both valves are rated to 1000 psi and are designed with non-metallic wetted parts and minimal internal dead volumes. For additional applications of this valve, see Section 4, Advanced System Applications.

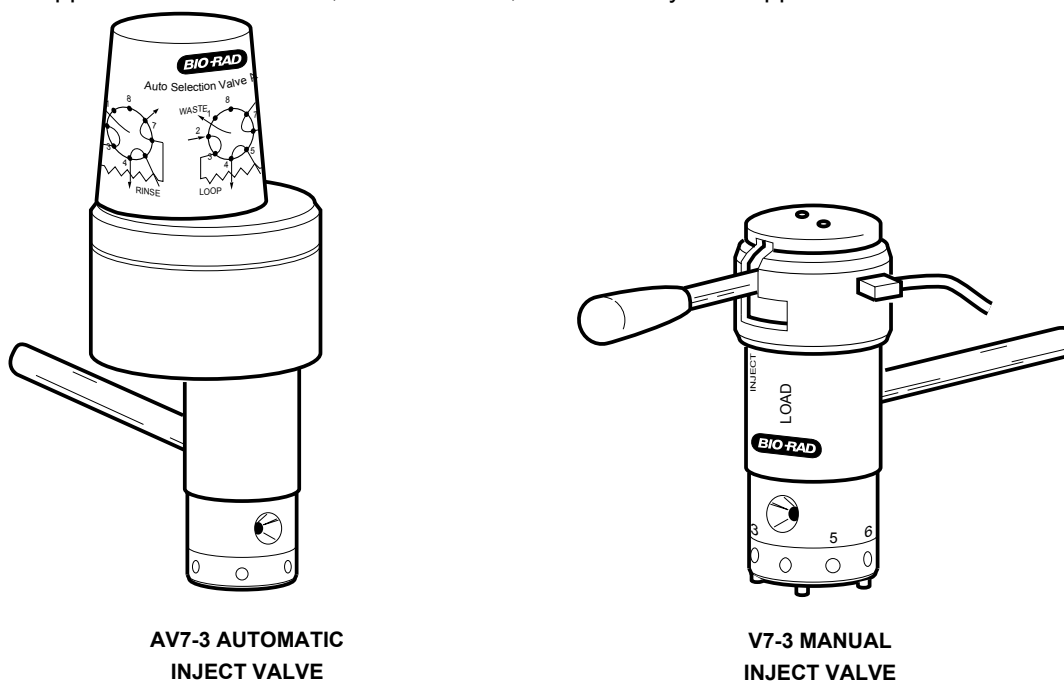


Figure 27. AV7-3 and V7-3 Inject Valves

To install and plumb the valve:

Note: Do not use an AV7-3 or V7-3 as a buffer select valve plumbed before the Gradient Pump.

1. Mount the AV7-3 or V7-3 valve to the System Rack. Select a location which will keep tubing lengths to a minimum and still allow easy access to the inject port for use and removal of a syringe.
2. Connect the AV7-3 valve Signal Cable to any of the available Automated Valve connectors on the back of the Workstation (ports 4, 5, 6).

Connect the V7-3 valve position-sensing switch (for communication with the Controller) to the Workstation's AUX connector at pins 1 and 9.

3. Use 1/4-28 fittings and 1/16" OD/0.020" ID Tefzel tubing when plumbing the inject valves. Keep tubing as short as possible.

See Figure 21 for a diagram of this plumbing.

4.4.2 High Pressure Stream Select Valves (AV9-8 and V9-8)

The BioLogic HR system has two types of high pressure stream-select valves available: an automated valve (the AV9-8) and a manual valve (the V9-8). These valves are bi-directional, with port 9 being the common port. Position 1 on the AV9-8 is the default position when the system is powered up or at the end of a Method run unless configured differently from the Edit User Preferences window, available from the Options menu. Both valves are rated at 1000 psi and are designed with non-metallic wetted parts for bio-compatibility and minimal internal dead volumes.

The stream-select valves have a number of uses in a chromatography system. The valve's use affects its plumbing. Refer to Section 4, Advanced System Applications, Chapters 8 - 10. For example:

- When two AV9-8 valves are used, up to eight columns may be run sequentially.
- When used in conjunction with an Auxiliary pump such as the Bio-Rad Model EP-1 Econo Pump for loop filling and an AV7-3 Inject valve, the AV9-8 automated valve may be used to select up to eight samples for consecutive chromatography runs.

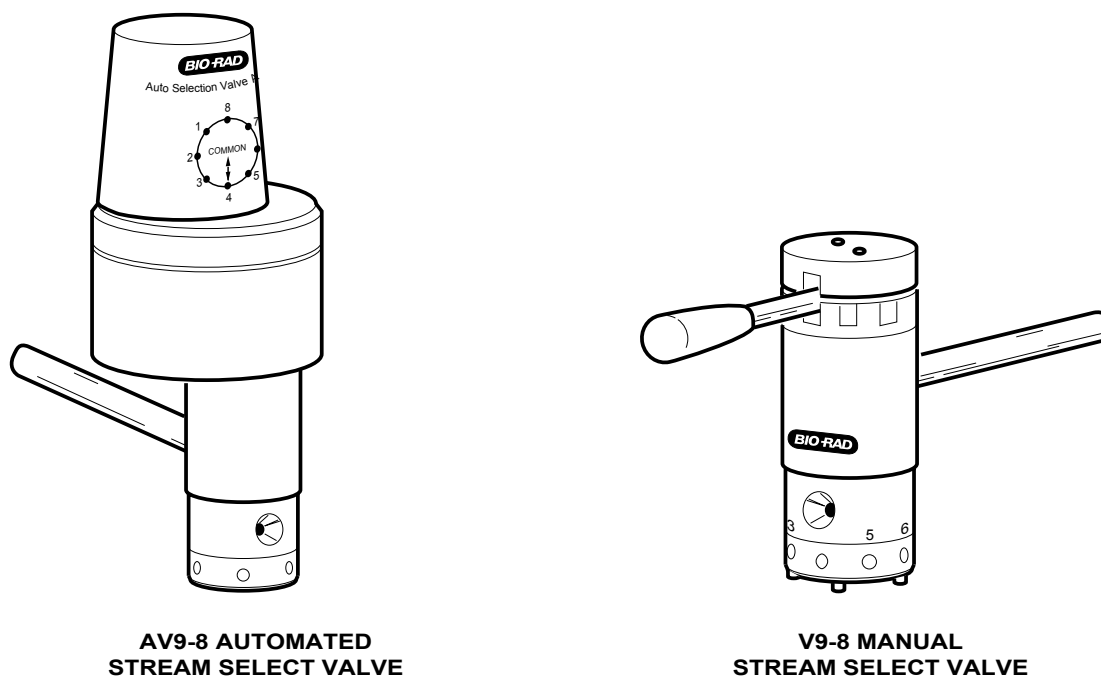


Figure 28. AV9-8 and V9-8 High Pressure Stream Select Valves

To install and plumb the valve:

1. Mount the AV9-8 or V9-8 valve to the System Rack. Select a location which will keep tubing lengths to a minimum.
2. Connect the AV9-8 valve Signal Cable to any of the available Automated Valve connectors on the back of the Workstation (ports 4, 5 or 6).
3. Use 1/4-28 fittings and 1/16" OD Tefzel tubing when plumbing these valves.

4.5 PRIMING THE SYSTEM

Before a method can be run, you must take proper steps to remove all air from the system.

1. Priming the Workstation Gradient Pump.

- Immerse the Gradient Pump A and B inlet lines in a container of HPLC grade (filtered, degassed) or other high quality water.
- Place a 10 ml luer syringe (supplied with the fittings kit) on the priming port of pump head A.
- Turn the priming port counter-clockwise to open the seal and gently withdraw the syringe plunger to draw water into the pump head.
- Repeat this operation several times until no air bubbles are visible in the inlet tubing.
- Tighten the priming port by turning it clockwise.
- Repeat this priming procedure for pump head B. **See Section 11.3.1 for additional priming procedures.**

2. Moving the Inject Valve to Purge Position.

- For a manual inject valve V7-3, move the arm to the Purge position (left, then up, then left). The figure in the valve position window displays "**P**."
Any flow from the Gradient Pump will now go directly to waste, bypassing any column or detector downstream of the valve.
- To change the status of an automated inject valve AV7-3, go to the Manual Screen displayed on the Controller Monitor.
If you plugged an inject valve into port #4, you will see a valve box designated "**AV7-3 at port 4**." The three radio buttons of this box correspond to valve positions as follows: **L** = Load position, **I** = Inject position, **P** = Purge position.
To move the AV7-3 valve to Purge position, click button **P**.
(**Note:** The default position at power up and at the end of a programmed method for the AV7-3 is position **L**, and for all other automated valves is position 1 unless changed from the Edit User Preferences window, available from the Options menu).

3. Purging the Gradient Pump.

With the inject valve in position **P** (Purge):

- Press the Purge buttons A and B on the front of the Workstation. The Gradient pump will run and the indicator lights will flash green. The default purge flow rate is 7ml/min. (This may be changed using the Manual Setup window, available from the Options menu.)
- Run both pumps for 2 minutes and then press the Purge buttons again to stop the pump.

4. Flushing the system through to the fraction collector.

With the Gradient Pump stopped, move the inject valve back to position **L** (Load) by either clicking radio button **L** (AV7-3) on the Manual Screen or moving the lever to the right (V7-3)

Using the Manual Screen, set the pump flow rate at 1.0 ml/min and start the pump. Water will now flow through the UV and Conductivity flow cells and ultimately to the fraction collector.

Note: If the system pressure variations exceed $\pm 10\%$, refer to section 11.3.1, Priming the Pump and Removing Trapped Air Bubbles.

5.0 INTRODUCTION TO THE SYSTEM SOFTWARE

The BioLogic HR system software is based on Microsoft Windows® 95, a widely used operating system for PC-compatible computers. This chapter discusses how to use the BioLogic HR system software.

5.1 USING THE SYSTEM INTERFACE

When you first turn on the system, the Manual Control screen is displayed (see Figure 29). Note that this screen, like all screens in the interface, is grouped into three sections, which consist of the Main (or System) Menus, System Control Window, and System Status Bar.

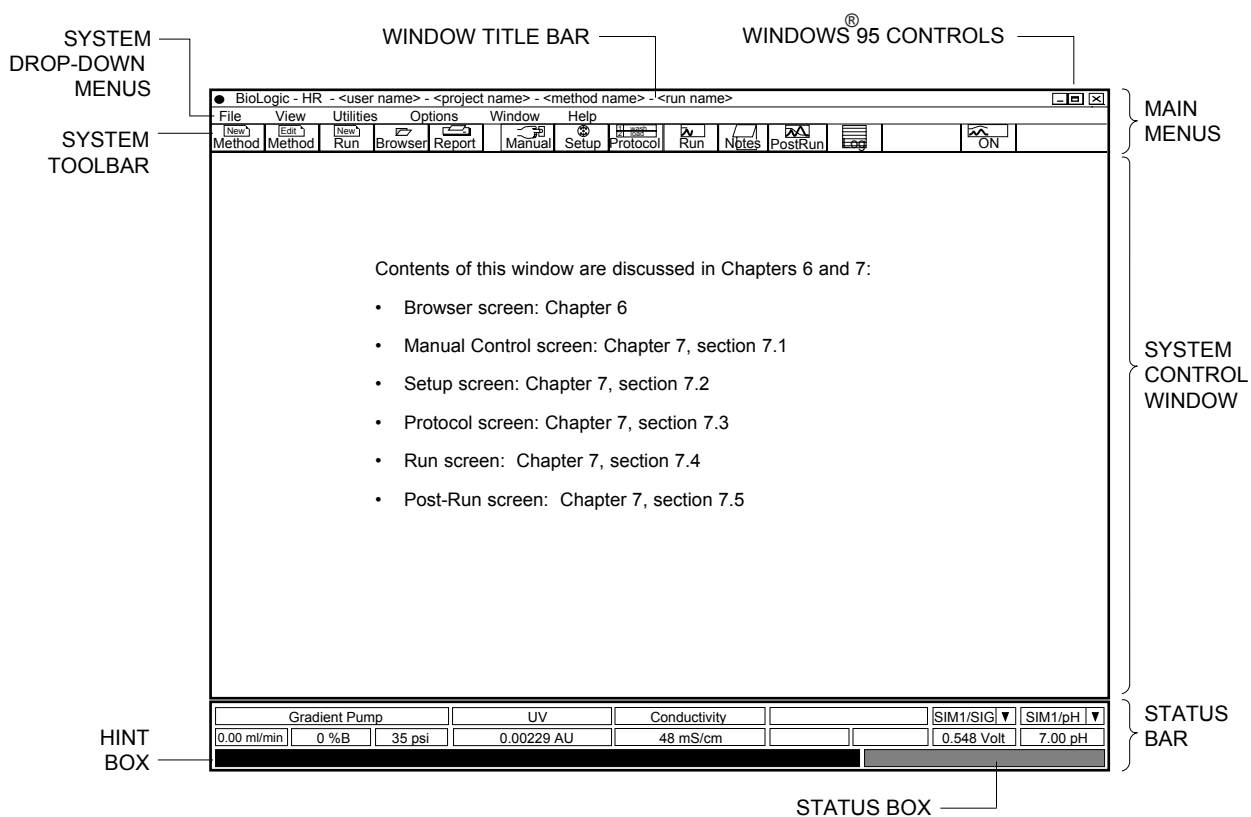


Figure 29. Layout of the Screen Display

The BioLogic HR system is controlled and monitored using the following:

- **System Toolbar:** This toolbar is the primary navigation tool for the system software.
- **System drop-down menus:** These menus are always displayed. Access to advanced functions are found only in the drop-down menus. **Note:** Some functions found on the System Toolbar are duplicated in the drop-down menus.
- **System Status bar:** This area shows information about components in the system, as well as providing information about software icons.

Note: Items shown in gray are not currently active or cannot be accessed because they are inappropriate to the current mode.

5.2 STANDARD MOUSE AND KEYBOARD FUNCTIONS

The BioLogic HR system is supplied with a standard mouse and the laboratory-safe (water resistant) BioLogic keyboard. The BioLogic HR will accept any standard PC-compatible mouse and any PS/2 keyboard. The left mouse button is used with system software, except as noted.

Table 7.
Special Function Keys

Special Function Keys	Description
	Running man: This is used during a run to advance the Method (i.e., satisfy the Hold) when the Method includes a Hold step which is programmed to “Hold until Keypress.” (Use the F2 key on a PC keyboard.)
	Help: Displays the Help menu for the currently displayed screen. (Use the F1 key on a PC keyboard.)
	Esc: Functions as an alternative to the Cancel selection in a Dialog box.
	Alt: Some system commands can be executed either by selecting them from a drop-down menu or by holding down the Alt key and then pressing the appropriate character key.

5.3 SYSTEM MENUS

The system menus consist of both System drop-down menus and the System Toolbar. In some cases, identical functions are found in both areas. Advanced features are located only in the System drop-down menu.

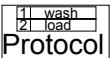
5.3.1 System Toolbar Buttons

Most of the System Toolbar buttons are continuously displayed across the screen. The function of each toolbar button is provided in Table 8.

Table 8.
Toolbar Buttons

Button	Description
File Buttons	Note: Similar functions can be found by selecting the File drop-down menu.
	Initiates the process of writing a new method. The new method will be saved into the current User and Project directories. The current user and project is shown in the window title bar. Note: To place a new method in a different user and project, choose Select User and Project under the File menu before clicking on the New Method button in the Toolbar. Choose the desired user and project from the drop-down list. To create a new user and project, you must use the Browser.
	This button is available when an open method has runs associated with it. It allows you to create a new method based on the open method. A copy of the open method is made, and the system software prompts you for a name change. Note: The original method remains unaltered.
	Initiates a new run for the currently open method. You will be prompted for a new run name prior to launching the run.
	Opens the database of existing methods and runs. See chapter 6 for a complete description of the browser.
	Allows you to define when to print and what is to be printed in the report for a run.

Table 8. (continued)
Toolbar Buttons

Button	Description
View Buttons	Note: Similar functions can be found by selecting the View drop-down menu.
 Manual	Displays the Manual screen, which allows individual control of instruments and devices in the system. In Manual mode, data is viewed from the electronic chart recorder; data cannot be saved.
 Setup	Displays the Setup screen, which allows you to choose components required for your method. From the Setup screen you can “alias” (add descriptions to) valve positions and buffers. The Setup screen is automatically displayed when you write a new method.
 Protocol	Displays the Protocol screen for creating and editing a method.
 Run	Displays the Run screen, from which you can initiate a run of the currently loaded method. You will be prompted for a new run name prior to launching the run.
 Notes	The notes section is a prompted section which allows you to enter additional information about your method and/or run. This section is always editable, even after a run is executed.
 PostRun	PostRun allows you to customize your chromatogram prior to printing. The PostRun screen is always editable, as the raw data is unaffected by these modifications.
 Log	As the run progresses, all events are recorded in the Log. This information cannot be edited.
 Integ.	Launches the EZLogic integration software. This button replaces Log when EZLogic is installed.

5.3.2 System Drop-down Menus

The system drop-down menus are discussed in the following tables.

Table 9.
File Drop-down Menu

• BioLogic HR - <user name> - <project name> - <method name>	
File	Edit View Utilities Options Window Help
New Method	
Copy and Edit Method	
New Run	
Open...	
Close	
Select User and Project	
Change Name, Author...	
Save setup as default	
Load default setup	
Print Report...	
Data management...	
Export Data...	
Export Chromatogram Image...	
Integration...	
Exit	
1 Method 1-	
2 Method 2-	
3 Method 3-	
4 Method 4-	

The File menu is always displayed. It consists of the following:

- **New Method:** Initiates the process of writing a new method. The new method will be saved into the open User and Project directories. To save to a different User or Project, first select **Select User and Project**. Selecting **New Method** during a run will automatically open the “BioLogic offline” function, allowing you to create a method while a run is in progress.
- **Copy and Edit Method:** Allows you to use an existing method as a template for a new method. A copy of the method is made, and the system software prompts you for a name change.
- **New Run:** Initiates a run for the currently open method. You will be prompted for a new run name prior to launching the run.
- **Open:** Displays the browser. Refer to Chapter 6, Browser Screen.
- **Close:** Closes the current method and/or run.
- **Select User and Project...:** Allows you to select the current user and project of the system. This action affects where new methods are automatically saved.
- **Change Name, Author...:** Allows you to change the method/run name, author, and description. This function is inactive once a run has been initiated.
- **Save setup as default:** When you create a method, the Setup screen displays a list of devices in setup. This is the default list of devices. If you have a method with a setup list that more accurately reflects your use of the system, you can open this method and then select **Save setup as default**. Then each time you create a method, the Setup screen will display this list of devices in setup.
- **Load default setup:** Recalls and loads the default setup.

Table 9. (continued)
File Drop-down Menu

-
- **Print Report:** Initiates the generation and printing of the currently open method, including its setup and run data, run results, and the run log report.
 - **Data Management:** Displays the browser. Refer to Chapter 6, Browser Screen.
 - **Export Data:** This feature establishes data export parameters, and is available from the PostRun screen.
 - **Export Chromatogram Image:** This feature exports a chromatogram image in a Windows Meta File (.WMF) format. This feature is available in the PostRun screen.
 - **Integration:** Launches the EZLogic integration software. Contact Bio-Rad for more information on EZLogic.
 - **Exit:** Exits the BioLogic application.
-

Table 10.
Edit Drop-down Menu

Setup Screen's Edit menu

● BioLogic HR - <user name> - <project name> - <method name>						
File	Edit	View	Utilities	Options	Window	Help
Delete						
Select All						

Protocol Screen's Edit menu

● BioLogic HR - <user name> - <project name> - <method name>						
File	Edit	View	Utilities	Options	Window	Help
Cut						
Copy						
Paste						
Delete						
Select All:						

Run Screen's Edit menu

● BioLogic HR - <user name> - <project name> - <method name>						
File	Edit	View	Utilities	Options	Window	Help
Multiple runs...						
Start run						
Pause buffer pump						
Hold gradient						

Post Run Screen's Edit menu

● BioLogic HR - <user name> - <project name> - <method name>						
File	Edit	View	Utilities	Options	Window	Help
Tag and Trace Options						
Full view in zoom chromatogram						
Print zoom chromatogram						
Copy zoom chromatogram to clipboard						
Delete Selected tag:						
Delete all tags						

The contents of the Edit menu depends upon the currently displayed screen. In most instances, the item in the drop-down menu also appears in the System Toolbar; exceptions are noted below.

Setup Screen

- **Delete:** Deletes the currently highlighted device in the Devices in Setup window of Setup screen. To delete all devices, first select Select All, as described below.
- **Select All:** Highlights all devices in the Devices in Setup window of the Setup screen. (Not available from the System Toolbar.)

Protocol Screen

- **Cut:** Cuts the currently highlighted step from the protocol. A cut step may be pasted elsewhere.
- **Copy:** Copies the currently highlighted step so that it can be repeated elsewhere in the protocol.
- **Paste:** Pastes the cut or copied step into the protocol.
- **Delete:** Deletes the currently highlighted step from the protocol.
- **Select All:** Highlights all protocol steps. Cut, Copy, and Delete then act on all steps. To remove the highlighting from individual steps, hold down the **Ctrl** key and click the mouse over the desired step. (Not available from the System Toolbar.)

Run Screen

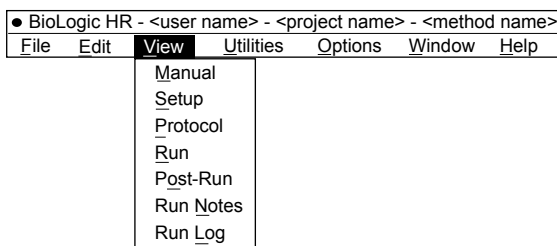
- **Multiple runs...:** Specifies the number of times the method is to run.
- **Start run:** Initiates the run.
- **Pause buffer pump:** Pauses a run and stops the pumps.
- **Hold gradient:** Holds a run at the current %B conditions and flow rate. It also holds method time, effectively holding the protocol step and the current fraction collection step if collection is programmed.

Table 10. (continued)
Edit Drop-down Menu

Post Run Screen

- **Tag and Trace Options...:** Displays the Post-Run Tags window that allows you to specify which traces are to be displayed, whether or not their tags are to be displayed, the tag style for the active trace, and the values for each tag in the active trace. (Also available from the **Tag** button in the System Toolbar.)
 - **Full View in zoom chromatogram:** Allows you to view the complete chromatogram. (Also available from the **Full View** button in the System Toolbar.)
 - **Print zoom chromatogram:** Allows you to print the complete chromatogram. (Also available from the **Print** button in the System Toolbar.)
 - **Copy zoom chromatogram to clipboard:** Copies to clipboard the complete chromatogram. From the clipboard, it can then be copied into other applications. (Not available from the System Toolbar.)
 - **Delete selected tag:** To delete a tag, highlight the selected tag before using this function. (Also available from the **Del. Tag** button in the System Toolbar.)
 - **Delete all tags:** Deletes all tags in the chromatogram. (Not available from the System Toolbar.)
-

Table 11.
View Drop-down Menu

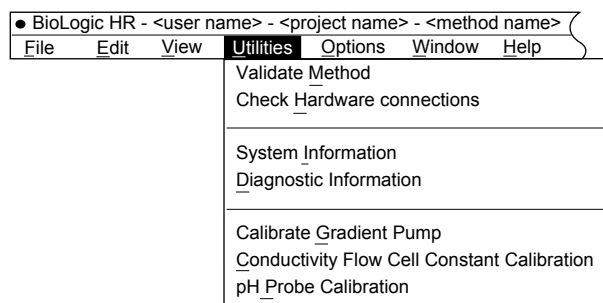


Each of these selections is available from the toolbar menu. The View menu consists of the following:

- **Manual:** Displays the Manual screen for individual control of installed instruments and devices in the system.
- **Setup:** Displays the Setup screen, which allows you specify the components required for your method.
- **Protocol:** Displays the Protocol screen for creating and editing chromatography steps in the method.
- **Run:** Displays the Run screen, from which you can initiate the running of the currently open method.
- **Post-Run:** When a run is completed, select **Post-Run** to annotate the chromatogram (for example, to apply tags to UV, Conductivity, and/or %B values).
- **Run Notes:** To keep notes regarding the run, select **Run Notes** and enter them into the Notes screen. There are fields for description of the sample, the column, the operator, the buffer(s), flow rates, gradients, chart speed, fraction size, and general text entry of variable length. These notes are printed with the report.
- **Run Log:** Displays a log of all events occurring during the run. This information cannot be edited.
Note: The Run Log may be disabled by selecting **Edit User Preferences** from the Option drop-down menu. This will save hard drive space. However, it is recommended that the Run Log be active for assistance in troubleshooting.

Note: If the Integration software option is installed, the **Log** toolbar button is replaced by the Integ toolbar button. To display the Log window, select **Run Log** from the View drop-down menu.

Table 12.
Utilities Drop-down Menu



These selections generally relate to system options, and it remains the same regardless of which screen is displayed. The Utilities menu consists of the following:

- **Validate Method:** This verifies that the devices required by the Protocol have been defined in the Setup screen. Validation is done automatically at the initiation of a new run.
 - **Check Hardware connections:** This checks that all devices appearing in the Setup screen and required for the run are connected to the system. It does not apply to a V7-3 Manual Inject valve, chart recorder, Model 2110 or generic fraction collector, or Auxiliary pump. It does not verify plumbing connections.
 - **System Information:** Displays the current system configuration, including each instrument in the system and its firmware version number, the Windows® version number, available hard disk space, and the number of methods and runs in the database.
 - **Diagnostic Information:** Displays information for service diagnostic purposes only.
 - **Calibrate Gradient Pump:** This selection is available only from the Manual screen, and it is typically used only after servicing of the pump. It allows the user to calibrate the gradient pump flow rate and zero the system pressure gauge. **Note:** You must exit the Calibration screen in order for the system to accept the calibration values.
 - **Conductivity Flow Cell Constant Calibration:** This utility allows the user to calibrate the conductivity flow cell. It is typically used to calibrate a new flow cell. **Note:** You must exit the calibration screen in order for the system to accept the calibration values.
 - **pH Probe Calibration:** This utility allows the user to calibrate the pH probe. It is typically used at the start of each day's use of the system. **Note:** You must exit the calibration screen in order for the system to accept the calibration values.
-

Table 13.
Options Drop-down Menu

● BioLogic HR - <user name> - <project name> - <method name>						
File	Edit	View	Utilities	Options	Window	Help
				Manual Setup		
				Edit User Preferences		
				Chromatogram Settings		
				Status lines		
				Browser Settings		
				View Chromatogram		

These selections generally relate to screen options.

- **Manual Setup:** From this screen you can set the pump's purge flow rate and the SIM-HR parameters.
- **Edit User Preferences:** This lets you specify Run Options (which determine whether or not the protocol's steps are listed in the Run Log and whether or not valves are automatically returned to position 1 at the end of the run) and the protocol program base (time or volume).
- **Chromatogram settings:** Allows you to select which traces will be visible in the chromatograms. For the Manual screen, the X-axis and the Y-axis ranges can be set. For the Run and Post-Run screens, only the Y-axis range can be set.
- **Status Lines:** Allows the status message at the bottom left of the screen to be toggled on or off.
- **Browser settings:** Displays the Browser & Database Options window. The Browser options allow you to select how the browser is to display users/projects/methods/runs. For example, select Projects to display a top level folder called "Projects" which contains all projects in the database.

The Options for Database selections typically should not be changed. They allow the system to test and optimize the database upon startup of the system. These settings check the integrity of the data and reduce the size of the database file.

- **View Chromatogram:** Allows the Manual screen chromatogram to be toggled on or off. When off, the valve panel is displayed.
-

6.0 INTRODUCTION TO THE BROWSER SCREEN

The browser is the organizational tool for chromatography data. It is a database that is displayed as a tree hierarchy which can be sorted by Users, Projects, Methods, and Runs. Figure 30 shows the layout of the Browser screen.

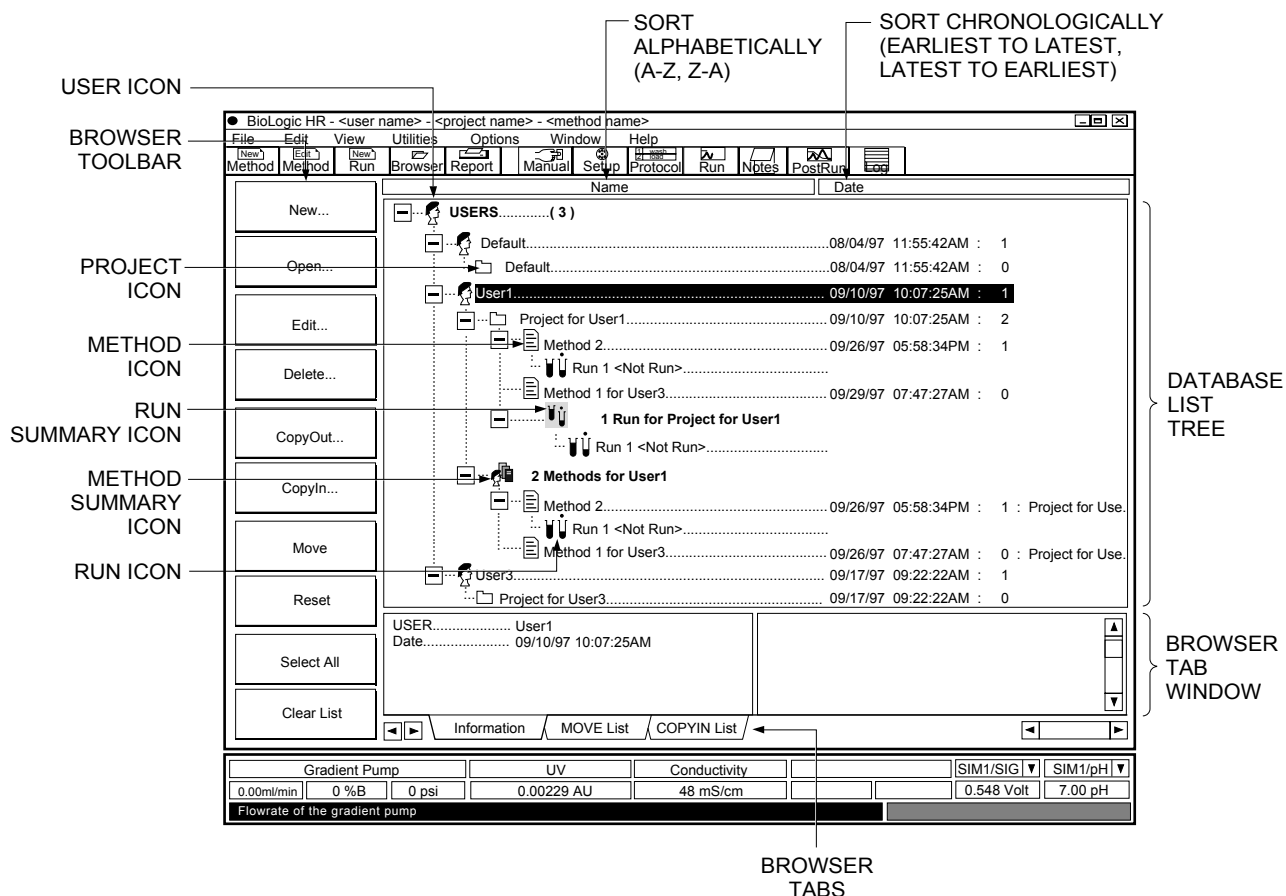


Figure 30. The Browser Screen

The Browser screen displays the following information and controls:

- **Collapsing/expanding the tree hierarchy:**



Click on this icon in the database tree to collapse the listing for a user, project, method, or run folder.



Click on this icon in the database tree to expand the listing for a user, project, method, or run folder.

RESET

Click on this button in the Browser toolbar to collapse the tree to its top level items.

- **Icons displayed in green or red in the database tree indicate the following:**

Green: Currently open in the active window. The active window may be the on-line window or the off-line window. For more on on-line and off-line, refer to the discussion in Chapter 7, section 7.4.2, Working Off-Line.

Red: Currently open in the window that is not active. This condition is displayed during off-line use: it applies to the method and run data for whichever window is **not** active (on-line or off-line). Items with a red check mark indicate that item is in the Move List or is marked for deletion.

- **Icon colors other than red and green in the database tree are as follows:**

Blue (profile): User folder

Yellow: Project folder

Blue (text page): Method folder

Blue (test tubes): Run folder

- **Use of bolded text in the database tree applies to the following:**

Top level folders: Typically this applies to the USERS folder, which lists all users of the system. However, by selecting **Browser Settings** from the Options menu, you can use the "Starting Browser Selections" to move Projects, Methods, and Runs to the top level of the tree hierarchy.

Summary information: The total number of Methods are listed for each user, and the total number of Runs are provided for each user project folder. **Note:** You cannot delete summary information.

In addition to the database tree, the Browser screen displays the following controls and information:

- **Browser toolbar:** These buttons control the different uses of the Browser screen. They are discussed in greater detail later in this chapter.
- **Name and Date buttons:** These buttons, located across the top of the tree hierarchy, allow you to sort the tree alphabetically or chronologically. These are toggle buttons. For example, if you click on the Name button to sort from a to z, then clicking again sorts from z to a. If you click on the Date button to sort from the latest listing, then clicking again sorts from the earliest listing.
- **Browser Tabs and Browser Tab window:** The Browser Tabs are listed at the bottom of the Browser screen, and they control what is displayed in the Browser Tab window. The following tabs are available:

Information: Clicking on this tab displays two panels in the Browser Tab window: The left panel shows details about the specific item selected; the right panel displays information about the next higher level item to which the item belongs.

MOVE List: Displays the list of all items selected to be moved from one place to another in the database list. Items selected to be moved are indicated by a red checkmark in the database tree. Projects can be moved from one user to another; Methods can be moved from one Project to another. When a Project is moved, it takes all Methods and Runs associated with it; when a Method is moved, it takes all Runs associated with it. The procedure for moving an item is discussed later in this chapter.

COPYIN List: Displays the list of all items selected to be copied in (restored) to the database from an archived data file. The procedure for copying in items is discussed later in this chapter.

The following steps provide procedures for using the browser.

1. Define a New User.

The first step is to define a new user. This allows you to group your methods and runs within the browser. To define a new user,

- a. Click once on a User icon in the browser.
- b. Click once on the **New...** button and select New User. You will be prompted for a new User Name. Enter the name you would like to use.
- c. Press **OK** to accept the new name.

Note: If you define a new user, you must also assign a project to the user before writing a new method.

2. Define a New Project.

The Browser is further segmented into Projects within a User's domain. To define a new project:

- a. With the new User Name highlighted, click once on the **New...** button.
- b. Select **New Project**. Enter the project name and description.
- c. Press **OK** to accept the new name.

3. Write a New Method.

To begin writing a new method from the browser:

- a. Select the Project folder or a Method in the Project folder where you want the new Method to be located.
- b. Click once on the **New...** button.
- c. Select **New Method**. Enter the method name and press **OK**.

Hint: You can also use the **New Method** button in the System Toolbar. By default the system assumes that the current user and project shown in the window title bar is the path for the new method.

4. Name a New Run.

To start a new run from the browser:

- a. Select the Method folder or a Run in the Method folder where you want the new Run to be located.
- b. Click once on the **New...** button.
- c. Select **New Run**. Enter the run name and press **OK**.

Hint: You can also use the **New Run** button in the System Toolbar. By default the system assumes that the current user, project, and method shown in the window title bar is the path for the new run.

The Browser toolbar buttons are described below:

New... The **New...** button creates a new method using the default setup. The new method is placed with the current user. You can select which project to place it in, as well as provide a method name, description, and author.

Open... The **Open...** button allows you to open a selected method or run from the database tree. If a selected method has no completed runs associated with it, it may be edited. Otherwise the Method is opened in view mode, where it can be examined but not changed. Opening a Run allows you to view, analyze, and print the run data.

If you wish to create a new method starting with the contents of an existing method, open the desired method (whether or not there are completed runs associated with it), and then use the Edit button, as described below.

Note: Opening a Method or Run while a run is in progress will result in the Run-in-progress remaining in the On-Line window and the selected method or run being opened in the Off-Line window. For more on On-Line and Off-Line, refer to the discussion in Chapter 7, section 7.4.2, Working Off-Line.

Edit... There are multiple functions of the **Edit...** button, depending upon what is highlighted when the button is pressed. Selecting a user or project name allows you to change that user name or project name and/or description. Selecting a Method or Run copies the contents of the selected item. When a method is selected, you create a new method when you enter the method name, author, and description. When a run is selected, you can change the run name, operator, and description. **Note:** Summary items are not editable.

Delete The **Delete** button eliminates Users, Projects, Methods and Runs from the Browser. This is useful after using the Copy Out function to archive files to floppy disk, as discussed below. In order to delete an element of the Browser, you must first delete data associated with the element. For example, to delete a Method, you must first delete all of the Runs associated with the Method.

Note: You cannot delete Methods or Runs from method and run summaries. Also, after deleting files, you may want to run the Windows 95 disk defragmenter. This is available from the Windows **Start** button: select **Programs** → **Accessories** → **System Tools** → **Disk Defragmenter**.

Copy Out... This function lets you copy methods and runs to floppy disk or to another location on your computer's hard disk. The copy out function is used for backup and archiving purposes and to transfer data to another BioLogic HR system.

Copied out files are stored in a file with the default name biologic.zib, which is a compressed file format to conserve disk space. Because the copy out function does not copy the user and project names, you may want to change the default filename - **but not its .ZIB extension** - to indicate the user and project names. **Note:** Initiating a copy out to floppy disk completely erases the contents of the floppy disk. Do not use a floppy disk containing data that you want to keep.

To CopyOut, highlight the Runs you wish to CopyOut (the Method will be automatically included with the Runs). Press the **CopyOut...** button on the Browser Toolbar. A dialog box appears which allows you to select the destination of the .ZIB file (either the hard drive or a floppy disk). Choose the destination and press **OK**.

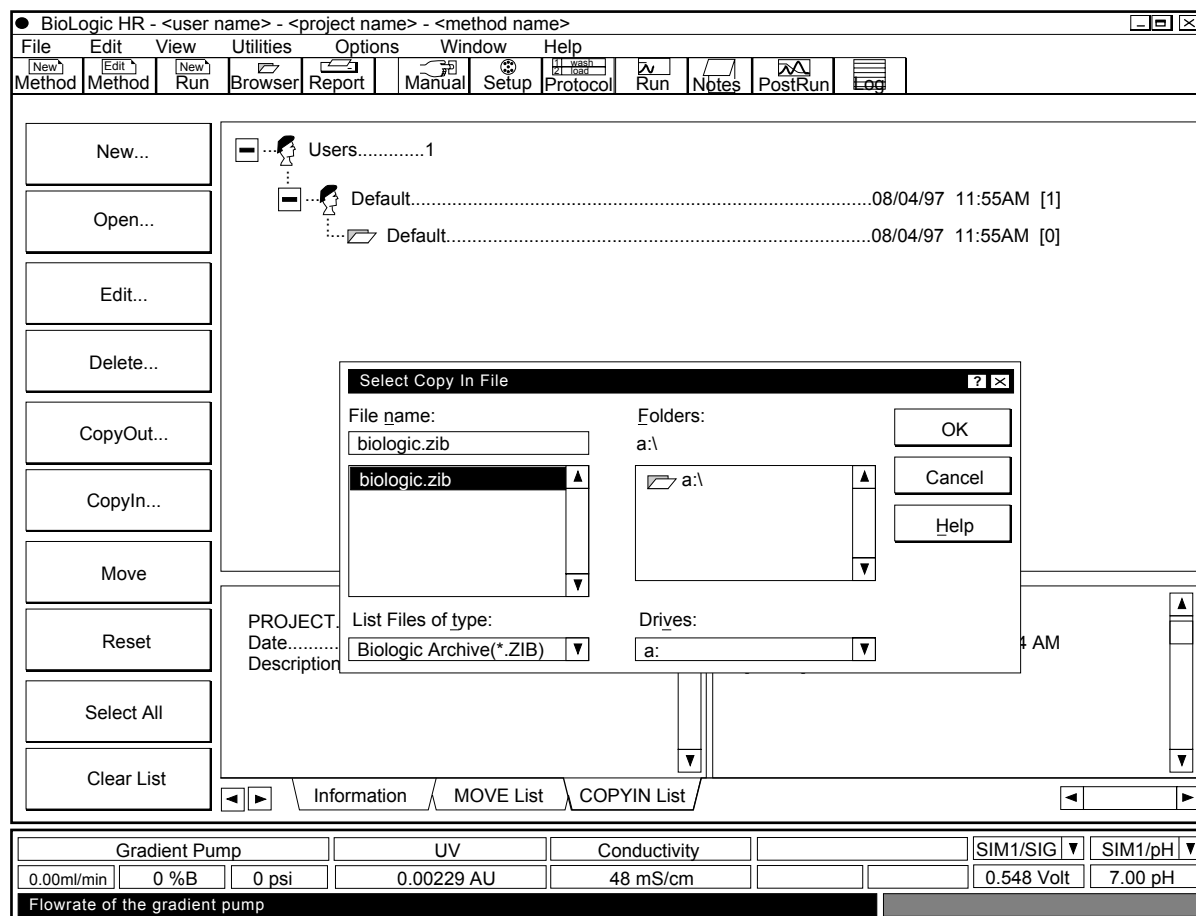


Figure 31. The Copy In Window

CopyIn...

This function allows you to copy in methods and runs from a backed-up or archived .ZIB file. To copy in,

- Press the **CopyIn...** button and select the .ZIB file you wish to CopyIn. (See Figure 31.)
- From the COPYIN List in the Browser Tab window, select the desired methods and runs. All methods and runs in the file are displayed in the COPYIN List. (See Figure 32.) From the database tree window, click the left mouse button to select the Project to which you want the methods and runs to be copied.
- Again, click on the **CopyIn...** button (or click on the right mouse button) and select **Copy to <project name>**.
Note: Clicking the right mouse button displays all the available Browser toolbar buttons for the *highlighted* selection.
- Once you finish the copy in procedure, use the **Clear List** button on the Browser toolbar to delete the remaining methods and runs in the COPYIN List.

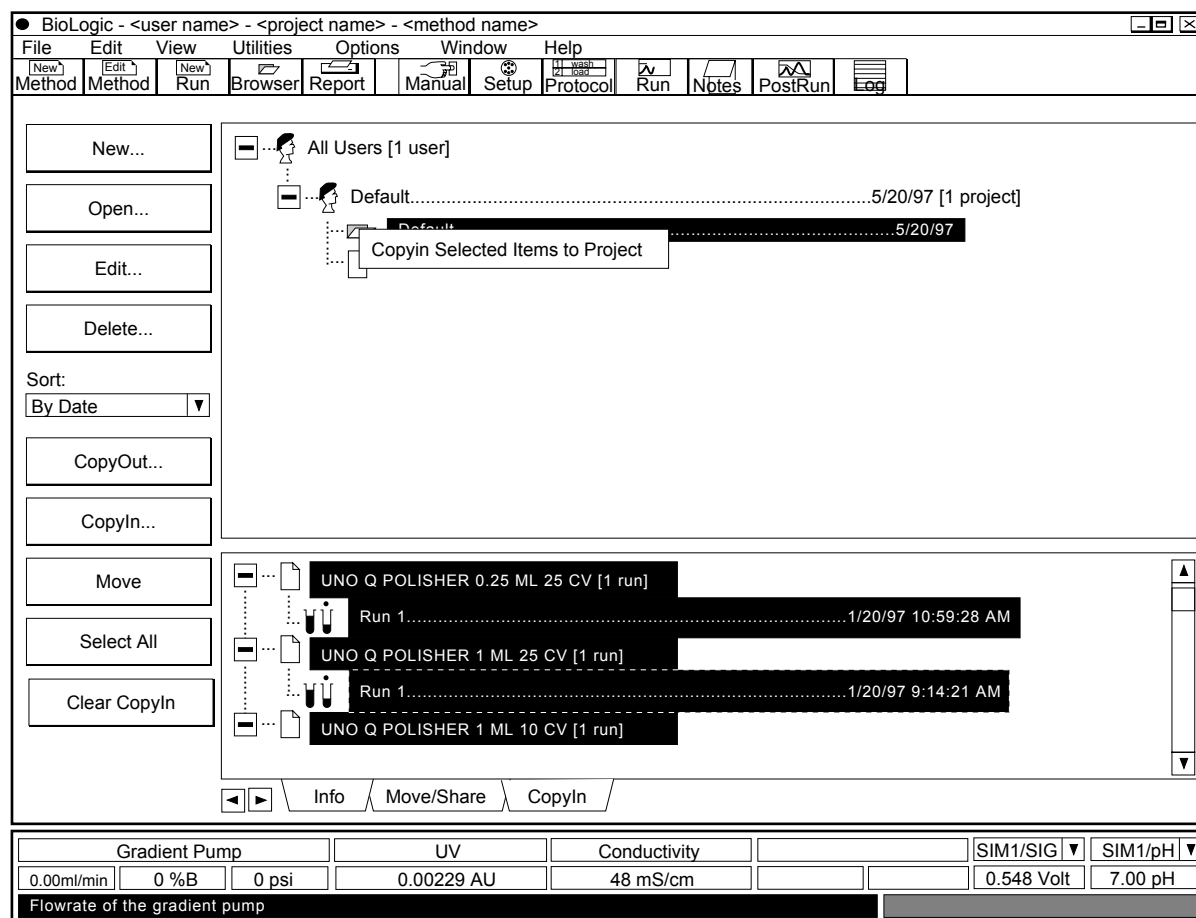


Figure 32. CopyIn with Information in Browser Tab Window

Move

Move allows you to transfer a selected item from one location to another. For example, a Project can be moved from one user to another, a Method can be moved from one Project to another and to another user. Runs can be moved, but they are always accompanied by their Method. To move an item,

- In the database tree, highlight the item you want to move. You may highlight a Project or a Method. Selecting a Run will select that run's methods and all its runs.
- Click once on the **Move** button on the Browser Toolbar. The Methods and Runs will appear in the MOVE List in the Browser Tab window.
- From the MOVE List, select what you want to move.
- From the menu tree, select the new location for the items you are moving.
- Again click on the **Move** button (Or click on the right mouse button) and select **Move to <project name>**.
- Once you finish the move procedure, use the **Clear List** button on the Browser toolbar to delete the remaining methods and runs in the MOVE List.

Reset

Collapses all of the folders in the Browser.

Select All

This button is active during the CopyIn and Move commands. It selects all contents of the Browser Tab window.

Clear List

Clears the listing in the Browser Tab window.

7.0 MODES OF OPERATION

There are four primary modes of operations, each represented by a screen display.

- Manual Control mode: To control the devices and instruments connected to the Workstation.
- Setup mode: Allows you define what instruments and devices will be used for any particular chromatography protocol, and if desired, to “alias” or name them for a specific task. For example, a SV3-2 valve may be aliased as a “Fraction collector diverter valve.”
- Protocol mode: To define each of the steps to be run during a Method. Most steps are programmed using a dialog box to enter all of the parameter values for that step. Note that the number of the step being programmed is displayed in the upper left corner of the dialog box.
- Run mode: To run a method or view a run in progress.

Note: You may move through the four modes of operation by activating buttons on the system toolbar as long as a run is not in progress. Those functions available during a run are discussed in section 7.4, Run Screen.

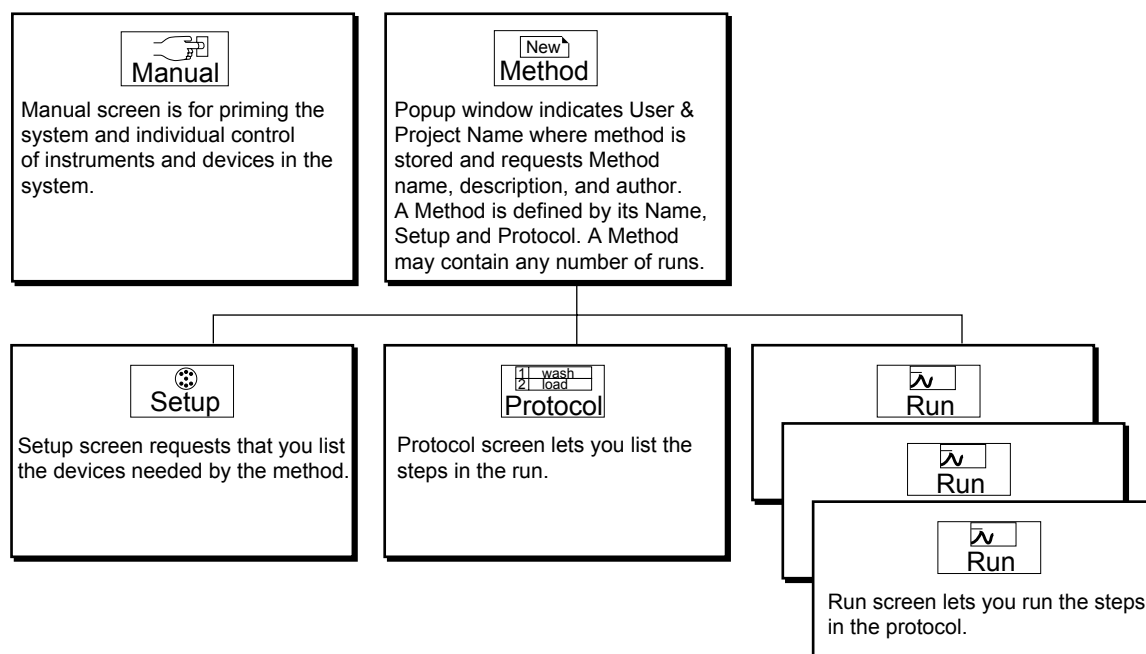


Figure 33. Relationships between Modes of Operation

7.1 MANUAL CONTROL SCREEN

The Manual mode screen lets you control different devices and instruments on an individual basis.

Note: If an instrument is connected to the BioLogic HR system on the instrument bus but does not appear in the Manual screen, check that its power switch is On and that the connection is secure.

BioLogic HR - <user name> - <project name> - <method name>

File View Utilities Options Window Help

Method Method Run Browser Report Manual Setup Protocol Run Notes PostRun Log ON

Gradient pump Flow rate: 1.00 ml/min Inlet A: 100 % Inlet B: 0 % High limit: 400 psi Low limit: 20 psi Set START STOP	Fraction collector Mode: ☒ System ☐ Local Rack: Rack 1 (128 tubes) Start Tube: 1 End Tube: 128 Fraction size: 1.00 ml Tube number: 1 Volume left: Advance START STOP	UV Detector Zero Baseline ON OFF Chart Recorder Conductivity range(mS/cm): 500 UV range (AU): 0.2 Event mark ON OFF Signal Import Module SIM 1 and SIM 2 connected
---	---	--

SV3-2 at port 1 1 ☒ 2	SV5-4 at port 2 1 ☒ 2 4 ☐ 3	AV7-3 at port 4 I ☐ L ☒ P	AV9-8 at port 5 I ☐ 8 ☐ 2 7 ☐ 3 6 ☐ 4 5 ☐
--	---	---	---

Gradient Pump	UV	Conductivity	SIM1/SIG	SIM2/pH
0.00 ml/min 0 %B 35 psi	0.00229 AU	48 mS/cm	0.548 Volt	7.00 pH

Figure 34. Manual Mode Screen, showing a Model 2128 Fraction Collector and Four Valves

From the Manual mode screen you can control the following:

- **Gradient Pump:** This control panel allows you to control the composition of the eluant being pumped, set the flow rate and pressure limits, and start/stop the pump. **Note:** Changing pressure limits requires pressing either the **Start** or **Set** buttons. When the pump is running, changing any parameters requires pressing the **Set** button.
- **Fraction Collector:** If you are using a Model 2110 or generic fraction collector, the fraction collector boxes allow you to set the **Fraction Size**. The system displays the “Tube number,” which is the number of the tube currently under the drop-head (assuming the run started at tube 1), and the “Volume left,” which is the volume remaining to be delivered. There are also toggle buttons for **Start** and **Stop** and a button for **Advance**. Note that the Tube number is a display field only.

If you are using a Model 2128 fraction collector, you have two modes of operation: **System**, in which the full functionality of the unit is available from the BioLogic HR system software, including the selection of a Model 2128 rack type, and **Local**, in which control of the fraction collector is from its front panel.

Note: Fraction collection is always by **Volume**. Collection by Drop or Time is not supported by the BioLogic HR system.

- **UV Detector:** Allows you to turn the UV Lamp on and off and to zero the baseline reading.
- **Chart Recorder:** If you are using the UV and Conductivity Detectors supplied with the system, you can set their chart recorder full scale ranges, start/stop the recorder, and send an event mark. The full scale ranges set here do not apply to the chromatograms displayed on the computer screen.
- **Signal Import Module (SIM 1 and 2):** This panel indicates the presence of up to two Signal Import Modules (SIM-HR). A SIM-HR is an analog-to-digital device. The signal from an analog instrument connected to the SIM is digitized so that the instrument can be monitored from the BioLogic HR system. Instruments that may be connected this way include any detection source, such as a variable wavelength UV detector, a pH meter, a refractive index detector, or a fluorescence detector. For more information, refer to Chapter 2.4.1 and to the separate instruction sheet for the SIM-HR module.
- **Valves:** Allows you to control any of the BioLogic HR solenoid or automated valves.

In addition, you can display the chromatogram window. From the System Toolbar, press the  button to display the Manual Control screen's electronic strip chart recorder. It will be placed over the valve control panels. Use the scroll bars or the **Chromatogram Settings** selection under the Options drop-down menu to increase or decrease the detection range.

7.2 SETUP SCREEN

The Setup screen is used to define what instruments/devices will be used by the Protocol for a particular Method. Each Method is thus accompanied by its own Setup. The information that you provide in Setup is necessary to properly operate the system. For example, if you plan to use a Model 2128 Fraction Collector, then you must select this particular collector in the Setup screen.

If you routinely use the same Setup, then it is recommended to select **Save setup as default** from the File drop-down menu. This setup is then loaded each time you create a New Method, thus eliminating the need to enter a new Setup each time.

Devices are chosen from the left hand “Available Devices” box. A dialog box will appear allowing you to define or alias the function of the device in the Method. For example, an AV7-3 may be selected and then defined as a Sample Inject Valve. Use the **Delete** button in the System Toolbar to remove a device from the Devices in Setup window. In order to **run** your Method, these devices must be connected to the rear of the Workstation and they must be powered on.

Note: Devices do not need to be connected to the system in order to include them in the Setup for a method. However, the Run will not start unless these devices are connected; an error message will appear on the Run screen.

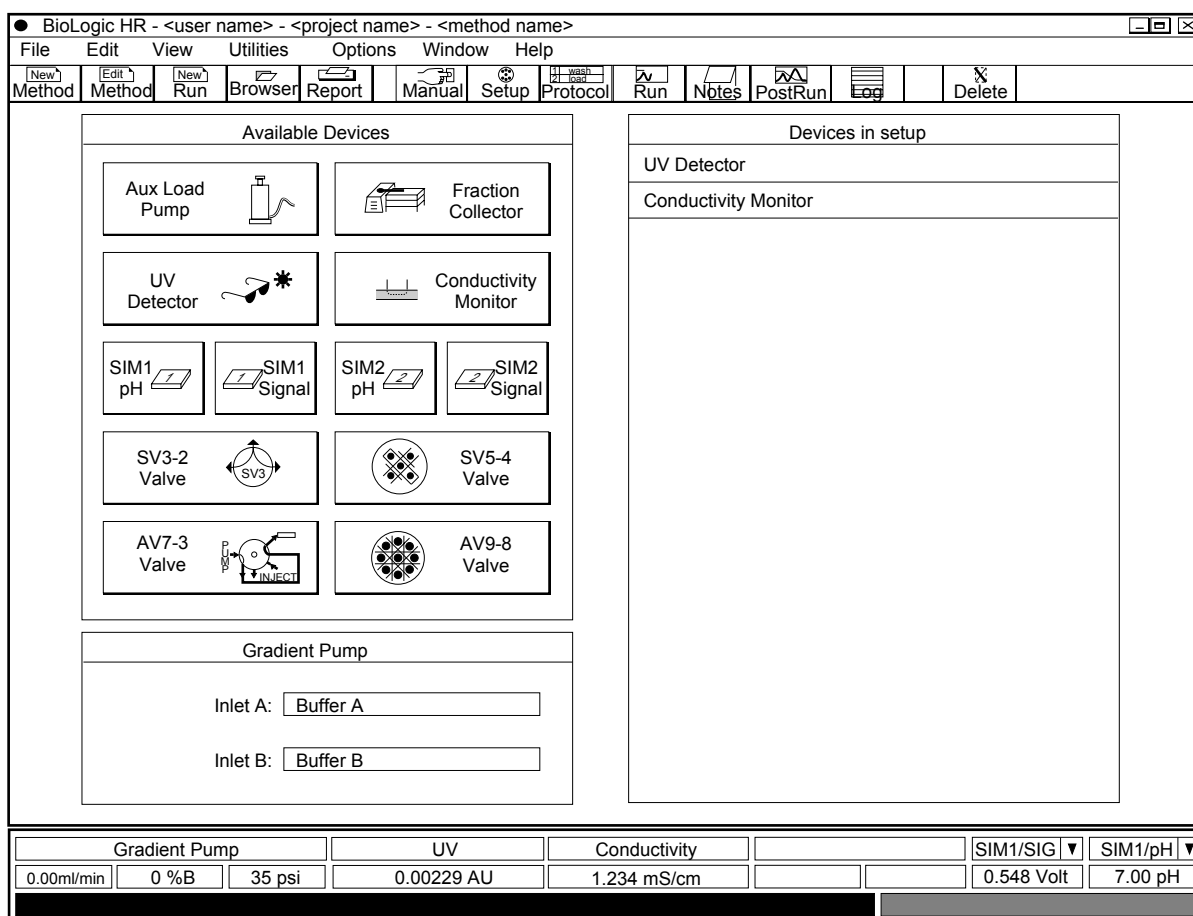


Figure 35. Setup Screen

Notes on using Valve Functioning and Aliasing:

One of the most powerful features of the BioLogic HR Software is the ability to define the specific purpose of the solenoid and automated valves and to “alias” or name each valve position. This greatly simplifies programming individual steps in the Protocol screen.

For example;

- The most common use of an AV7-3 valve is as a Sample Inject Valve. Once an AV-7-3 valve has been assigned to this role, the software controls all of its valve changes based on the programmed sample injection step.
- The most common use of a SV3-2 valve will be as a Fraction Collector Diverter valve with a Model 2110 or generic collector. Once an SV3-2 valve has been assigned to this role, the software controls eluant diversion based on the programmed fraction collection scheme.

In summary, you can identify the following as applying to all valves:

- **Valve Name/Function.** What is the valve to be used for? Select from the drop-down list box or input a unique function that you will recognize when the particular valve is selected in the Protocol screen.
- **Connector.** Which connector at the rear of the workstation is the valve connected to?
- **Position Names.** These fields let you name (“Alias”) each valve position.

The following devices are listed in the Setup screen:

- **Auxiliary Load Pump:** This button should be selected if you are using an Auxiliary Load Pump, such as the Model EP-1 Econo Pump. See Section 4, Advanced System Applications, for ways to incorporate an auxiliary load pump into a method.
- **Fraction Collector:** Select this button if you are using a fraction collector with the system. A dialog box appears, allowing you to select which collector is to be used. If you use threshold detection to initiate fraction collection, choose the detector source (UV, conductivity, pH, or another SIM signal) in this screen.

Note: To enable the use of collection windows or threshold collection for a Model 2110 or generic fraction collector, it is necessary to assign an SV3-2 valve as the fraction collector diverter valve.

- **UV Detector:** Typically, the UV Detector is always listed in the Devices in Setup panel, as it is standard with the BioLogic HR system.
- **Conductivity Monitor:** Typically, the Conductivity Monitor is always listed in the Devices in Setup panel, as it is standard with the BioLogic HR system.
- **Signal Import Modules (SIMs):** The analog devices which may be used with the Signal Import Module-HR include any detection source, such as a variable wavelength UV detector, a pH meter, a refractive index meter, an RI detector, or a fluorescence detector.

Pressing the **SIM pH** button adds the device to the Devices in Setup list. Pressing the **SIM Signal** button displays the **Edit Signal Import Module** dialog box. Set the **Units** and the minimum and maximum for the **Units Range** and **Device Output Range (volts)**. Select the device number on the SIM-HR to match the number displayed in the dialog box. Refer to the analog device’s documentation for information on output voltages.

- **Valves (SV3-2, SV5-4, AV7-3, and AV9-8):** The BioLogic HR system’s automatic valves can be configured in a number of different ways depending on their function in a method. Table 14 discusses how each of the valves can be set up. See Section 4, Advanced System Applications, for ways to use and alias valves.
- **Gradient Pump:** Inlets A and B may be aliased by overwriting “Buffer A” and “Buffer B” with the description of the buffer.

Table 14.
Valve Setup Information

Valve Type	Valve Name/ Function	Position Names	Notes
SV3-2 Low pressure solenoid valve	Fraction Collector Diverter	1 Waste 2 Collect	Eliminates the need for a “Change Valve” step in the Protocol screen, since its function as a fraction collection diverter is determined by the actual fraction collection parameters chosen.
	Inlet A	Aliased by user	When used before the inlet to Pump A or B, the valve enables buffer selection. The buffer name/alias specified for each position will appear in the Protocol screen’s “Isocratic Flow”, “Linear Gradient”, and “Change Valve” dialog box. Refer to Chapter 8, Sample Loading, for examples.
	Inlet B	Aliased by user	
	Aux Pump Inlet	Aliased by user	Used for auxiliary pump load selection to select one of two solutions. Refer to Chapter 8, Sample Loading, for examples.
	User-assigned name.	Aliased by user	When used for a purpose other than that described above. The name/alias specified for each position will appear in the Protocol screen’s “Change Valve” dialog box.
SV5-4 Low pressure solenoid valve	Inlet A	Aliased by user	When used before the inlet to Pump A or B, the valve provides preparative sample loading or buffer selection. The buffer or sample name/alias specified for each position will appear in the Protocol screen’s “Isocratic Flow”, “Linear Gradient”, and “Change Valve” dialog box. Refer to Chapter 8, Sample Loading, for examples.
	Inlet B	Aliased by user	
	Aux Pump Inlet	Aliased by user	Used for Auxiliary Pump load selection to select one of four solutions. Refer to Chapter 8, Sample Loading, for examples.
	User-assigned name.	Aliased by user	When used for a purpose other than that described above. The name/alias specified for each position will appear in the Protocol screen’s “Change Valve” dialog box.
AV7-3 High pressure valve	Sample Inject	1 Load Sample 2 Inject Sample 3 Purge	For automatically loading a sample.

Table 14. (continued)
Valve Setup Information

Valve Type	Valve Functions	Valve Positions	Notes
AV7-3 (cont'd)	User-assigned name.	Aliased by user	When used for a purpose other than that described above. The name/alias specified for each position will appear in the Protocol screen's "Change Valve" dialog box. Refer to Section 4, Advanced System Applications, Chapters 8 through 10.
AV9-8 High pressure valve	Aux Pump Inlet	Aliased by user	Used for Auxiliary Pump load selection to select from up to eight samples, buffers, or rinse solution.
	Inlet A	Aliased by user	When used before the inlet to Pump A or B, the valve provides preparative sample loading or buffer selection. The buffer or sample name/alias specified for each position will appear in the Protocol screen's "Isocratic Flow", "Linear Gradient", and "Change Valve" dialog box. Refer to Section 4, Advanced System Applications, Chapters 8 and 9.
	Inlet B	Aliased by user	
	User-assigned name.	Aliased by user	When used for purposes other than those described above. The name/alias specified for each position will appear in the Protocol screen's "Change Valve" dialog box. Refer to Section 4, Advanced System Applications, Chapters 8 and 9.

7.3 PROTOCOL SCREEN

The Protocol screen provides a means to program step-by-step instructions to the system regarding what to do and when to do it.

Note: The programming base is either Volume (ml) or Time (min). The default programming base is set by selecting **Edit User Preferences** from the Options drop-down menu.

The Protocol screen contains the following types of buttons, which are described in detail in this section.

- **Add Step buttons:** These buttons are used to define the steps in the protocol. They are located in a vertical box on the left side of the screen. To insert a step, select the step *below* where the step is to be added, and then press the appropriate button to begin defining that step.
- **Fraction Collector button:** This button is used to define the fraction collection scheme. It is located below the Add Step buttons.
- **Edit buttons:** These buttons, located in the System Toolbar, are used to cut, copy, paste, and delete steps from the method. To cut, copy, or delete a step, select that step and then press the appropriate button. To paste, select the step below where the step is to be pasted and then paste.
- **File Buttons:** Use the **New Method**, **Edit Method**, and **Browser** System Toolbar buttons to create a new method or to open an existing Method.

The system checks the Setup information before allowing you to add steps or run the Method. This ensures that each device you need is present and available for the required function.

ADD STEP BUTTONS

EDIT BUTTONS

COLLECTOR BUTTON

Step	Time	Action
1	0.0	Collect Fractions during entire method
2	0.0	Chart Recorder Turn ON
3	2.0	Isocratic Flow 25mM Tris 100% Time: 2.0 min
4	3.0	Load: Sample 25mM Tris + Salt 0% Flow: 2.0 ml/min
5	7.0	Isocratic Flow Load: standard Volume: 1.0 ml Auto Inject Valve
6	23.0	Linear Gradient 25mM Tris 100% Volume: 4.0 min
7	25.0	Linear Gradient 25mM Tris + Salt 0% -> 55% Flow: 2.0 ml/min
8	28.0	Isocratic Flow 25mM Tris + Salt 55% -> 100% Volume: 2.0 min
End of Protocol		

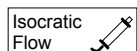
Gradient Pump: 0.00ml/min, 0 %B, 0 psi UV: 0.00229 AU Conductivity: 1.234 mS/cm SIM1/SIG: 0.548 Volt SIM1/pH: 7.00 pH

Figure 36. Protocol Screen

The following tables describe each button in the “Add Step” column of the Protocol screen.

Table 15.
Isocratic Flow Button

Isocratic Flow		Buffers:		% Composition	Volume (ml)	OK
Start Volume (ml)	0.00	Buffer A	▼	A 100	A 1.0	
		Buffer B	▼	B	Flow (ml/min)	Cancel
					B 2.0	

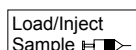


To deliver a buffer under isocratic conditions. The composition may be chosen by altering either Inlet A or Inlet B percent values which increase or decrease in tandem. Use the spin arrows or input values from the keyboard when this field is highlighted.

- **Start Time/Volume:** Calculates the elapsed time/volume from all previous steps. This is not user editable.
- **Buffers:** Displays all available buffers for separation. Buffer A and Buffer B can be aliased to actual buffer names by editing the Gradient Pump fields in the Setup screen. If pump inlet valves are assigned in the Setup screen, all additional buffers will be displayed in the drop-down boxes.
- **Composition:** This allows you to choose the buffer composition (e.g., 75% A and 25% B).
- **Volume/Time:** Allows you to choose the duration of the step.
- **Flow rate (ml/min):** Allows you to choose the flow rate of the step.
- **OK:** Adds the step to the protocol. This is the same as pressing the Enter key on the keyboard.
- **Cancel:** Does *not* add the step to the protocol. This is the same as pressing the Esc key on the keyboard.

Table 16.
Load/Inject Sample Button

4 Load/Inject Sample		☒ Static Loop ☐ Fill before Inject ☐ Rinse after Inject ☐ Dynamic Loop ☐ Fill before Inject ☐ Direct Inject		OK									
Start Volume (ml) 2.0				Cancel									
Inject Sample Auto Inject Valve will move to INJECT at Start of step and return to LOAD at end of step.		<table> <tr> <td>Injection Buffers:</td> <td>% Composition</td> <td></td> </tr> <tr> <td> Buffer A </td> <td>A 100 </td> <td>Volume (ml) 1.0 </td> </tr> <tr> <td> Buffer B </td> <td>B 0 </td> <td>Flow (ml/min): 2.00 </td> </tr> </table>			Injection Buffers:	% Composition		Buffer A	A 100	Volume (ml) 1.0	Buffer B	B 0	Flow (ml/min): 2.00
Injection Buffers:	% Composition												
Buffer A	A 100	Volume (ml) 1.0											
Buffer B	B 0	Flow (ml/min): 2.00											

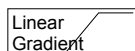


To program the sample injection. The Load Sample dialog box allows you to specify how the sample is to be introduced onto the column, as well as the volume of sample to be loaded. (If programming in time mode, the system calculates the time required for this step.)

- **Start Time/Volume:** Calculates the elapsed time/volume from all previous steps. This is not user editable.
- **Static Loop:** One of three options for sample loading. A static loop is a standard fixed volume sample loop. By adding an Auxiliary (AUX) pump and an auxiliary pump valve (such as the SV3-2, SV5-4, or AV9-8), the system can fill the sample into the static loop prior to injecting it into the column, and it can rinse the sample loop after the sample has been injected. **Note:** An isocratic step of sufficient length must precede a "fill before inject" step to allow sufficient time to fill the loop. Refer to Section 4, Advanced System Applications, for examples.
- **Dynamic Loop:** This is a second sample loading option which uses Bio-Rad's DynaLoop or other sliding-piston sample loop. Both partial loop and full loop injections can be programmed. By adding an Auxiliary (AUX) pump and an auxiliary pump valve (such as the SV3-2, SV5-4, or AV9-8), the dynamic loop can be filled prior to injection onto the column. No Rinse function is available. The dynamic loop can also be filled manually through Inject valve port #5. Refer to Section 4, Advanced System Applications, for examples.
- **Direct Inject:** Allows direct injection of sample through either the Gradient Pump or Auxiliary Load Pump. Pre-pump valves (such as the SV3-2, SV5-4, or AV9-8) automate direct injection. Use of an auxiliary pump (such as the Bio-Rad Model EP-1 Econo Pump) allows direct injection of sample onto a low pressure column (≤ 15 psi). Use of an Auxiliary (AUX) pump and an auxiliary pump valve allows multiple direct injections onto a low pressure column. Refer to Section 4, Advanced System Applications, for examples.
- **Inject Sample:** Explains the action of the inject valve. The AV7-3 auto inject valve will move to the Inject position at the start of the step and return to the Load position at the end of the step. The V7-3 manual inject valve will cause an alarm to sound and the Protocol to Hold for Start of Injection; the protocol will continue when the manual valve lever is moved to the inject position.
- **Injection Buffers and % Composition:** These fields allow you to choose the buffer type and composition which will push the sample through the sample loop.
- **Volume:** Allows you to specify the volume of the sample to be injected. The smallest volume allowed is 0.1 ml; the largest volume allowed is 9999 ml (for larger volumes, use another Load/Inject Sample step).
- **Flow Rate:** Allows you to specify the flow rate for the sample injection step.
- **OK:** Adds the step to the protocol. This is the same as pressing the **Enter** key on the keyboard.
- **Cancel:** Does **not** add the step to the protocol. This is the same as pressing the **Esc** key on the keyboard.

Table 17.
Linear Gradient Button

Linear Gradient		Buffers:	% Composition	Initial	Final	Volume (ml)	
Start Volume (ml)	2.0	Buffer A	A	0	0	A	0.00
		Buffer B	B	100	100	Flow (ml/min)	
						B	1.0
							OK
							Cancel

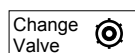


To deliver buffer gradients. The Linear Gradient dialog box sets the initial and final composition of the buffers and the period over which the change in composition is to occur. There is no limit to the number of gradient steps in a protocol.

- **Start Time/Volume:** Calculates the elapsed time/volume from all previous steps. This is not user editable.
- **Buffers:** Allows you to choose two buffers to make a binary gradient. Drop-down menus display all buffers that are set up using Gradient Pump Inlet valves.
- **% Composition:** Allows you to choose initial and final values that will form a linear binary gradient. Notice that changing one value in the “Initial” column affects the other value in that column. The “Final” column behaves identically.
- **Volume:** Allows you to choose the duration of this step.
- **Flow:** Allows you to choose the flow rate of this step.
- **OK:** Adds the step to the protocol. This is the same as pressing the **Enter** key on the keyboard.
- **Cancel:** Does **not** add the step to the protocol. This is the same as pressing the **Esc** key on the keyboard.

Table 18.
Change Valve Button

Change Valve		Valve Name:	Valve Position:	
Start Volume (ml)	0.00	Sample Inject	Load Sample	OK
				Cancel



To select a valve and change its position.

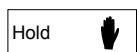
Note: This step does not affect manual valves such as the V7-3 and V9-8.

- **Start Time/Volume:** Calculates the elapsed volume from all previous steps. This is not user editable.
- **Valve name and Valve position:** Allows you to choose a valve and to make a change in valve position. Note that certain valve functions defined in the Setup screen (such as Fraction Collector Diverter, Aux Pump Inlet, Sample Inject, Inlet A, Inlet B) are “tied” to other Protocol steps, so valve position changes will be made automatically. An example would be an AV7-3 valve defined as a Sample Inject valve in the Setup screen, which is then tied to the Load/Inject Sample step in the Protocol screen.

However, valves assigned a “User-defined name” during Setup require the “Change Valve” step at the desired point in the Protocol.

Table 19.
Hold Button

Hold Until Event		Event:	☐ Sound Alarm	OK
Start Volume (ml)	0.0	Hold until Keypress		Cancel



Inserts a Hold step into the protocol. During a Hold, the method's current Gradient Pump composition (%B) remains constant. The method time (or volume) does not advance. When the run resumes, the current fraction collection condition is maintained and the volume counter advances. The hold condition is maintained until a specified activity occurs (e.g., "Hold until Start of Inject" or "Hold until Keypress"). For example, you may want a hold so that the V7-3 Manual Inject Valve can be moved to its Inject position. If the hold starts at time 2 (minutes), then the method resumes at time 2, no matter how long the Hold was in effect.

- **Start Time/Volume:** Calculates the elapsed time/volume from all previous steps. This is not user editable.
 - **Event:** Choose one of three events to discontinue the hold:

Hold until Keypress: Press the "Running Man" key on the keyboard (or the **F2** key on a standard keyboard) to end the programmed hold.

Hold until Start of Inject: The run will continue when the V7-3 manual inject valve is moved to the Inject position.

Hold until End of Inject: The run will continue when the V7-3 manual inject valve is moved back to the Load position from the Inject position.
 - **Sound Alarm:** When this box is checked, an alarm will sound at the beginning of the step to remind you that the system requires an action on your part that will then allow the rest of the Method to advance.
 - **OK:** Adds the step to the protocol. This is the same as pressing the **Enter** key on the keyboard.
 - **Cancel:** Does *not* add the step to the protocol. This is the same as pressing the **Esc** key on the keyboard.
-

Table 20.
Miscellaneous Buttons

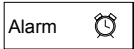
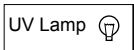
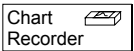
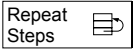
Button	Description
	To sound a 10 second audible alarm at the programmed time. The progress of the run will not be affected.
	To zero the UV baseline at the programmed time.
	At system start-up, the default condition is Lamp ON. If the lamp is OFF, it takes about 20 minutes for the lamp to warm up prior to use. If the system will not be used for an extended period after the run, you can use this button to insert a step turning the UV Lamp OFF at the end of the run.
	To turn on/off the Chart Recorder at the programmed time. Inserts a step that tells the Chart Recorder to start/stop the paper feed and lower/lift the pen. Note that the recorder will be automatically turned off at the end of a Method.
	To repeat the highlighted step(s) a specified number of times. To highlight more than one step, hold down the Ctrl key while selecting steps with the mouse.

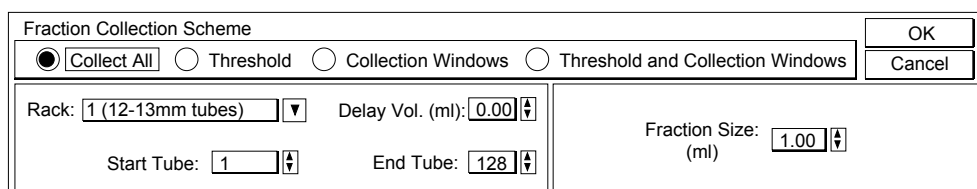
Table 21.
Fraction Collection Button

Button	Description
	To set up fraction collection. Fraction collection schemes are discussed below.

IMPORTANT NOTE:

The BioLogic HR system is designed to control the Model 2128 Fraction Collector, which is the only collector that can provide a choice of racks and the ability to overlay fractions from consecutive runs. It is recommended to use the optional Model 2128 diverter valve which sits next to the drop-head. The BioLogic HR system automatically registers the presence of this valve when it is plugged in. However, the Model 2128 will function in all collection modes without an additional diverter valve, as the drop-head will move back and forth between the tubes and the drain trough.

The BioLogic HR system will also accommodate the Bio-Rad Model 2110 Fraction Collector as well as generic collectors. To use a fraction collection scheme other than “Collect All” with these collectors, it is **essential** to assign a SV3-2 valve as a “Fraction Collector Diverter” in the Setup screen. Rack choice and Overlay features are **not** available with these collectors.



Collect All: The eluant from the entire run will be collected by the fraction collector.

- **Rack:** This applies only to the Model 2128 Fraction Collector. The following racks may be specified:
 - 1 128 tubes; 12-13 mm tube diameter
 - 2 78 tubes; 16-18 mm tube diameter
 - 3 Microplates, 3 total for 288 wells (96 wells/plate, arranged 8x12)
 - 4 Microtubes (128 capless microtubes)
 - 5 Prep-Adapter (10 bottles of any size)
- **Start Tube** and **End Tube:** This applies only to the Model 2128 Fraction Collector. Identify the first and last tubes to receive fractions.
- **Delay Vol. (ml):** The Delay function precisely synchronizes fraction collection with the UV signal on the chart recorder. It is defined as the tubing volume between the UV detector and the fraction collector drop-head. If you are using the recommended 1/16" OD, 0.020" ID Tefzel tubing, 1 cm of tubing has a volume of 2 µl. The volume of the supplied back-pressure regulator is 70 µl.
- **Fraction Size:** Enter the fraction size as volume (ml).

Table 21. (continued)
Fraction Collection Button

Description

Fraction Collection Scheme				OK
☐ Collect All	☒ Threshold	☐ Collection Windows	☐ Threshold and Collection Windows	Cancel
Rack: 1 (12-13mm tubes) ▼	Delay Vol. (ml): 0.00 ▼	Fraction Size: 1.00 ▼	Non-peak Parameters:	
Start Tube: 1 ▼	End Tube: 128 ▼	Threshold: 0.0100 ▼	Destination	Fraction Size (ml)
			☒ Waste	
			☐ Tubes	1.00 ▼

Threshold: This method allows the flow to be sent to the collector if the detector signal of the eluant is above a certain signal level. **Note:** You can select any detection source (UV, conductivity, pH, SIM data) to initiate threshold-based fraction collection. Non-peak eluant can be sent to waste or to the fraction collector.

- **Rack, Start Tube, End Tube, Delay Vol. (ml), and Fraction Size:** See description for **Collect All**.
 - **Threshold:** Enter the global threshold value in absorbance units.
 - **Non-peak parameters Destination and Fraction Size:** Enter the non-peak parameters. If the buffer stream is diverted to Waste, there is no need to enter a fraction size. But if you want to collect non-peak material, you must enter a fraction size.
-

Table 21. (continued)
Fraction Collection Button

Description

Fraction Collection Scheme			
☐ Collect All ☐ Threshold ☒ Collection Windows ☐ Threshold and Collection Windows			
Rack: 1 (12-13mm tubes)	☐ Link Windows	Start (min)	End (min)
Start Tube: 1	Delete Window	0.0	4.0
End Tube: 128		1	1.00
Delay Vol. (ml): 0.00	Add Window		

Collection Windows: This method specifies collection during specified parts of the run (Time or Volume). Each collection window can have a unique fraction size (ml). **Note:** Collection windows programming should be done after all other steps have been written.

- **Rack, Start Tube, End Tube, Delay Vol. (ml), and Fraction Size:** See description for **Collect All**.
 - **Link Windows:** Select this to join windows, so that the end of one window is the beginning of another. It is particularly useful as an editing feature which, if selected, will automatically change the time. For example, it will change the end time of one window if the beginning time of the next collection window is altered.
 - **Start (ml or min), End (ml or min), and Fraction Size (ml):** Each window is defined by these parameters. Up to three windows can be displayed at any one time; use the scroll bar to the right of the window display to scroll through the list of window. Enter the fraction size as volume.
 - **Add Window and Delete Window:** The **Add Window** button inserts a window after the highlighted window; **Delete Window** deletes the highlighted window.
-

Table 21. (continued)
Fraction Collection Button

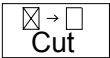
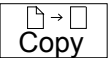
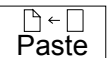
Description

Threshold and Collection Windows: Each Collection Window can also contain a threshold value to further discriminate when eluant can be collected. For each collection window, non-peak effluent can be directed to waste or to the fraction collector.

- **Rack, Start Tube, End Tube, Delay Vol. (ml), and Fraction Size:** See description for **Collect All**.
 - **Link Windows:** Select this to join windows, so that the end of one window is the beginning of another. It is particularly useful as an editing feature which, if selected, will automatically change the time. For example, it will change the end time of one window if the beginning time of the next collection window is altered.
 - **Start (ml), End (ml), and Fraction Size (ml):** Each window is defined by these parameters. Up to three windows can be displayed at any one time; use the scroll bar to the right of the window display to scroll through the list of window. Enter the fraction size as volume (ml).
 - **Add Window and Delete Window:** The **Add Window** button inserts a window after the highlighted window; **Delete Window** deletes that window.
 - **Threshold:** Enter the threshold value for each collection window.
 - **Non-peak parameters Destination and Fraction Size:** Enter the non-peak parameters. When collecting non-peak material, you must enter a fraction size.
-

The following Toolbar menu buttons are available only when creating and editing a protocol.

Table 22.
Protocol Screen's Editing Toolbar Buttons

Button	Description
Edit buttons:  Cut	These buttons are available from the System menu. To cut the currently highlighted step from the protocol. It places the step in the clipboard, which means it can be placed elsewhere by using the Paste button.
 Copy	To copy the currently highlighted step in the protocol.
 Paste	To paste the cut or copied step before the currently highlighted step.
 Delete	To delete the currently highlighted step from the protocol.

7.4 RUN SCREEN

The Run screen starts and displays the progress of the Method Run. All data associated with the run are automatically saved. An example of a run in progress is shown in Figure 37 below. Table 23 discusses the buttons which may be used to control the run. In addition, there are toolbar buttons that display additional screens associated with a completed run (Post Run and Integ.). These are discussed in the following sections.

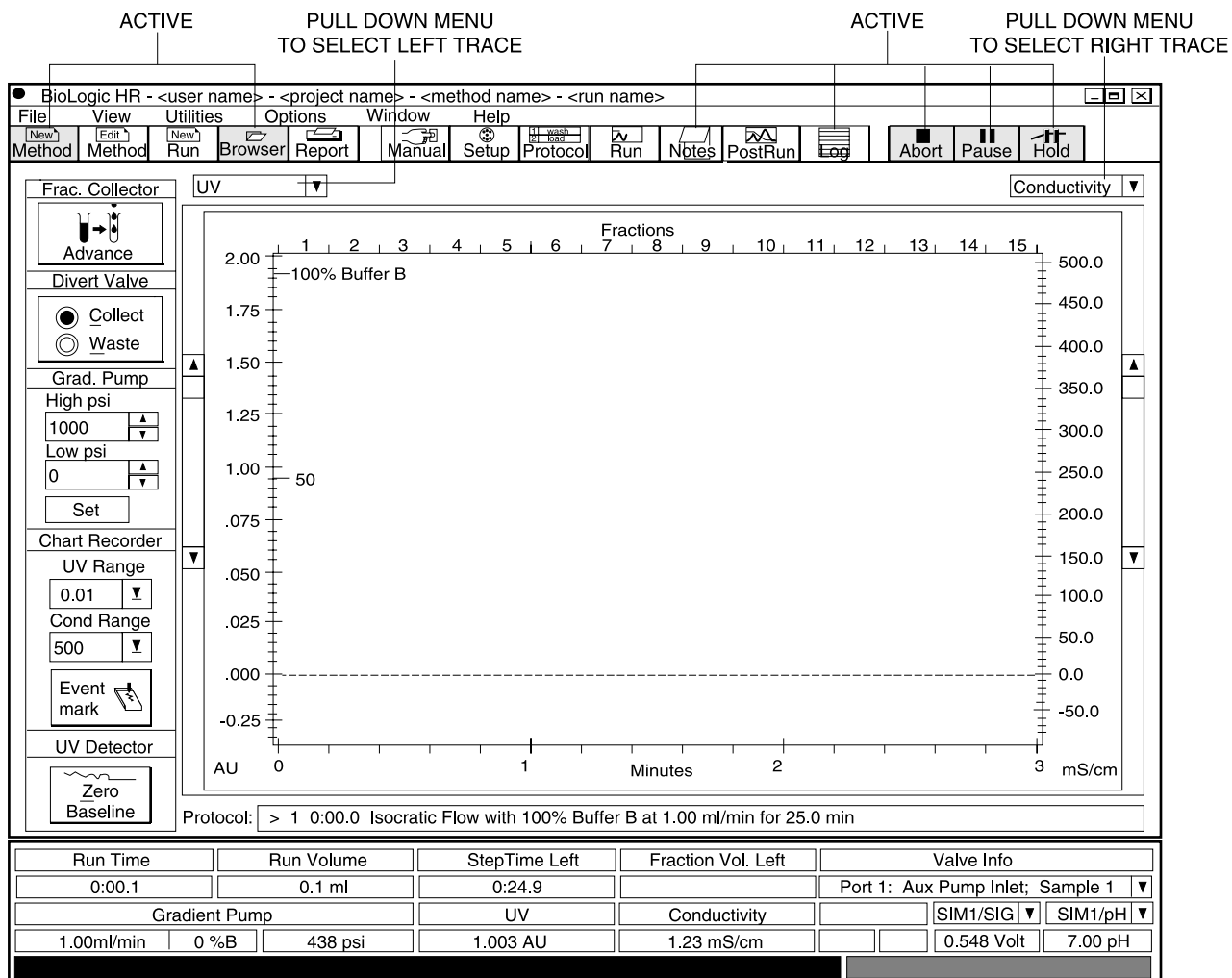
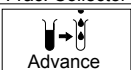
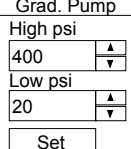
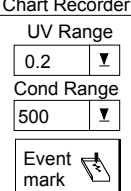
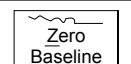
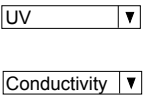
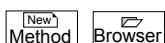


Figure 37. Run Screen showing a Run in Progress

Table 23.
Run Screen's Control Buttons

Button	Description
	Advances the fraction collector to the next tube. Pressing this button does not modify the method. The event is recorded in the Run Log.
	Immediately changes the position of the diverter valve. Pressing this button does not modify the method. The event is recorded in the Run Log.
	To set the pressure limits of the Gradient Pump. This is not part of a protocol; it is a system setting.
	To set the UV Detector range for the Chart Recorder output. The settings are: 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0, 2.0 AUFS. To set the Conductivity range on the Chart Recorder output. The settings are: 500, 200, 100, 50, 20, 10, 5 mS/cm. To set an Event Mark on the chart recorder UV trace output only. It will not record event marks on the screen.
	Use the Zero Baseline button to reset the UV detector absorbance value to zero. Note: Use care when using this feature during a collection scheme which uses a Threshold value.
	To select which of the active data trace scales are viewed during a run. As each trace scale is selected, the scroll bars can be used to adjust the scale setting.
	To work offline while a run is in progress. (Refer to section 7.4.2, Working Offline.)
	Notes displays the Run Notebook screen, which allows you to enter notes regarding the run. Log display the Run Log screen, which provides information about the run.
	To stop, pause, or hold a method in progress. (Refer to Figure 38.)

7.4.1 Pausing/Stopping a Method in Progress

The Run screen provides three buttons for pausing or stopping a method in progress. These buttons are discussed in the figure below.

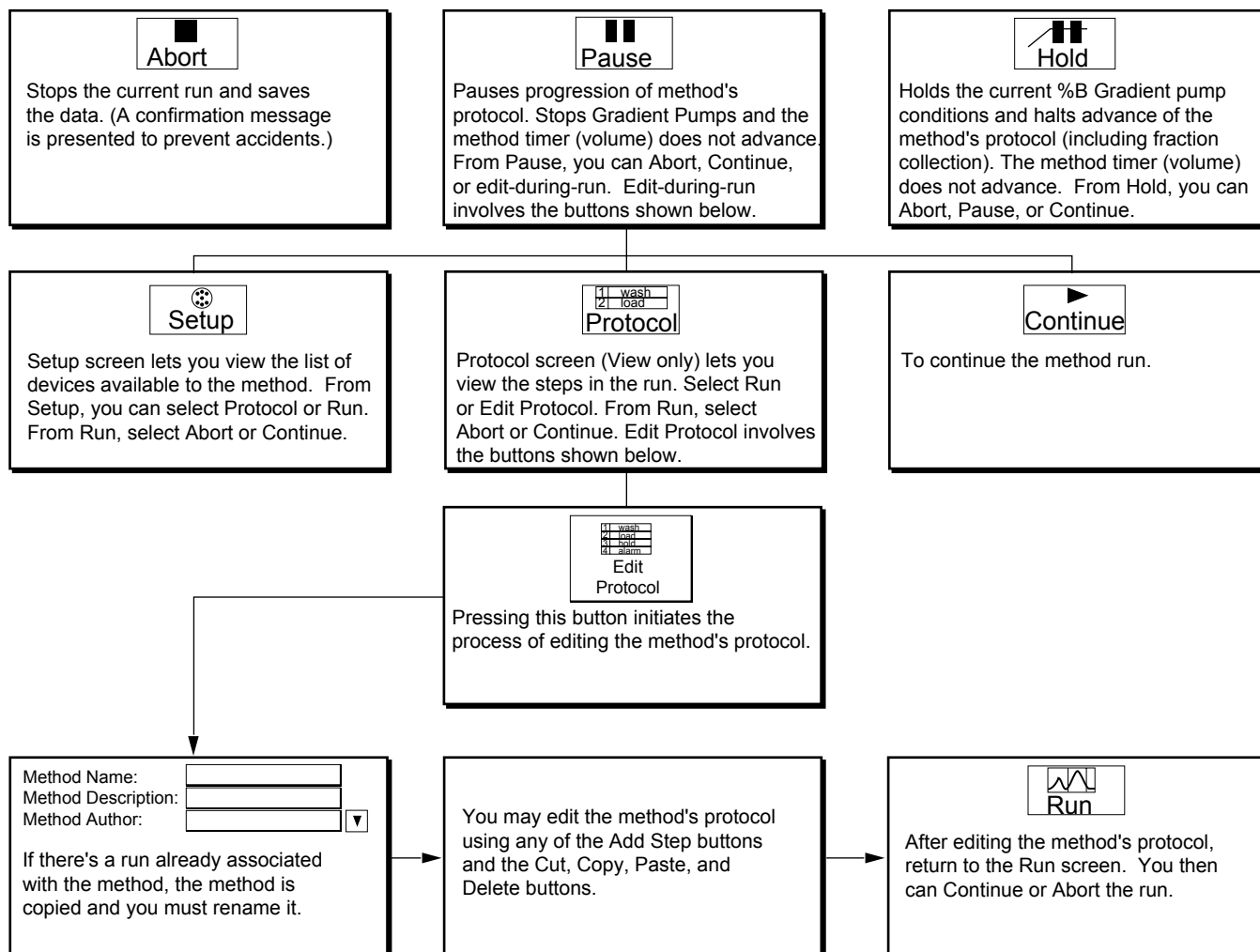


Figure 38. Run Screen's Abort, Pause, and Hold Buttons

7.4.2 Working Offline

The BioLogic HR system software allows you to work offline while a run is in progress. The following actions can be performed:

- Write a new method
- Edit an existing method
- View results from a completed run
- Print a report from a completed run
- Integrate run data from a completed run
- Perform PostRun analysis on a completed run
- Copy In / Copy Out data
- Export Data or Export Chromatogram Image of a completed run
- Access the HELP screen

These functions are not available while a run is in progress:

- PostRun analysis of the Run in progress
- Integration of the data from the Run in progress
- Copy Out data from the Run in progress
- Initiation of a new Run
- Utilities functions, including calibrating pH probe, gradient pumps, conductivity
- Manual mode functions

During a run, the following toolbar buttons activate the offline window:

- **New Method** button
- **Browser** button

The online and offline windows are distinguished as follows:

- Online: The BioLogic HR icon in the upper left of the screen is green, and the title bar does not indicate online.
- Offline: The BioLogic HR icon in the upper left of the screen is yellow, and the title bar indicates offline.

To change between the online and offline windows, you can use any of the following:

- Windows drop down menu: The online and offline windows are listed.
- Task bar: The Windows® 95 taskbar lists the BioLogic HR windows.
- Alt-Tab: Simultaneously hold down these two keys to list the open windows.

7.4.3 Editing A Protocol While a Run is In Progress

During a run, you may find you want to change some of the run parameters. The figure below shows how this is handled by the system and some of the restrictions.

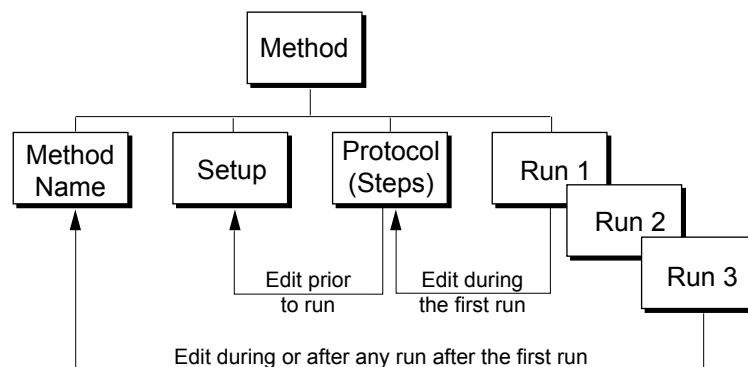


Figure 39. Editing during a Run

- During the first run of a method, you can **pause** the method and return to the Protocol screen to edit steps that have not already been started.
- Any run after the first can be paused and then edited, but you will first be asked to rename the method (or accept the default new name) when the **Edit Protocol** button is clicked. This is to ensure the integrity of the database in terms of which Method Protocol particular runs are tied to.

For example, assume you make three runs under a method named “Ion-exchange Bio-Scale Q2.” During the fourth run, you may decide to pause the run, and enter the Protocol screen to increase the length of the salt gradient. In this case, you will first be prompted to rename the Method (e.g., Ion-exchange Bio-Scale Q2 Rev. 1) before proceeding with the edit.

Upon completion of the run, the database of Methods and Runs will show three runs associated with the method “Ion-exchange Bio-Scale Q2” and one run associated with the method “Ion-exchange Bio-Scale Q2 Rev. 1.”

- A fraction collection scheme cannot be added to a Protocol once a run has been started. In such cases, you should abort the run and select **Edit the Method**.
- A currently programmed fraction collection scheme may be edited during a run, but note that changing the fraction size of a “Collect All” scheme will first require selection of “Collection Windows.” This is because the editing process effectively turns the initial “Collect All” scheme into the “first” collection window.

● BioLogic HR - <user name> - <project name> - <method name>

File Edit View Utilities Options Window Help

New Method Edit Method New Run Browser Report Manual Setup Protocol Run Notes PostRun Log Cut Copy Paste Delete

Add Step:

1 0.0 Collect Fractions during entire method

2 0.0 Chart Recorder Turn ON

3 2.0 Isocratic Flow 25mM Tris 100% Time: 2.0 min
25mM Tris + Salt 0% Flow: 2.0 ml/min
Static Loop Auto Inject Valve

4 3.0 Load: Sample Load: standard Volume: 1.0 ml Flow: 2.0 ml/min

5 7.0 Isocratic Flow 25mM Tris 100% Volume: 4.0 min
25mM Tris + Salt 0% Flow: 2.0 ml/min

6 23.0 Linear Gradient 25mM Tris 100% --> 55% Volume: 16.0 min
25mM Tris + Salt 0% --> 45% Flow: 2.0 ml/min

7 25.0 Linear Gradient 25mM Tris 55% --> 100% Volume: 2.0 min
25mM Tris + Salt 45% --> 0% Flow: 2.0 ml/min

8 28.0 Isocratic Flow 25mM Tris 100% Volume: 3.0 min
25mM Tris + Salt 0% Flow: 2.0 ml/min

End of Protocol

View-Only Mode

Fraction Collection

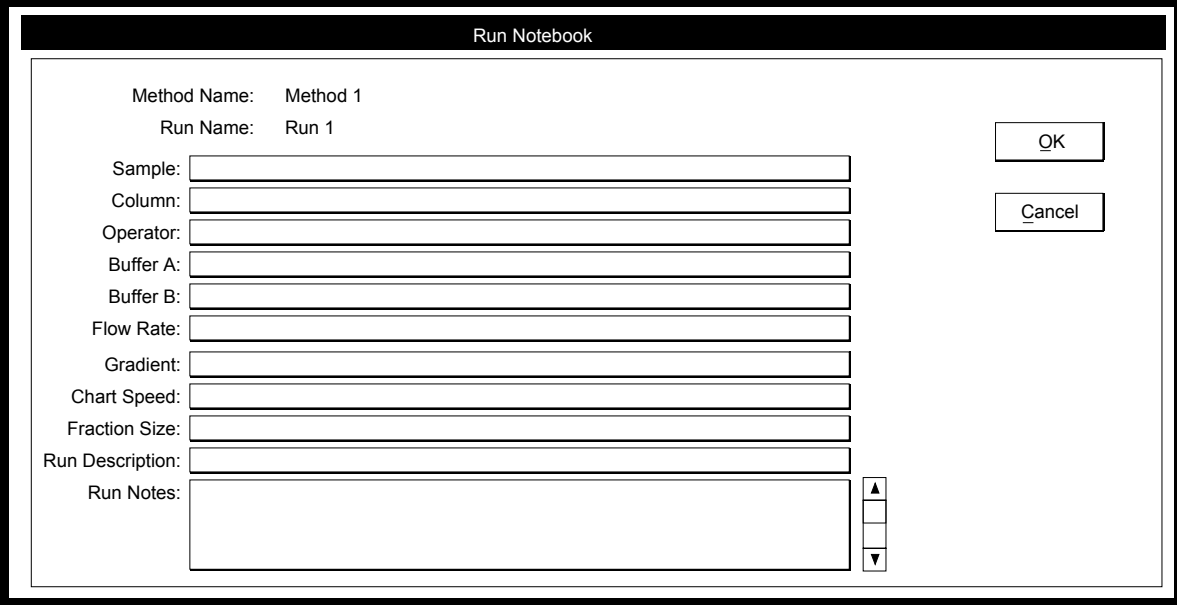
Gradient Pump			UV		Conductivity		SIM1/SIG ▼		SIM1/pH ▼	
0.00ml/min	0 %B	35 psi	0.00229 AU		48 µS		0.989 Volt		4.00 pH	

Select item(s) for copy/cut/delete. Double-click on step to edit.

Figure 40. Protocol Screen during a Run

7.4.4 Run Notebook Screen

Use the Run Notebook screen to maintain any information you want regarding the run. It contains fields for description of the sample, the column, the operator, the buffer(s), flow rates, gradients, chart speed, fraction size, and any other information you might want to enter.



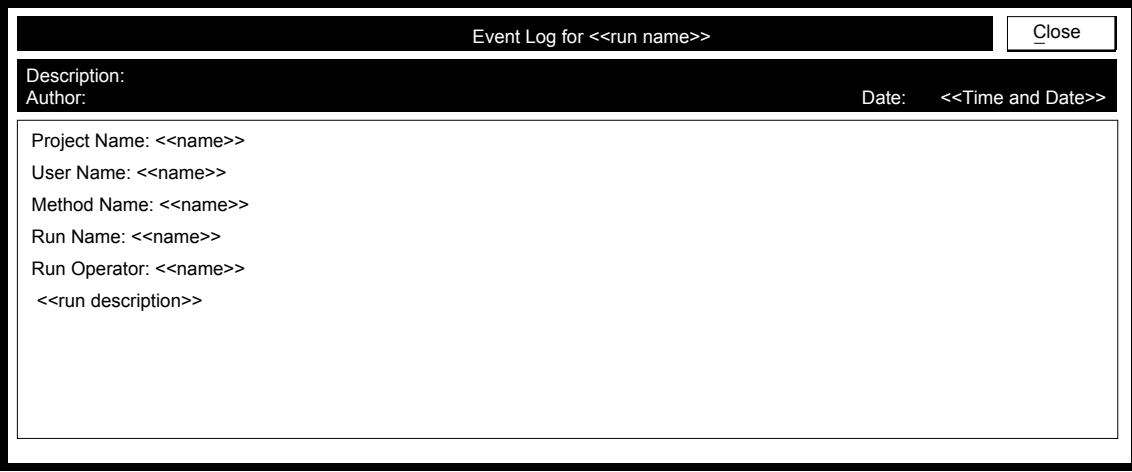
The Run Notebook screen is a dialog box titled "Run Notebook". It contains the following fields and controls:

- Method Name: Method 1
- Run Name: Run 1
- Sample: [Text Field]
- Column: [Text Field]
- Operator: [Text Field]
- Buffer A: [Text Field]
- Buffer B: [Text Field]
- Flow Rate: [Text Field]
- Gradient: [Text Field]
- Chart Speed: [Text Field]
- Fraction Size: [Text Field]
- Run Description: [Text Field]
- Run Notes: [Text Area]
- OK button
- Cancel button
- Up, Down, and Enter arrow buttons next to the Run Notes field.

Figure 41. Run Notebook Screen

7.4.5 Run Log Screen

The Run Log screen details the time and order of the execution of each step, error, and/or event of the run. This information cannot be edited. It is possible, although not recommended, to turn off the Run Log by deselecting its checkbox in the Edit User Preferences window, available from the **Options** menu.



The Run Log screen is a dialog box titled "Event Log for <<run name>>". It contains the following fields and controls:

- Close button
- Description: [Text Field]
- Author: [Text Field]
- Date: <<Time and Date>>
- Project Name: <<name>>
- User Name: <<name>>
- Method Name: <<name>>
- Run Name: <<name>>
- Run Operator: <<name>>
- <<run description>>

Figure 42. Run Log Screen

7.5 POST-RUN SCREEN

The Post-Run screen enables peak annotation (“tagging”) and data export. To enter the Post-Run screen, open a Run file using the **Browser** button on the button bar, or when a run is finished choose **PostRun** from the Toolbar buttons. Two chromatograms are viewed: the main chromatogram in the lower large window and the reference chromatogram in the upper right window.

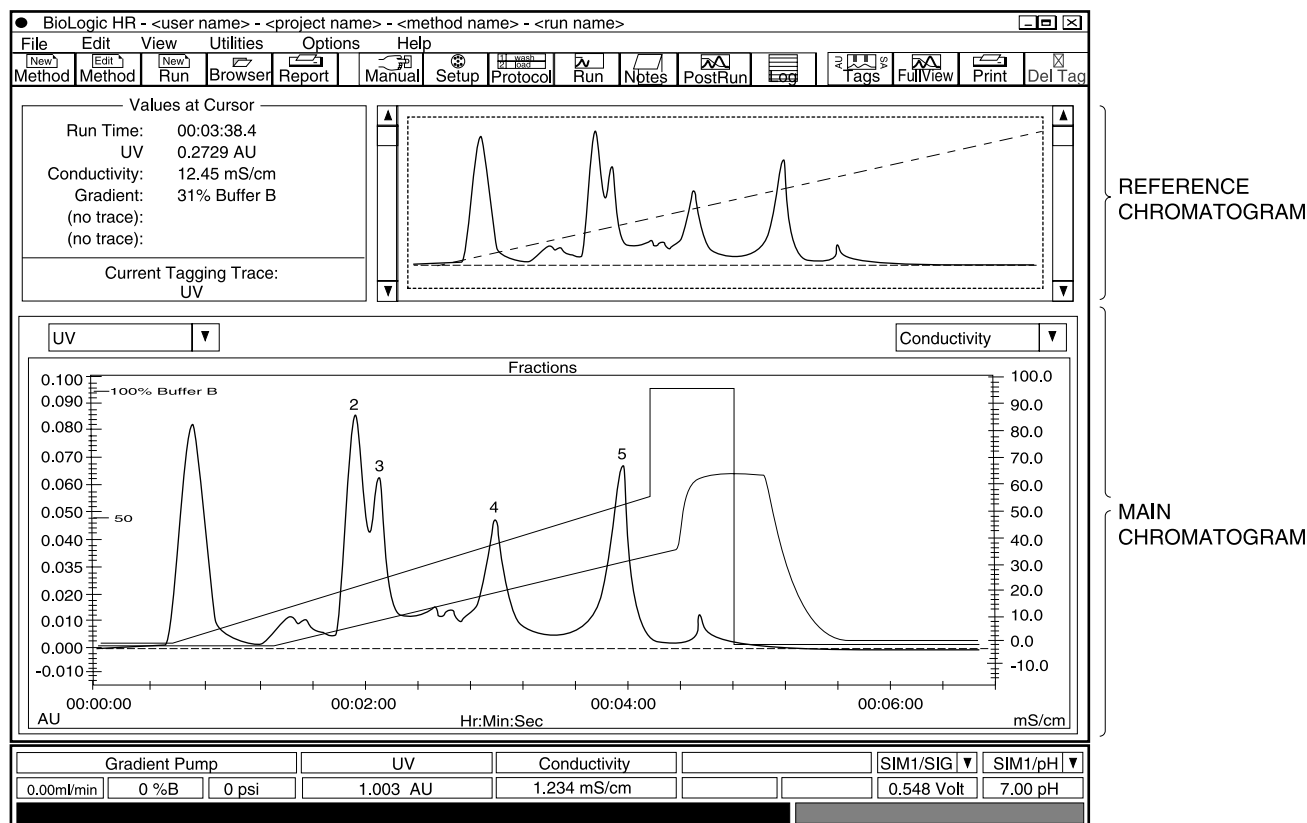


Figure 43. Post-Run Screen

7.5.1 Resizing

The main chromatogram shows the current “Zoom” region. To resize the image (or “zoom in” to a particular region of the chromatogram), either change the axes scale in the reference chromatogram, or use the “rubber band” controls on either chromatogram. To use the rubber band controls, click and hold the cursor within the chromatogram region and move the cursor so that the area of interest is enclosed. Release the cursor and the main chromatogram will be resized. You can then pan across the area by clicking the right mouse button and moving the mouse.

7.5.2 Chromatogram Information (Values at Cursor)

The top left corner of the PostRun screen contains information about the chromatogram. The position of the vertical bar within the main chromatogram dictates the displayed information. Run time, the absorbance (AU) and conductivity (mS/cm) values, gradient progression (%B), and the current tagging trace are displayed. To view the vertical bar, move the cursor inside the main chromatogram.

7.5.3 Annotating (“tagging”) the Chromatogram

Tags can be placed on any trace. To set up the assignment of tags to a trace, and to view all assigned tags and the data associated with them, use the Post-Run Tags window, shown below. To display the Post-Run Tags window, click once on the **Tags** button in the toolbar or from the Edit drop down menu, select **Tag and Trace Options**.

In the Post-Run Tags window, select a trace by clicking on an Active Trace radio button. Trace visibility and tag visibility can also be selected from this box. Tag styles are specified in the upper right box. Peaks can be annotated with a sequential tag number (1,2...), a user-defined name (user tag name), with the trace value, or the run time.

Trace	Trace Visibility	Tag Visibility	Active Trace
UV (AU)	☒	☒	☒
Cond. (mS/cm)	☒	☐	☐
(%) Buffer B	☒	☐	☐
(no trace)	☐	☐	☐
(no trace)	☐	☐	☐

Tag Style for active trace

☒ Sequential Tag Number
☐ User Tag Name
☐ Trace Value (Y axis value)
☐ Run Time (X axis value)

Ok

Cancel

Tag #	Tag name	Retention time (h:m:s)	Volume (ml)	Buffer B (%B)	UV (AU)	Cond. (mS/cm)	(no trace)	(no trace)	Tube #
1	Myoglobin	00:00:32	2.2	0	0.07992	1.496	N/A	N/A	0.07992
2	Conalbumin, peak 1	00:01:41	6.7	18	0.08617	7.999	N/A	N/A	0.08617
3	Conalbumin, peak 2	00:01:48	7.2	20	0.05921	9.025	N/A	N/A	0.05921
4	Ovalbumin	00:02:44	11.0	34	0.03799	16.472	N/A	N/A	0.03799
5	Soybean Trypsin Inhibitor	00:03:30	14.0	46	0.06844	22.308	N/A	N/A	22.308

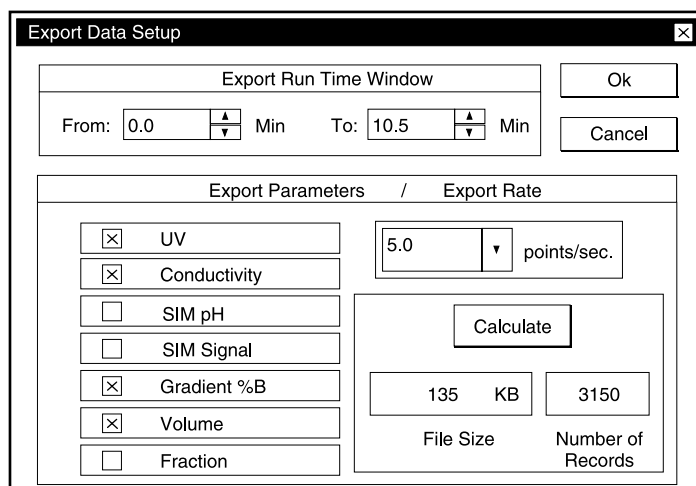
Figure 44. Post-Run Tags

To change a tag name in the Post-Run screen chromatogram, move the cursor over the tag and note that the cursor changes. Double click on the tag and in the window that appears, enter the tag name. To deselect a tag, highlight the undesired tag and press DelTag from the button bar. To remove all tags, select “Delete all tags” from the **Edit** drop-down menu.

7.5.4 Transferring Data into other Programs

To transfer data, go to the Post-Run screen of the desired run. Select **Export Data** from the File drop-down menu (See Figure 45, Export Data Setup screen). From this screen, choose the run time and the data trace(s) that will be exported. Although data is collected at a rate of five points per second, it can be exported at a user-defined rate. Data is exported in an ASCII.TXT file format. The file name is user-defined and can be saved anywhere on the hard drive. Note that the default file is called "Export" and is located in the following directory, "Biologic\data/export".

When importing the data into Excel, choose comma delimited.



The **Export Data Setup** dialog box is used to configure data export settings. It features a title bar with a close button. The main area is divided into two sections: **Export Run Time Window** and **Export Parameters / Export Rate**. The first section contains 'From' and 'To' time fields with spinners, both set to 0.0 and 10.5 minutes respectively, and 'Ok' and 'Cancel' buttons. The second section contains a list of parameters with checkboxes: UV (checked), Conductivity (checked), SIM pH (unchecked), SIM Signal (unchecked), Gradient %B (checked), Volume (checked), and Fraction (unchecked). To the right of this list is an 'Export Rate' field set to 5.0 points/sec. Below these is a 'Calculate' button and two display fields: 'File Size' (135 KB) and 'Number of Records' (3150).

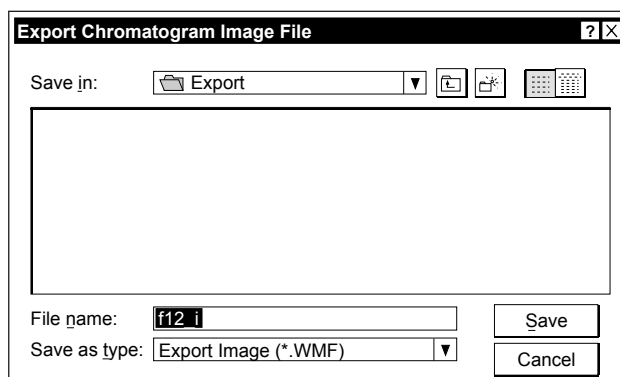
Figure 45. Export Data Setup Screen

7.5.5 Exporting Chromatogram Images

Chromatographic images can be exported into other applications, such as word processing programs. Before transferring an image, modify the chromatogram in the PostRun screen since changes can not be made to the file once it is transferred out of the BioLogic HR program. The image in the main chromatogram will be the exported image.

Select Export Chromatogram Image from the File drop-down menu (see Figure 46). The image is exported as a Windows Meta File (*.wmf) format. In general, Windows Meta File format is used for transfer into a Windows-based application.

Alternatively, a chromatogram can be copied to the Windows clipboard and pasted into another Windows program. Choose "Copy zoom Chromatogram to clipboard" from the Edit drop-down menu.



The **Export Chromatogram Image File** dialog box is used to save a chromatogram image. It has a title bar with a help icon and a close button. The 'Save in:' field shows the 'Export' folder. Below this is a large empty rectangular area representing the image. At the bottom, the 'File name:' field contains 'f12.i' and the 'Save as type:' dropdown is set to 'Export Image (*.WMF)'. 'Save' and 'Cancel' buttons are located at the bottom right.

Figure 46. Exporting a Chromatographic Image

8.0 SAMPLE LOADING

The BioLogic HR system supports several methods for sample loading. A typical method for loading sample is through a static fixed volume loop, as shown in chapter 4. When large volume sample injections are required, the following options are also available:

- sample loading through the Workstation Gradient Pump
- sample loading using the DynaLoop sliding-piston loop, available in both 25 and 90 ml volumes
- sample loading onto a low pressure column using the Econo Pump as an auxiliary (Aux) pump
- multiple sample loading using either an AV9-8 valve or DynaLoop
- multiple sample loading using Bio-Rad's Model AS-100 Autosampler

8.1 SAMPLE LOADING THROUGH THE WORKSTATION GRADIENT PUMP

Samples may be loaded directly through the Workstation Gradient Pump, as long as the sample is first filtered through a 0.45 μm filter. Use a sanitizing solution to remove protein residues after the sample has been loaded; protein residues might otherwise reduce piston seal lifetime. Consult the On-Line Help for additional information.

8.1.1 System Setup

When loading sample through the Workstation Gradient Pump, use an SV3-2, SV5-4, or AV9-8 valve to facilitate sample loading and cleaning. Whichever valve is selected, it should be plumbed to Inlet A on the Gradient Pump. To prevent air from entering the system, unused ports should be closed with the 1/4-28 plugs which are supplied in the BioLogic HR system fittings kit. In the Setup screen, where the Gradient Pump is aliased, Inlet A can be identified as "Sample Inject Valve."

In this example, one port of the SV5-4 valve is dedicated to the sample, another to running/equilibration buffer, a third to storage buffer (such as 20% EtOH), and the fourth to a sanitizing solution (such as 1M NaOH). The elution buffer is plumbed directly into Gradient Pump B and is configured as Inlet B in the Setup screen.

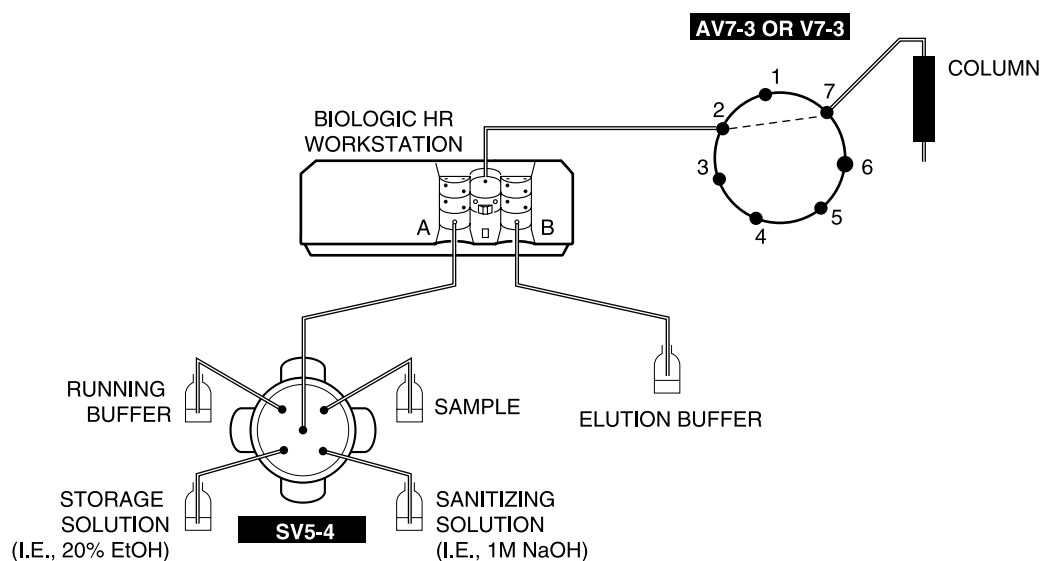


Figure 47. Sample Loading Through the Gradient Pump

8.1.2 Writing the Protocol

The following steps should be programmed within the Method Protocol, as shown in the figure below:

1. At the Load/Inject Sample step, specify Direct Inject and choose Gradient Pump A as the inlet through which the sample will be delivered. Enter the volume of sample and the flow rate for the injection step. The step starting time or volume appears to the left of the options buttons.
2. Add an Isocratic Flow Step with Running buffer to ensure that all the sample has been moved out of the pre-pump inlet lines and onto the column. Three to five volumes of inlet tubing length is recommended. There is approximately 20 µl in 1 cm of 1/8" OD tubing. For example, for 10 cm of 1/8" OD tubing, program an isocratic flow step of 1.0 mL.
(10 cm) (0.020 mL/cm) 5 = 1mL
3. Add a Change Valve step to move the AV7-3 Sample Inject valve to the Purge position, taking the column off line, and add an Isocratic Flow step to run 10mL of buffer B through Gradient Pump B. This will ensure that the mixer has been removed of any protein residue. Add an Isocratic Flow step to run 10mL of Running buffer through the Gradient Pump A to establish Running Buffer conditions.
4. Add a Change Valve step to move the AV7-3 Sample Inject valve back to the Load position to begin the run.
5. Continue writing the separation protocol.
6. For sanitization, 1.0M NaOH may be used. Be sure to wash with water to remove the base. Refer to Chapter 12, Maintenance, for information on how to store the BioLogic HR system.

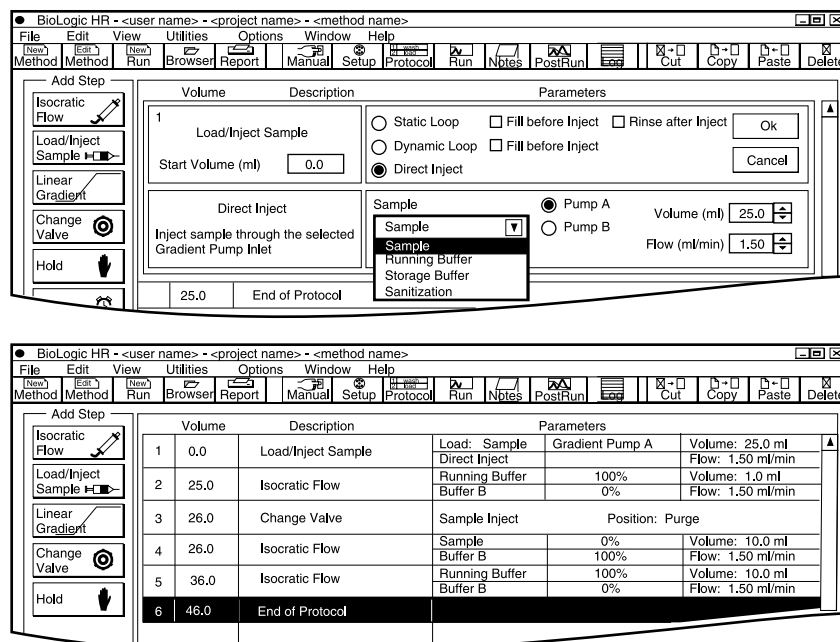


Figure 48. Writing a Protocol for Sample Injection Through the Gradient Pump

8.2 SAMPLE LOADING USING THE DYNALOOP SLIDING-PISTON LOOP

The DynaLoop, which is available in 25 ml and 90 ml sizes, is used to load large volumes of sample directly onto chromatography columns. These loops have a sliding piston and function very much like a syringe. The sample is inserted into the DynaLoop's Sample End Connector. The loading of the sample pushes the DynaLoop's sliding seal assembly towards the Buffer End Connector. **Note:** While other dynamic loops may be used, the application discussed in this section applies specifically to the DynaLoop.

8.2.1 System Setup

When plumbed directly to the AV7-3 or V7-3 Inject valves, the DynaLoop functions just like a static sample loop. The inlet (top) of the DynaLoop should be plumbed to port 3 and the outlet (bottom) to port 6 of the Inject valve. When used with the AV7-3 Inject valve, the valve's operation and sample loading are controlled automatically, simplifying the sample injection process and insuring precise sample loading and gradient formation.

The DynaLoop may be filled either manually with a syringe or automatically using an auxiliary pump such as the Econo Pump, which is controlled by the BioLogic HR system.

- When using a syringe to fill a DynaLoop, the syringe can be connected to port 5 of the AV7-3 valve. However, using a 1/4x28 to female Luer adapter between the syringe and the valve makes filling the loop much easier.
- When using the auxiliary pump to fill the loop, the pump's outlet plumbing should be connected directly to port 5 of the inject valve.

Consult the DynaLoop user's manual or on-line help for additional information. The figure below shows how to set up a DynaLoop.

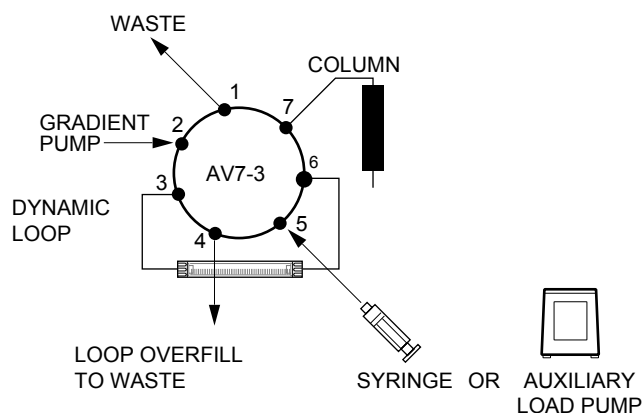


Figure 49. Plumbing the DynaLoop for use with an Inject Valve

8.2.2 Writing the Protocol

Before running a protocol which involves the use of a DynaLoop, be sure to purge air from the lines, as discussed in the DynaLoop Instruction Manual, Chapter 2, Installation. Then follow the guidelines which apply to your application.

The following steps should be programmed within the method protocol to manually load a sample:

1. From the Manual screen, set the AV7-3 Sample Inject Valve to its Inject (I) position and allow the sliding seal to contact the sample end fitting on the DynaLoop. Then set the valve to the Load (L) position. If you are using the V7-3, you will need to do this manually.

2. Load sample into the DynaLoop by using a syringe. As sample is loaded into the loop, it displaces the buffer that was used to purge the unit. **Note:** The DynaLoop cannot be overfilled.

When sufficient volume of sample for a run or series of runs is loaded into the DynaLoop, stop the filling process. For a series of partial volume injections after one loading sequence, fill the DynaLoop with about 2 percent extra sample. Leave the syringe in the Inject Port to minimize the introduction of air into the DynaLoop.

The sample is now ready for injection, and the protocol can be programmed.

3. In the Protocol screen, select **Load/Inject Sample**. In the Load/Inject Sample window, note that the AV7-3 valve will automatically move to the "Inject" position at the start of the step and to the "Load" position at the end of the step.
 - a. In the Load/Inject Sample window, specify Dynamic Loop.
 - b. In the "Injection Buffer" area of the window, select the buffer(s) and percent (%) composition to be used to inject the sample onto the column.
 - c. In the "Volume (ml)" field, enter the sample volume.
 - d. In the "Flow (ml/min)," enter the flow rate of the BioLogic HR gradient pump that will be used to inject the sample onto the column.
4. Continue writing the separation protocol.

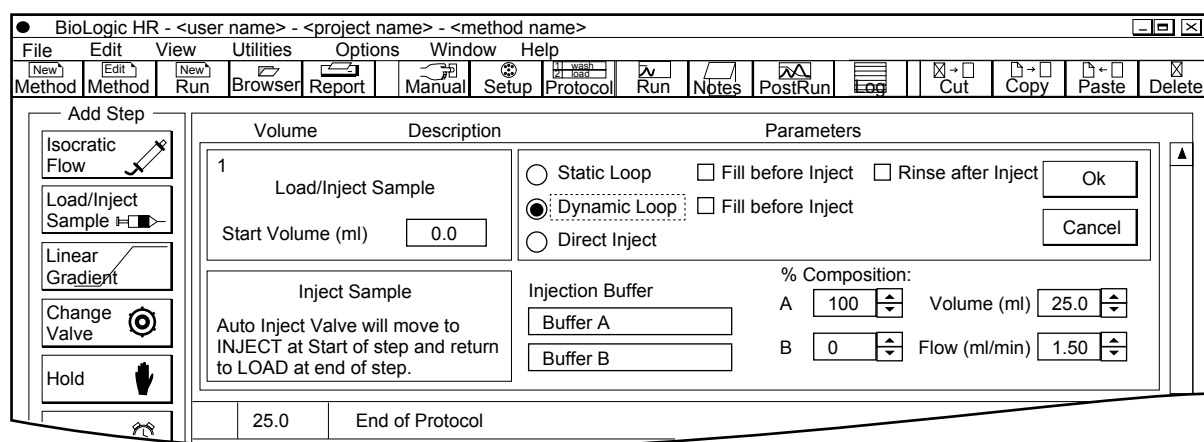


Figure 50. Load/Inject Sample Window for a Manual Load/Inject of the Sample

The following steps should be programmed within the method protocol to automatically inject a sample using an auxiliary pump:

1. Consult Section 8.3.1 for connecting the EP-1 Econo Pump as an Aux pump.
2. From the Setup screen, select Aux Load Pump and AV7-3 valve.
3. In the Protocol screen, program an Isocratic step that is long enough for the auxiliary pump to load the desired quantity of sample into the DynaLoop. For example, if you are loading 25 ml of sample at an Aux pump flow rate of 5 ml/min, you will need at least a 5 minute step prior to the sample loading step. Otherwise the protocol will fail the automatic validation done by the BioLogic HR prior to the run. Because this is a timing issue, the flow rate of the BioLogic HR gradient pump is not critical. Therefore, set a low flow rate (e.g., 0.1 ml/min) to minimize buffer waste, but be certain that either the time or volume length of this first step is of sufficient duration to allow the Aux pump to fill the DynaLoop.
4. In the Protocol screen, select Load/Inject Sample to program the BioLogic HR system to automatically fill and inject the DynaLoop sample. From the Load/Inject Sample window,
 - a. Select the Dynamic Loop as the type of loop to be used.
 - b. Select Fill Before Inject. This instructs the Aux pump to load the sample into the DynaLoop. Note that the Aux pump flow rate is not under BioLogic HR control. The flow rate is used by the system when validating the protocol before the run. Because the BioLogic HR system only starts/stops the Aux pump, you must remember to set the correct flow rate at the Aux pump. The flow rate of the Econo pump is set from the pump and is recorded in the yellow data entry boxes of the Load/Inject Sample dialog box.
 - c. In the Fill Sample Loop area of the window, select the sample to be loaded and enter its volume and flow rate of the Aux pump.
 - d. In the Inject Sample area of the window, select the Injection Buffers, the buffer composition, the flow rate of the BioLogic HR pump, and the sample volume to be injected onto the column.

The BioLogic HR system will now automatically control the loading and injection of the sample. Note that the rinse function is not available when the DynaLoop is being used.

5. Continue writing the separation protocol.

Figure 51. Load/Inject Sample Window for an Automatic Load/Inject of the Sample

During the run, the valve functions as follows:

- **LOAD.** While in this position the valve connects ports 2 and 7 for equilibration of the column and for sample elution. In this position, sample loop is loaded to the desired volume via port 5 and buffer is expelled from the dynamic loop through port 4.
- **INJECT.** While in this position the valve connects ports 2 and 3 and ports 6 and 7 for applying the sample onto the column. The flow from the Workstation's Gradient Pump forces the sliding piston to expel the sample onto the column.
- **PURGE.** While in this position the valve connects ports 1 and 2 and allows purging or buffer changes of the Workstation's Gradient Pump without the need to remove the column from the system.

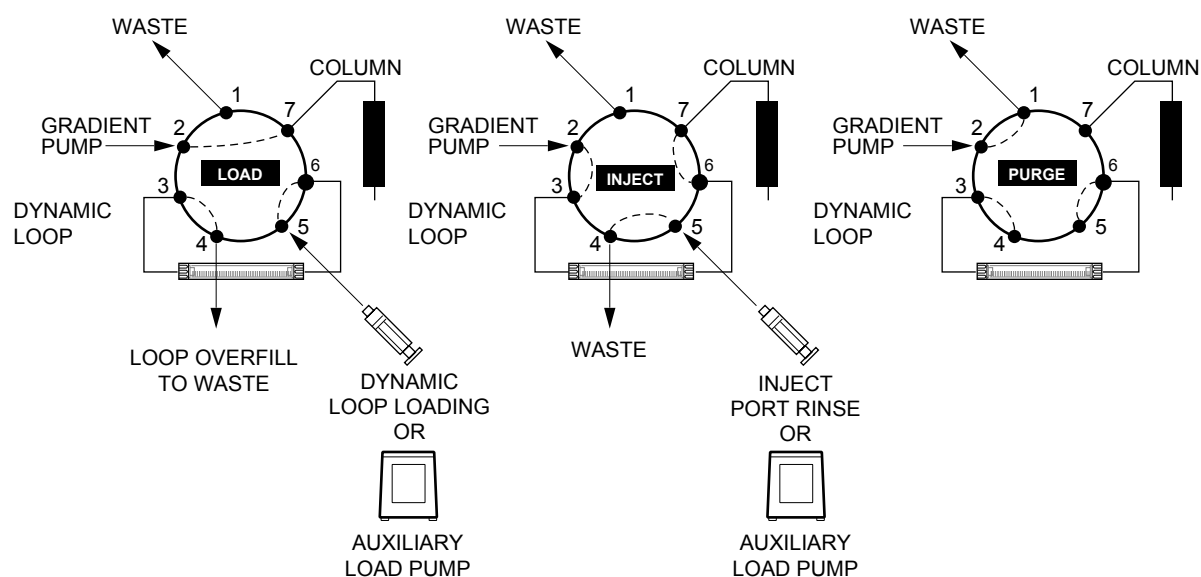


Figure 52. Valve Positions During a Run using the DynaLoop

8.3 SAMPLE LOADING ONTO A LOW PRESSURE COLUMN USING THE ECONO PUMP AS AN AUX PUMP

The EP-1 Econo Pump is ideal for loading samples onto a low-pressure column. However, when using a low-pressure column, such as an Econo-Pac cartridge, remove the 40-psi back-pressure device from the post-column position. Place it between the Workstation's Gradient Pump and the injection valve. This will allow the check valves to properly seat and ensure pump flow performance. In addition, low pressure columns may burst if the back pressure device is placed after the column.

8.3.1 System Setup

The BioLogic HR can be used with the following auxiliary pumps:

- Model EP-1 Econo Pump.

The Econo Pump must contain firmware version 2.12 or higher to function with the BioLogic HR system. To confirm the firmware version, simultaneously press and hold down the Direction key and the down arrow key on the front panel of the Econo Pump. If the Econo Pump does not have the correct firmware version, contact Bio-Rad for information on how to upgrade the pump.

The Econo Pump receives start/stop signals only. The flow rate of the Econo Pump is set from the pump, and the flow rate is recorded in the yellow data entry boxes of the Load/Inject Sample dialog box. **Note:** The BioLogic HR system does not control the flow of the Econo Pump. Because the BioLogic HR system only starts/stops the Aux pump, you must remember to set the correct flow rate at the Aux pump.

- Other external pumps.

The BioLogic HR system sends a TTL signal to control an external pump. This signal is normally TTL high (5 volts). The BioLogic HR system commands the external pump to run by holding Pin 4 "low". Any pump accepting this signal may be used with the BioLogic HR system as a sample loading pump. Refer to your pump's separate documentation when connecting the pump to the BioLogic HR.

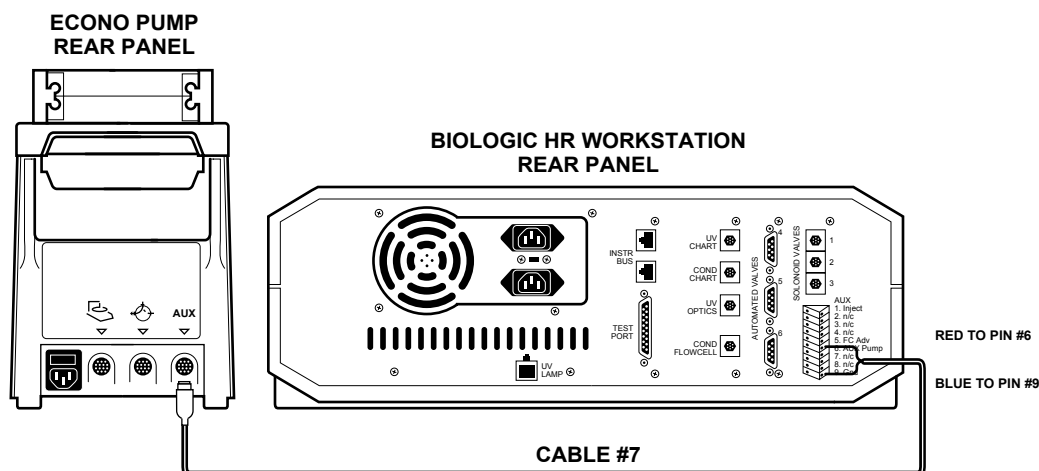


Figure 53. Connecting an EP-1 Econo Pump to the BioLogic HR Workstation

To connect the Econo Pump with the BioLogic HR, make the following cable connections:

1. Obtain **System Cable #7**, Catalog # 731-8267
2. Locate the red and blue wires at the wire end of the cable and connect them to the BioLogic HR Workstation.

Note: Cut short all other wires and isolate with tape.

 - a. Connect the red wire to pin 6 of the Aux connector.
 - b. Connect the blue wire to pin 9 of the Aux connector.
3. Plug the 8-pin mini-DIN connector into the Aux connector on the rear panel of the Econo Pump.

When plumbing the Econo Pump, be sure to use PharMed tubing. For additional information on the Econo Pump, consult its user manual. The injection valve (either AV7-3 or V7-3) is plumbed without a sample loop. Plumb the valve as shown below, with the EP-1 Econo Pump connected to port 6 of the Inject Valve.

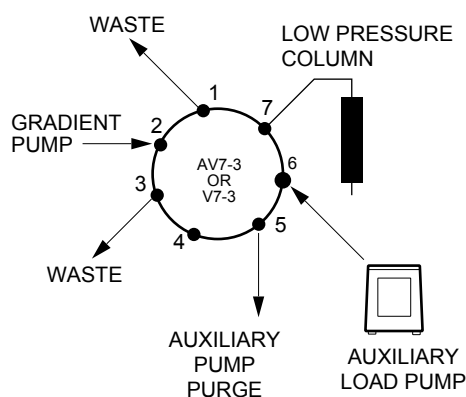


Figure 54. Plumbing an AV7-3 Inject Valve with an EP-1 Econo Pump as an Auxiliary Load Pump

In the Setup screen, select the AV7-3 valve as the Sample Inject Valve and add the Aux Pump to the Setup List. When using an auxiliary pump to load sample directly to a Sample Inject (AV7-3 or V7-3) valve, it is possible to have a valve, such as an SV3-2 valve, assigned as the Auxiliary Pump Inject Valve to allow switching between samples or rinse solutions.

8.3.2 Writing the Protocol

The following steps should be programmed within the Method Protocol, as shown in the next figure:

1. At the Load/Inject Sample step, specify Direct Inject and choose Aux Pump as the inlet through which the sample will be delivered. Note the message displayed in yellow, "Aux Pump Flow Rate not under Biologic control," and that the Flow rate also is displayed in yellow. This is to serve as a reminder that the BioLogic HR only starts/stops the Aux pump and that the actual flow rate must be set at the Aux pump.
2. Enter the volume of sample and the flow rate for the injection step. The step starting time or volume appears to the left of the options buttons.

Note: The BioLogic HR system does not control the flow of the Econo Pump. The correct flow rate must be entered at the Econo Pump.
3. Continue writing the separation protocol.

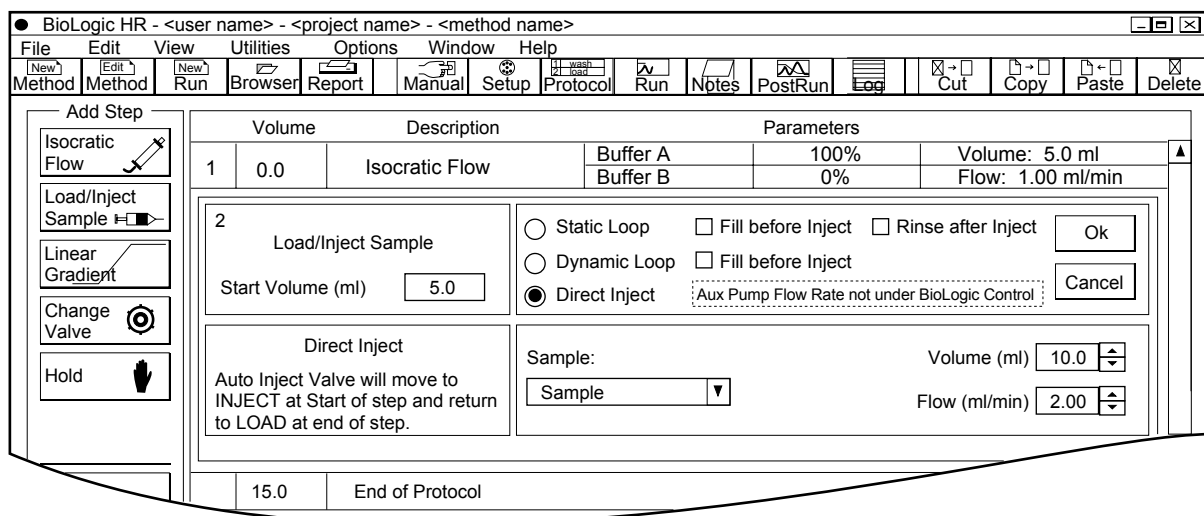


Figure 55. Writing the Protocol: Direct Sample Loading

In the configuration shown below, the valve operates as follows:

- **LOAD.** While in this position the valve connects ports 2 and 7 for equilibration of the column and for column elution.
- **INJECT.** While in this position the valve connects ports 6 and 7 to load sample onto the column.
- **PURGE.** While in this position the valve connects ports 1 and 2 and allows purging or buffer changes without the need to remove the column from the system. The same is true for ports 5 and 6.

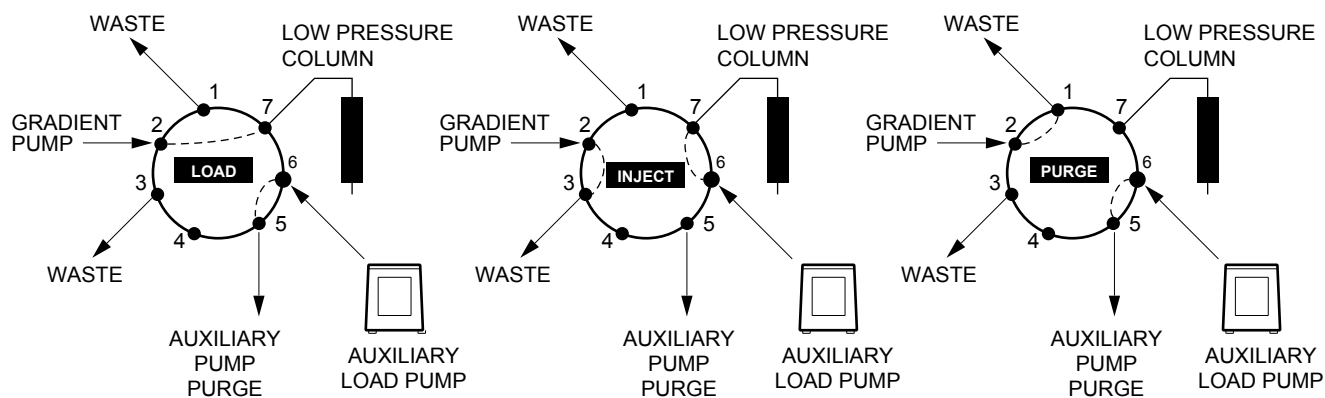


Figure 56. Valve Positions During a Run with Direct Sample Loading and Injection

8.4 LOADING MULTIPLE SAMPLES

With the addition of the Econo Pump as an Aux pump and an AV9-8 as an Aux pump inlet valve, the system can load up to 7 samples sequentially.

8.4.1 System Setup

Connect the Econo Pump as described in section 8.3.1. In the Setup screen, alias the AV9-8 as an Aux Pump Inlet. Name seven positions to identify each sample, and name one position "Rinse." The rinse solution will clean the sample loop between injections. Add the Aux pump to the Setup List.

Note: You will lose some sample in the process of pulling sample from a remote beaker. Be sure to factor in the volume that is required to fill all buffer lines leading up to the top of the column when programming the injection step.

When using a low-pressure column such as an Econo-Pac cartridge, remove the 40 psi back-pressure device from post-column. Place it between the pump and injection valve. This will allow the check valves to properly seat. In addition, low-pressure columns may burst if the back-pressure device is placed after the column.

See the illustration below to assist in setting up the system. Be sure to keep tubing lengths to a minimum when plumbing the system. This will minimize sample loss.

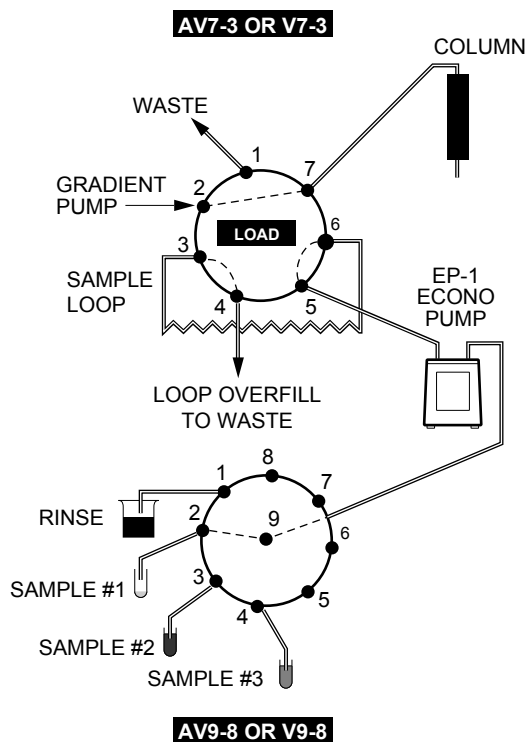


Figure 57. Using a Peristaltic Pump and an AV9-8 or V9-8 Valve to Load Multiple Samples

8.4.2 Writing the Protocol

The following steps should be programmed within the method protocol:

1. Remember that the Econo Pump flow rate is not under BioLogic HR control. Because the BioLogic HR system only starts/stops the Aux pump, you must remember to set the correct flow rate at the Aux pump. Set the correct flow rate on the Econo Pump and enter the flow rate in the yellow boxes of the Load/Inject Sample dialog box.
2. If multiple injections are performed in a single Run, you may view or print each separation independently by opening the Post-Run window and highlighting the section of the chromatogram you wish to view.
3. At the Load/Inject Sample step, specify Static Loop and choose "Fill before Inject" and "Rinse after Inject." This automates the process of filling the sample loop using the Aux pump, injecting the sample using the gradient pump, and then rinsing the sample loop using the Aux pump. Note the message displayed in yellow, "Aux Pump Flow Rate not under BioLogic control," and that the Flow rates also are displayed in yellow. This is to serve as a reminder that the actual flow rate must be set at the Aux pump. Refer to the figure below.
4. Continue writing the separation protocol.

BioLogic HR - <user name> - <project name> - <method name>

File Edit View Utilities Options Window Help

New Edit New Run Browser Report Manual Setup Protocol Run Notes PostRun Log Cut Copy Paste Delete

Add Step

Isocratic Flow

Load/Inject Sample

Linear Gradient

Change Valve

Hold

Alarm

Zero Baseline

UV Lamp

Chart Recorder

Repeat Steps

Volume	Description	Parameters
1 0.0	Isocratic Flow	Buffer A 100% Buffer B 0% Volume: 5.0 ml Flow: 1.00 ml/min
2 5.0	Load/Inject Sample	☒ Static Loop ☐ Dynamic Loop ☐ Direct Inject ☒ Fill before Inject ☐ Fill before Inject ☒ Rinse after Inject Aux Pump Flow Rate not under BioLogic control Sample: Sample 1 Volume (ml) 5.0 Flow (ml/min) 5.00 Injection Buffers: Buffer A Buffer B % Composition: A 100 B 0 Volume (ml) 10.0 Flow (ml/min) 2.00 Rinse Buffer: Rinse Volume (ml) 5.0 Flow (ml/min) 5.00
25.0	End of Protocol	

Figure 58. Load/Inject Sample Window for Loading Multiple Samples

8.5 MULTIPLE INJECTIONS OF A SINGLE SAMPLE

The BioLogic HR system can be used in conjunction with the DynaLoop to run repeated injections of the same sample.

8.5.1 System Setup

Install the DynaLoop (either the 25 ml or 90 ml loop) as described in section 8.2.

8.5.2 Writing the Protocol

The following should be programmed within the method protocol:

1. In the Load/Inject Sample step, choose Dynamic Loop and enter the volume of sample to be injected during each run. At the initiation of the run, select Multiple Runs and the number of times you want to repeat the separation.

Note: The BioLogic HR system does not sense how much sample is available in the sample loop, and will not indicate a programming error if you program more runs than you have sample for.

2. Continue writing the separation protocol.

8.6 USING THE AS-100 AUTOSAMPLER WITH THE BIOLOGIC SYSTEM

This section describes how to connect Bio-Rad's AS-100 Autosampler to a BioLogic HR Workstation and to program for both single and multiple injections.

8.6.1 System Setup

To connect the AS-100 to the BioLogic HR, make the cable connections shown in the figure below.

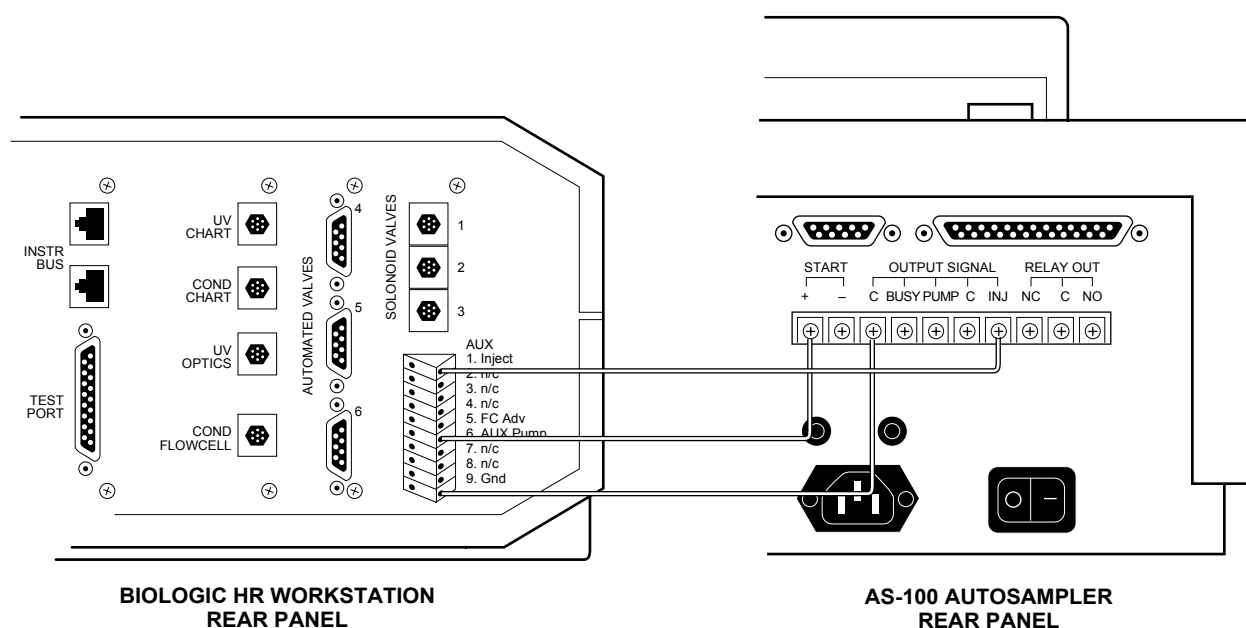


Figure 59. Rear view of AS-100

- Using three wires, connect as described below (see Figure 59):
Note: This cable is not provided with either the AS-100 or the BioLogic HR system.

BioLogic HR Workstation's Aux Port*	AS-100's Aux Port
Inject (pos 1)	INJ (inject output signal)
AUX Pump (pos 6)	START +
GND (pos 9)	C (common output signal)

*The AUX connector on the rear of the BioLogic HR Workstation is configured as follows:

- Position 1 inject, for use with a manual valve;
 - Position 6 Auxiliary Pump, for use with either the Econo Pump or another external sample loading pump, and
 - Position 9 GND (ground).
- Configure the mode selector switches on the AS-100 as follows:
 - down
 - up
 - down
 - down

Follow the procedure below to program the AS-100:

- Select **Function** and **8** for Set System, and set the parameters you prefer according to the directions in the AS-100 instruction manual.
 If the AS-100 is to be started by the BioLogic HR system, then select the Remote feature:
REMOTE: [ON]/OFF
- Select **Function** and **7** for Sample Table. Select a Sample Table and input parameters, such as multiple injections, as needed.

8.6.2 Writing The Protocol

When programming the BioLogic HR system, note the following:

- The Aux Pump should be listed in the Setup screen. Since the AS-100 start signal originates from the same connector as that used by the Aux Pump, programming must proceed as if an Aux Pump were connected.
- In the Protocol screen, select the "Dynamic Loop" and "Fill before Inject" options for the **Load/Inject Sample** step. A message stating that the Aux Pump Flow Rate is not under BioLogic control will appear. This also means that the injection time of the AS-100 is not under BioLogic control.
- The "Fill before Inject" option starts the AS-100 when the run is initialized by sending a signal through the AUX connector to the AS-100. The time required for the "Fill Sample Loop" step must be less than or equal to the time programmed in the previous isocratic/gradient step. The information in the "Inject Sample" portion of this screen is irrelevant because the AS-100 injects the sample onto the column.

When programming for a single injection, note the following:

- Locate the Status bar at the upper left-hand side of the AS-100 display. Prior to the start of a run on the BioLogic HR system, press the green **START** button on the AS-100. This puts the AS-100 in STATUS: IDLE mode for remote operation.
- For a single injection, select **RUN**, then **START** in the BioLogic HR screen. Because the (manual) inject connector of the BioLogic HR Workstation's AUX connector is used, the run conditions are held until the BioLogic HR receives an injection signal from the AS-100. Depending on the selected parameters for the AS-100, the time to load and inject the sample is approximately two minutes.

- c. The AS-100 status will progress from Idle to Load status when the run is initialized.
- d. When the Method protocol progresses to the point of the Load and Inject step, the Run screen displays these messages: "Please move the Manual Inject Valve to Inject Position" and "Gradient Pump is Holding." When the AS-100 injects the sample, both messages disappear and the message "Running" is displayed at the bottom right-hand side of the screen.
- e. The Run Log file will document the exact time of injection.
- f. Continue writing the separation protocol.

When programming for multiple injections, note the following:

- a. Locate the Status bar at the upper left-hand side of the AS-100 display. Prior to the start of a run, press the green **START** button on the AS-100. This puts the AS-100 in STATUS: IDLE mode for remote operation.
- b. For multiple injections, enter the number of injections in the AS-100 Sample Table. From the Run screen, select the "Multirun" icon and enter the same number of runs as were entered for the AS-100. Press **START**. Because the (manual) inject connector of the BioLogic HR Workstation's AUX connector is used, the run conditions are held until the BioLogic HR receives an injection signal from the AS-100. Depending on the selected parameters for the AS-100, the time to load and inject the sample is approximately two minutes.
- c. The AS-100 status will progress from IDLE to LOAD when the Run is initialized.
- d. When the Method's protocol progresses to the point of the Load and Inject step, the Run screen displays these messages: "Please move the Manual Inject Valve to Inject Position" and "Gradient Pump is Holding." When the AS-100 injects the sample, both messages will disappear and the message "Running" will be displayed at the bottom right-hand side of the screen.
- e. The AS-100 status will display IDLE, in preparation for the subsequent injection, and will display STATUS: STOP when all injections have been performed.
- f. A separate Run Log file will document the exact time of injection for each run.
- g. Continue writing the separation protocol.

9.0 MULTIPLE COLUMNS

The BioLogic HR system supports the use of multiple columns. Refer to Chapter 2, section 2.6, Columns and Column Fittings, for a description of columns and fittings offered by Bio-Rad. Several examples are discussed below.

9.1 COLUMN SWITCHING

For this application two AV7-3 valves will be used: one as a sample injector, and the second used downstream as a column switcher.

9.1.1 System Setup

Figure 60 below shows the two valve positions when the injection valve is used for column switching. In this configuration, the sample from an upstream injection valve enters via port 2 and is diverted to either column using the Load and Inject positions of the valve. Although the purge position is not described in this application, it may be used in the conventional manner.

Column 1 has its inlet connected to port 7 and its outlet to port 6. Column 2 has its inlet connected to port 3 and its outlet to port 4. Port 5 functions as a common exit port.

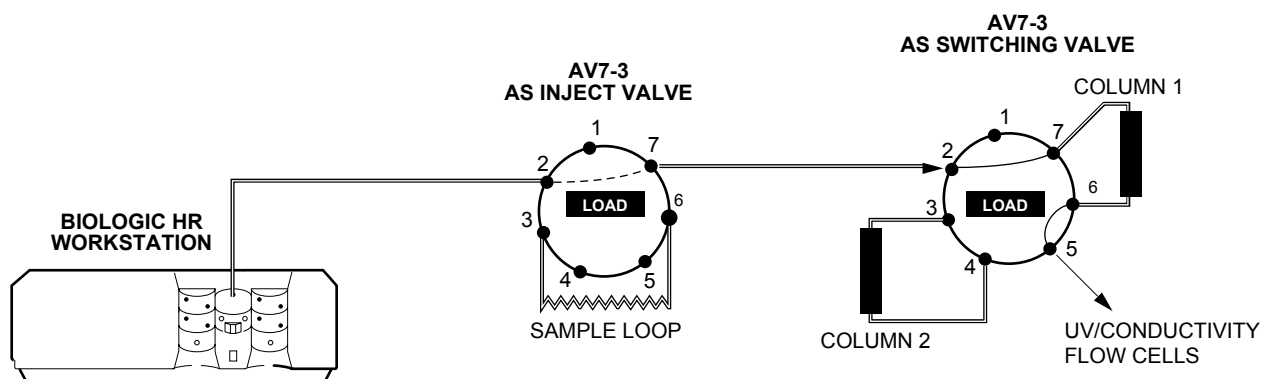


Figure 60. Column Switching using Two AV7-3 Valves

In the Setup screen, use the default name of "Sample Inject" for the first AV7-3 valve. Name the second AV7-3 "column switcher." Enter the column names into positions 1 and 2, and name position 3 as "Purge."

Note that when solution is directed to Column 1, this is similar to the "Load" position. When solution is directed to Column 2, this is similar to the "Inject" position.

9.1.2 Writing the Protocol

The following steps should be programmed within the Method Protocol, as shown in the figure below.

1. Equilibrate both columns with the appropriate equilibration buffer.
2. Use the "Change Valve" step to ensure that the column switcher valve is in the appropriate position for loading the sample onto either Column 1 or Column 2.

- Use the Load/Inject Sample step to load the sample on to the appropriate column and continue to write the separation protocol for the sample and column.
- Use the "Change Valve" step to ensure that the column switcher valve is in the appropriate position for loading sample onto either Column 1 or Column 2.
- Use precautionary washes/rinses to prevent cross-contamination by selecting the purge position on the column switcher valve. This is used to clean the tubing between the two AV7-3 valves. Alternatively, performing an isocratic flow step before directing flow to the other column may also be used. This later step is depicted in the figure below.
- Continue writing the separation protocol.

BioLogic HR - <user name> - <project name> - <method name>

File Edit View Utilities Options Window Help

Method [New] [Edit] [Run] [Browser] [Report] [Manual] [Setup] [Protocol] [Run] [Notes] [PostRun] [Log] [Cut] [Copy] [Paste] [Delete]

Add Step						
Isocratic Flow Load/Inject Sample Linear Gradient Change Valve Hold Alarm Zero Baseline UV Lamp Chart Recorder Repeat Steps Fraction Collection	1	0.0	Isocratic Flow	Buffer A	100%	Volume: 5.0 ml
				Buffer B	0%	Flow: 1.00 ml/min
	2	5.0	Change Valve	Column Switcher	Position: Column #1	
	3	5.0	Load/Inject Sample	Load: Sample		Volume: 5.0 ml
				Static Loop	Auto Inject Valve	Flow: 2.00 ml/min
	4	10.0	Linear Gradient	Buffer A	100% --> 30%	Volume: 6.0 ml
				Buffer B	0% --> 70%	Flow: 1.00 ml/min
	5	16.0	Isocratic Flow	Buffer A	0%	Volume: 10.0 ml
				Buffer B	100%	Flow: 2.00 ml/min
	6	26.0	Isocratic Flow	Buffer A	100%	Volume: 10.0 ml
				Buffer B	0%	Flow: 2.00 ml/min
	7	36.0	Change Valve	Column Switcher	Position: Column #2	
8	36.0	Isocratic Flow	Buffer A	100%	Volume: 10.0 ml	
			Buffer B	0%	Flow: 2.00 ml/min	
9	46.0	Load/Inject Sample	Load: Sample		Volume: 5.0 ml	
			Static Loop	Auto Inject Valve	Flow: 2.00 ml/min	
10	51.0	Linear Gradient	Buffer A	100% --> 30%	Volume: 6.0 ml	
			Buffer B	0% --> 70%	Flow: 1.00 ml/min	
11	57.0	Isocratic Flow	Buffer A	0%	Volume: 10.0 ml	
			Buffer B	100%	Flow: 2.00 ml/min	
12	67.0	Isocratic Flow	Buffer A	100%	Volume: 10.0 ml	
			Buffer B	0%	Flow: 2.00 ml/min	
	77.0	End of Protocol				

Gradient Pump		UV		Conductivity		Aux. Pump		SIM1/SIG ▼	SIM1/pH ▼
0.00ml/min	0 %B	0 psi	0.2869 AU	29.4 mS/cm		Stopped	-0.000 Volt	7.00 pH	

Select item(s) for copy/cut/delete. Double-click on step to edit.

Figure 61. Protocol for using an AV7-3 Inject Valve for Column Switching

9.2 RUNNING ONE SAMPLE SEQUENTIALLY OVER SEVERAL DIFFERENT COLUMNS

With the addition of two AV9-8 valves, up to eight columns can be tested with a single sample.

9.2.1 System Setup

In the Setup screen, select "User-assigned name" when naming each of the AV9-8 valves. Name the first AV9-8 valve "column inlet," and type in the names of the columns in positions 1-8. Name the second AV9-8 valve "column outlet," and type in the names of the columns in positions 1-8.

The two column switching valves will be plumbed after the inject (AV7-3) valve. Position both valves to a convenient location on the rack to minimize tubing lengths. Connect port 7 of the inject valve to the common port of the "column inlet" valve. Connect the common port of the "column outlet" valve to the UV flow cell.

As with plumbing any column, be certain that all column inlet lines are free of air to avoid introducing air into the columns.

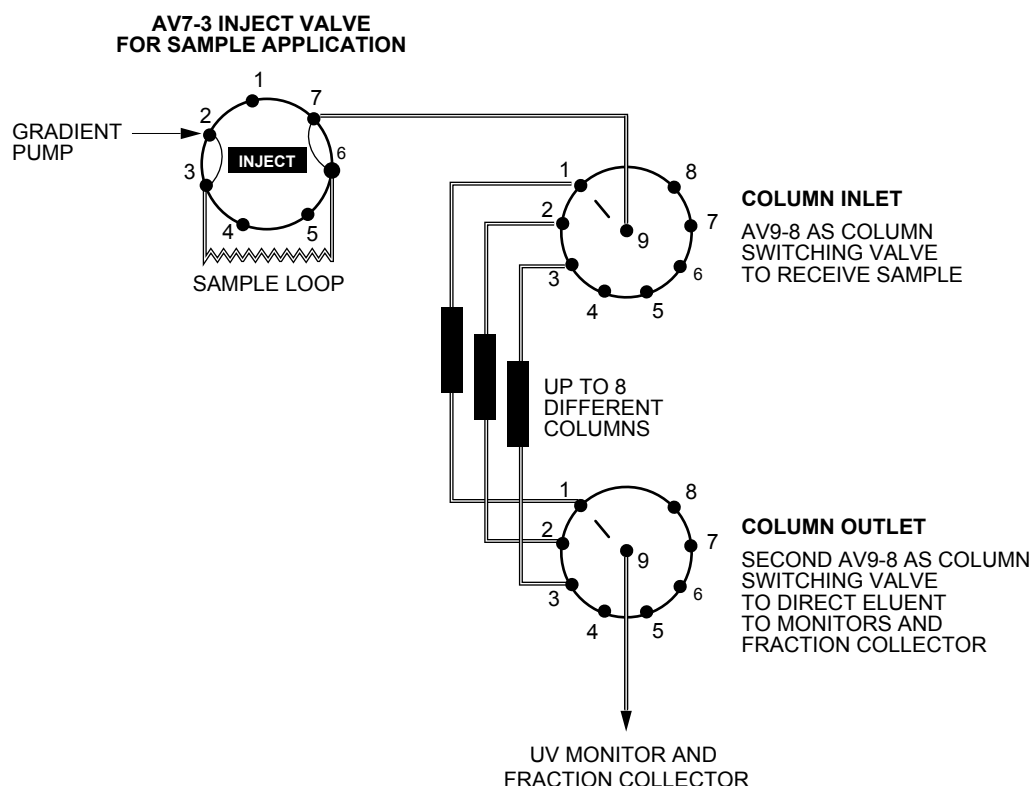


Figure 62. Column Switching using Two AV9-8 Valves

9.2.2 Writing the Protocol

The following should be programmed within the Method Protocol:

1. Begin the protocol by adding two Change Valve steps: one to position the column inlet at starting position, and the second to position the column outlet at the starting position. Be sure that these two positions are identical. For example if you wish to equilibrate column #1 with buffer A, the protocol should look as shown in the figure below.

Note: Reduce the flow rate to 0.1 ml/min for a short amount of time before the "change valve" steps. This is done to minimize high pressure surges as the valves rotate from one port to another.

2. Continue writing the separation protocol.

If multiple injections are performed in a single run, you may view or print each separation independently open up post run and highlight the section of the chromatogram you wish to view.

BioLogic HR - <user name> - <project name> - <method name>

File Edit View Utilities Options Window Help

Method Method Run Browser Report Manual Setup Protocol Run Notes PostRun Log Cut Copy Paste Delete

Add Step

Isocratic Flow

Load/Inject Sample

Linear Gradient

Change Valve

Hold

Alarm

Zero Baseline

UV Lamp

Chart Recorder

Repeat Steps

Fraction Collection

Time	Description	Parameters			
1	0:00.0	Change Valve	column inlet	Position: Column 1	
2	0:00.0	Change Valve	column outlet	Position: Column 1	
3	0:00.0	Isocratic Flow	Buffer A	100%	Time: 10.0 min
			Buffer B	0%	Flow: 2.00 ml/min
			Load: Sample	Volume: 6.0 min	
4	10.0	Load/Inject Sample	Static Loop	Auto Inject Valve	Flow: 2.00 ml/min
5	13.0	Linear Gradient	Buffer A	100% --> 30%	Time: 6.0 min
			Buffer B	0% --> 70%	Flow: 1.00 ml/min
6	19.0	Isocratic Flow	Buffer A	0%	Time: 5.0 min
			Buffer B	100%	Flow: 2.00 ml/min
7	24.0	Isocratic Flow	Buffer A	100%	Time: 5.0 min
			Buffer B	0%	Flow: 2.00 ml/min
8	29.0	Isocratic Flow	Buffer A	100%	Time: 0.1 min
			Buffer B	0%	Flow: 0.10 ml/min
9	29.1	Change Valve	column inlet	Position: Column 2	
10	29.1	Change Valve	column outlet	Position: Column 2	
11	29.1	Isocratic Flow	Buffer A	100%	Time: 10.0 min
			Buffer B	0%	Flow: 2.00 ml/min
12	39.1	Isocratic Flow	Buffer A	100%	Time: 0.1 min
			Buffer B	0%	Flow: 0.10 ml/min
13	39.2	Change Valve	column inlet	Position: Column 3	
14	39.2	Change Valve	column outlet	Position: Column 3	
15	39.2	Isocratic Flow	Buffer A	100%	Time: 10.0 min
			Buffer B	0%	Flow: 2.00 ml/min
49.2	End of Protocol				

Gradient Pump UV Conductivity SIM1/SIG ▼ SIM1/pH ▼

0.00ml/min 0 %B 0 psi -0.00191 AU 0.000 mS/cm

Figure 63. Programming a Sequential Run

10.0 REVERSE FLOW CHROMATOGRAPHY

Using two AV7-3 inject valves for reverse flow chromatography is advantageous for certain applications such as affinity chromatography, where minimal dilution during sample elution is important.

10.1 SYSTEM SETUP

In the Setup screen, select one AV7-3 valve as the Sample Inject valve. Select a second AV7-3 valve and name (alias) it as "Change Flow Valve." On the "Change Flow Valve," alias position 1 as Forward Flow, and alias position 2 as Reverse Flow. Leave Position 3 as is.

Connect the sample loop to ports 3 and 6 of the AV7-3 inject valve. To the Reverse Flow valve, connect the column inlet to port 7, the column outlet to port 3. The reverse flow valve will require a connection between ports 4 and 5. Use 15 cm of 1/16" OD tubing to make this connection. Keep this tubing length to a minimum to minimize sample dilution. Use minimal length of 1/16" OD tubing to connect port 7 of the AV7-3 inject valve to port 2 of the Reverse Flow valve. Figure 64 shows how to set up this configuration.

If a low-pressure column is used, remove the 40-psi back-pressure device and place it between the pump and the AV7-3 valve. This will allow the check valves to properly seat and ensure pump flow performance. In addition, low-pressure columns may burst if the back-pressure device is placed after the column.

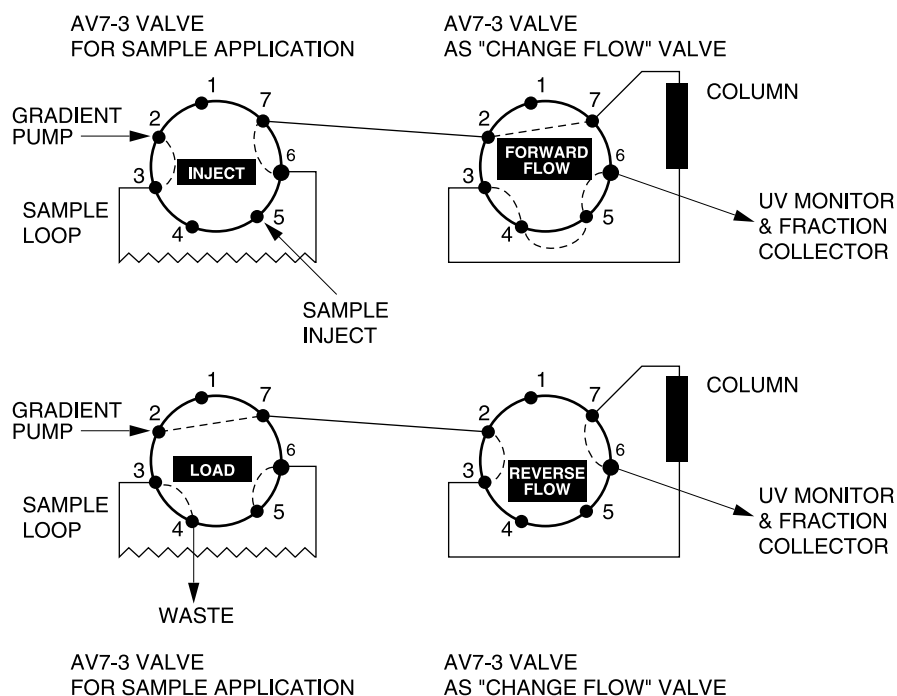


Figure 64. Injection Valve Plumbing for use in Affinity Chromatography

10.2 WRITING THE PROTOCOL

The following steps should be programmed within the method protocol.

1. Ensure that the column is properly equilibrated by using an Isocratic step.
2. Use a Change Valve step to place the Change Flow Valve in the Forward Flow orientation.
3. Use the Load/Inject Sample step to load the sample loop and subsequently inject it onto the column. The AV7-3 valve will automatically move to the INJECT position at the start of the step and return to the LOAD position at the end of the step.
4. Use the appropriate buffer to wash all nonbinding components from the column.
5. When the sample is to be eluted, first use the Change Valve step to select the "Reverse Flow" valve position of the Change Flow valve. This directs the flow of buffer through the bottom of the column to elute sample components of interest in a concentrated form.
6. Continue writing the separation protocol.

● BioLogic HR - <user name> - <project name> - <method name>

File Edit View Utilities Options Window Help

Method Method Run Browser Report Manual Setup Protocol Run Notes PostRun Log Cut Copy Paste Delete

Add Step

Isocratic Flow Load/Inject Sample Linear Gradient Change Valve Hold Alarm Zero Baseline UV Lamp Chart Recorder Repeat Steps Fraction Collection	1	0.0	Isocratic Flow	Buffer A	100%	Volume: 5.0 ml
				Buffer B	0%	Flow: 1.00 ml/min
	2	5.0	Change Valve	Change Flow Valve	Position: Forward Flow	
	3	5.0	Load/Inject Sample	Load: Sample		Volume: 1.0 ml
				Static Loop	Auto Inject Valve	Flow: 2.00 ml/min
	4	6.0	Linear Gradient	Buffer A	100% --> 30%	Volume: 10.0 ml
			Buffer B	0% --> 70%	Flow: 1.00 ml/min	
	5	16.0	Change Valve	Change Flow Valve	Position: Reverse Flow	
		16.0	End of Protocol			

Gradient Pump UV Conductivity Aux. Pump SIM1/SIG SIM1/pH

0.00ml/min 0 %B 0 psi 0.2869 AU 29.4 mS/cm Stopped -0.000 Volt 7.00 pH

Select item(s) for copy/cut/delete. Double-click on step to edit.

Figure 65. Programming for Reverse Flow Chromatography

11.0 MAINTENANCE

The BioLogic HR system requires minimal maintenance to assure reliable operation.

11.1 CLEANING THE OUTER SURFACES OF THE INSTRUMENTS

During normal operation, spills and splashes may cause residues to form on the component cases.

To clean the case of an instrument, first unplug the instrument. Use a damp cloth to wipe down the outer case. Avoid wetting the power switch located below the front panel and the connectors on the rear of the unit.

The System Rack's tray should be mopped up and any residues rinsed away with water via the drain.

11.2 STORAGE OF THE BIOLOGIC HR SYSTEM

For overnight storage, please note the following:

1. The system tolerates low salt buffer overnight, but sample and high salt buffers need to be washed out of the tubing and the column.
2. Use DI water and the 10mL syringe provided in the BioLogic HR System Fittings Kit to wash behind each pumphead with at least 10mL of water.
3. If the system is in a cold environment, keep it powered on. If it is at room temperature, it may be left off or on. (When it is at room temperature, the BioLogic HR Color Monitor can be turned off, to conserve energy and prolong monitor life.)

For long-term storage, please observe the following precautions:

1. Remove the column from on-line. Follow the column manufacturer's instructions to clean and store the column.
2. Run DI water through the entire system. Be sure to wash through all parts of both the solenoid and the automated valves to remove any contaminants. **Note:** It is particularly important to wash out solenoid valves. Failure to do so may result in costly replacements.
3. Use DI water and the 10mL syringe provided in the BioLogic HR System Fittings Kit to wash behind each pumphead with at least 10 mL of water.
4. Store the entire system with either 20% EtOH or 0.05% sodium azide to inhibit bacterial growth. If necessary, fit the pump inlets, UV and conductivity monitors, and all solenoid valves with 1/4-28 plugs which are provided in the BioLogic HR System Fittings Kit.

11.3 CARE AND MAINTENANCE OF THE GRADIENT PUMP

The following section describes how to maintain the gradient pump.

11.3.1 Priming the Gradient Pump and Removing Trapped Air Bubbles

Warning: Do not run the BioLogic HR Gradient Pump dry (without buffers in line); damage to the pumpheads may result.

1. Check that both of the inlet tubings are securely placed into buffer bottles A and B. Inlet tubing should be finger-tightened. Inlet filters are highly recommended.
2. Remove air from inlet buffer lines and prime the Gradient Pump.
 - a. Insert a 10 ml syringe into the luer fitting of the priming port (See Figure 19) for Pump A. Open the priming valve by rotating it counterclockwise. Draw buffer into the inlet buffer line and into the syringe. Close the priming valve (rotate clockwise) prior to removing the syringe, and expel the buffer into a waste container.
 - b. Repeat several times until no more bubbles are seen and then fill the syringe with 10 ml of buffer. Close the priming valve and remove air from the syringe by tapping the sides of the syringe to bring air bubbles to the top.
 - c. Remove the Pump A outlet tubing (See Figure 19). Attach the syringe filled with buffer A to the priming port and open the priming valve. Cover the pump A outlet with a paper towel to catch expelled buffer. Pulse the syringe plunger vigorously to expel any bubbles trapped behind the pump head piston. Do not push all of the buffer into the pump because air might be introduced in the last milliliter of buffer. Tighten the outlet fitting and close the priming port while the syringe is still in place.
 - d. Repeat for Pump B.

Note: Air bubbles trapped behind the pumphead pistons will not damage the pumphead but will produce an erratic flow. This problem is easily diagnosed by watching the pressure readout on the BioLogic HR Controller status bar. If it is extremely erratic and unstable, then suspect a trapped bubble and repeat the above steps.

Some pressure fluctuation ($\pm 10\%$) is normal due to the response of the pressure transducer over the available flow rate of the Gradient Pump.

11.3.2 Daily Maintenance

Washing behind the piston seal extends seal life. Insert a 10 ml syringe filled with deionized water in the Washout Port luer fitting (at top of washout block, Figure 66). A 10-ml syringe is provided in the BioLogic HR System Fittings Kit. Each pumphead should get a 10 ml washout at the end of the day's operation. Run off water exits through the Washout Drain between the two pumpheads. It can be collected in a small beaker.

11.3.3 Routine Maintenance of the Gradient Pump

The BioLogic HR System Gradient Pump requires minimal maintenance to stay in good working condition. A maintenance kit includes the piston seals, check valves, and piston wipers for routine maintenance of the pumps. Indications that maintenance is required include flow rates that are below the specified rate, an erratic delivery of liquid, liquid running out of the Washout drain during normal operation (often manifested as salt crystal accumulation) and poor gradient performance.

1. **Replacing a piston seal.** (Refer to Figure 66.)
 - a. Disconnect all tubing from the pump head.
 - b. Using the supplied Allen wrench, remove the four Pump Head Module Assembly screws on the **face** of the Pump Head Module in the order shown. Pull the pump head away from the Workstation.
 - c. Remove the washout block from the back of the pump head module by gently pulling it away.

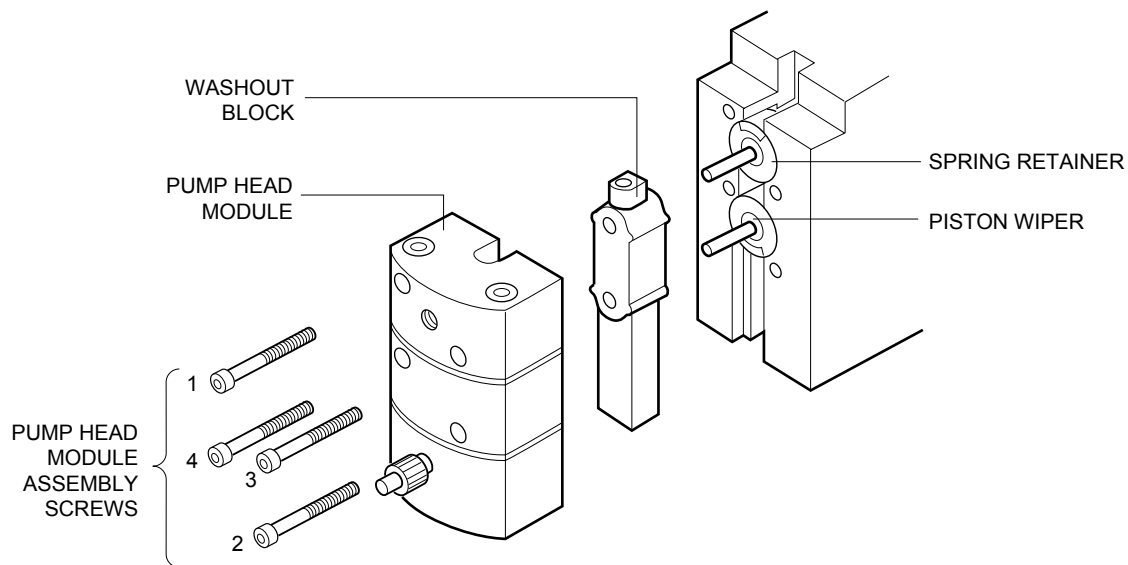


Figure 66. BioLogic HR Workstation Pump Mechanism Parts

- d. Note that there are two piston seals per pumphead. Always replace both seals at the same time. Once a piston seal is removed from its gland, it cannot be reused. It is recommended to pre-condition the new seals by soaking or sonicating them in water for several minutes. Always shake off or blot excess moisture from the seals before installation.

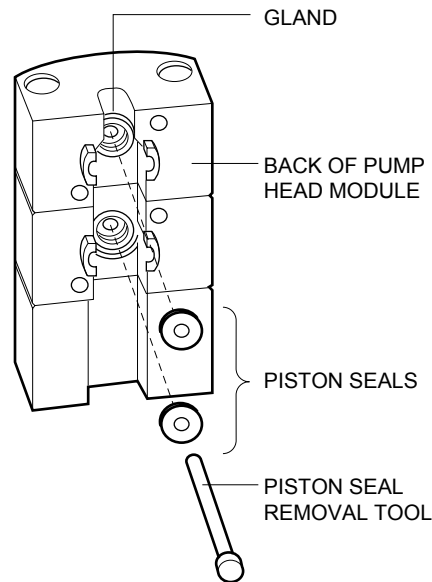


Figure 67. Removing the Piston Seals (view showing the back of the Pump Head Module)

- e. Using the Piston Seal Removal Tool, pry the seal from the gland. To do this, you insert the tip of the Seal Removal Tool into the seal and gently pry the seal away. (See Figure 67.)
- f. To install a new seal, line it up with the gland and use the Washout Block to seat the seal.
- g. When both seals have been replaced, reassemble the pump head by reversing the sequence shown in Figure 66. Firstly, wipe the pistons with a dampened cloth to ensure that no crystallized salt residues are present which could damage the new seals. Then, position the Washout Block onto the pistons over the piston wipers, and set the pump head module back onto the Washout Block on the BioLogic HR Workstation. (The Washout Block guides the pump head module in the reattachment process and protects the pistons.)
- h. Replace the four screws until the pump head module is secure on the Workstation but do not over tighten.

2. Cleaning and replacing a check valve.

- a. Remove the Pump Head Module from the Workstation. Refer to Figure 66 and the previous procedure.
- b. Remove the two Pump Head Module Assembly screws at the **top** of the module. See Figure 68.

Note: Never remove the screws without first removing the Pump Head Module from the Workstation.

The pump head can now be separated into three parts and the two check valves will be accessible.

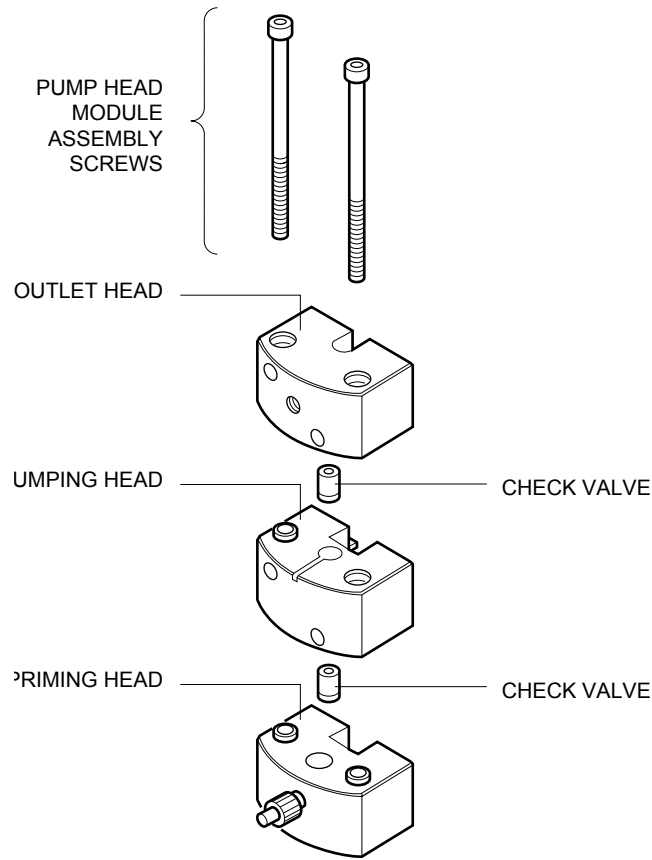


Figure 68. Pump Head Module Assembly

- c. Reassemble the pump head with a new check valve. Because the orientation of the check valve is critical, they are designed to fit into the pump head module in one direction only.

Note: If spare check valves are not available, you may sonicate the existing check valves and continue using them until they can be replaced. Sonication is only a short-term solution.

- d. Reassemble the pump head as shown in Figure 68. Replace the two pump head module assembly screws and tighten firmly.
- e. Position the Washout Block over the pistons and set the pump head module back onto the Washout Block on the BioLogic HR Workstation. See Figure 66. (The Washout Block guides the pump head module in the reattachment process and protects the pistons.)
- f. Replace the four screws until the pump head module is secure on the Workstation but do not over tighten.

3. Replacing the Piston Wiper

Piston wipers are available in the event that they are lost during any of the pump maintenance procedures. If the piston wiper has fallen out of the spring retainer, simply place a new piston wiper onto the piston inside the spring retainer. See Figure 66.

11.4 MAINTENANCE OF THE UV OPTICS MODULE AND THE CONDUCTIVITY FLOW CELL

The following two sections discuss the cleaning of the UV Optics Module and the Conductivity Flow Cell and the Replacing the Lamp in the UV Optics Module

11.4.1 Cleaning the UV Optics Module and the Conductivity Flow Cell

Problems with UV detection during medium pressure chromatography are often solved by a few simple precautions. Air bubbles trapped in the flow cells (especially the “Z” cell) give rise to a “sawtooth” signal pattern that will never stabilize. We recommend always running with at least a 40 psi backpressure regulator after the optics module (if using low pressure columns, place back pressure regulator between the pump outlet and inject valve). Equilibrate the column for at least 30 minutes to allow all bubbles to clear. Always degass buffers extensively before use.

Make sure all fittings are tight to prevent leaks. If liquid leaks down between the flow cell and the optics module case, the cell should be removed and dried with a stream of pure nitrogen (never use compressed air). The case should be dried using a warm air blower or left for several hours to completely dry.

Air bubbles trapped in the Gradient Pump give rise to erratic flow rates that are sometimes interpreted as problems with the UV optics module. Refer to the previous section.

Cleaning of the UV flow cell is indicated when the baseline is very noisy or unstable yet air bubbles have been removed from the flow cell. There are several components to clean in the UV Optics module:

- **Filters:** The standard filter tray contains both a 280 nm and a 254 nm filter. To clean the filters, loosen the filter’s thumbscrew and lift out the filter holder. The filters should be cleaned only when necessary using a dry lens tissue. To remove greasy fingerprints, use dilute ethanol or isopropanol solutions.
- **UV Flow Cell:** To remove the flow cell, undo the flat bottom fittings at the top and bottom of the flow cell. Then loosen the thumbscrews and push out the flow cell from the bottom. The flow cell is keyed to mount in one way only, so that it cannot be incorrectly mounted. Avoid touching the quartz windows with fingers.

If the flow cell is visually dirty, it should be cleaned with one or more of the following solvents: dilute SDS, 1M HCl, 1M NaOH, ethanol, or acetone. Run the solvent through the flow cell using a syringe and leave for no more than 5 minutes. Rinse extensively with water and then blow dry using a gentle stream of pure nitrogen. **Never** dry with compressed air from a “house” line as this will contain microdroplets of oil that will coat the cell.

When the Optics module is not in use, disconnect the flow cell and use a syringe filled with distilled water to clean out traces of salts and protein.

Before storing the flow cell, inject a dilute solution (10% to 25%) of ethanol or isopropanol into the cell to prevent microbial growth. Use the end caps provided with the flow cell to seal the flow cell’s inlet and outlet lines.

- The conductivity cell requires little maintenance other than rinsing routinely with deionized water.

11.4.2 Replacing the Lamp in the UV Optics Module

The following information covers the replacement procedures both for a new mercury lamp unit and a new zinc lamp unit. Note that the zinc lamp assembly has cooling fins and is larger than the mercury lamp assembly.

Indications that a lamp replacement is necessary include an unstable baseline and a decreased response to a standard concentration of a test chromophore. In this latter case, it is advisable to ensure that the flow cell is clean and that the correct wavelength filter is chosen before replacing the lamp. Refer to the previous section.

Note: Before performing the following procedure, make sure the BioLogic HR Workstation is turned off. This will eliminate the risk of electric shock to you or damage to the Optics module during cable connection.

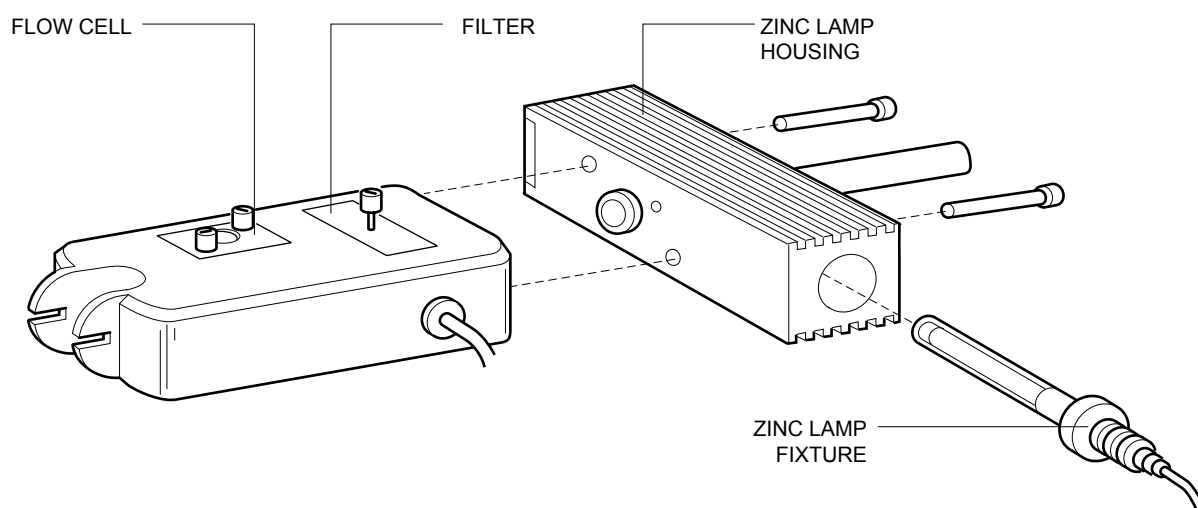


Figure 69. Replacing a Mercury Lamp in a Model OM-II Optics Module

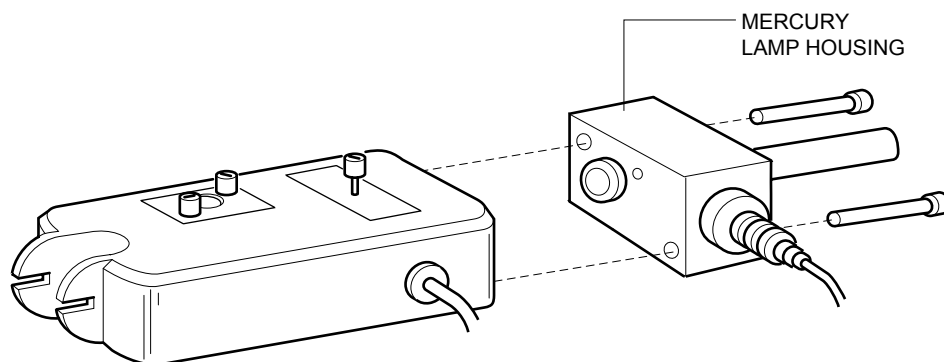


Figure 70. Replacing a Zinc Lamp in a Model OM-II Optics Module

1. Turn off the lamp from the BioLogic HR Controller Manual screen. Turn off power to the Workstation and unplug the UV Lamp cable, which is plugged into the UV Lamp connector on the rear of the Workstation.
2. Without disconnecting the lamp housing from the UV bench, undo the two small Allen screws securing the lamp fixture and gently pull the lamp out of the housing.
3. Remove the lamp unit from its packing material.
WARNING: Do not touch the glass part of the lamp. Oils from your fingers will degrade the lamp over a period of time.
4. Carefully insert the replacement lamp fixture. The replacement lamp can be installed only in one orientation due to a small guide which fits into the housing. Secure it using the Allen screws for that purpose. Note that extra screws are provided with the kit.

Note: Check that the O-ring seal at the Lamp Housing aperture is present and undamaged. If there is any indication of damage, or if it is missing, be sure to replace it using the O-ring supplied with the kit.

5. Reconnect the UV Lamp cable to the Workstation.
6. Turn on electrical power to the Workstation and check that the lamp is turned on at the Controller Manual Screen. Allow at least 30 minutes for the mercury lamp to warm up and at least several hours for the zinc lamp.

12.0 TROUBLESHOOTING THE BIOLOGIC HR SYSTEM

The following section lists potential problems and some suggested solutions for the different instruments in the system.

12.1 TROUBLESHOOTING THE BIOLOGIC HR CONTROLLER AND SOFTWARE

Problem	Possible Cause	Solution
Software response is slow.	Database size may need to be reduced.	1. Archive unnecessary methods and runs. a. Open the Browser window. b. Use COPY OUT to archive seldom used methods and runs. You may archive to a floppy disk, the BioLogic "C" drive, or any other drive selectable from the Drives drop down menu. 2. From the Browser & Database Options window (available by selecting Browser Settings from the Options menu), make sure that the Optimize on Startup button is selected. 3. Use the Windows 95 Disk Defragmenter utility. (Minimize the BioLogic HR application, and then from Windows Start, select Programs, Accessories, System Tools, Disk Defragmenter. Follow the on-line instructions to defragment files on your disk.
A method or run you expect is not displayed in the Browser.	There may be no apparent cause for this. Use the Reset button in the Browser toolbar.	1. From the Browser toolbar, click on the Reset button, and then check the database tree for your method or run. Note: Using Reset collapses the database tree.
While in offline mode, the offline window is not displayed.	The Windows® 95 minimize button may have been selected.	1. Use any of the following three means of activating the offline window: a. Use the Windows drop-down menu. b. Use the Windows® 95 taskbar. c. Simultaneously hold down the Alt-Tab keys. 2. Click on the right mouse button and from the menu that is displayed, select Restore.
Problems using Browser functions.	If you are using the right mouse button rather than the toolbar buttons, make sure you have first used the left mouse button to highlight the method or run name.	1. Using the left mouse button, click on a method or run name. 2. Using the right mouse button, select a function from the drop-down menu that is displayed.

Problem	Possible Cause	Solution
Instrument face-plates (eg. pump, UV, fraction collector) “drop-off” the screen when in Manual mode.	A communication problem has occurred.	1. Exit the BioLogic application and turn off the Controller. Turn off the Workstation. 2. Make sure all Instrument Bus (phone jack style) connections are secure in all the instruments. Turn the Controller and Workstation back on. 3. If the problem persists, contact Bio-Rad Technical Service.
Controller monitor displays unusual colors or characters.	The Monitor or the Controller video card may be defective and need replacement.	1. Exit the BioLogic application and restart the Controller. 2. If the problem persists, contact Bio-Rad Technical Service.
Controller displays “General Protection Fault” message or “locks up” during normal operation.		1. Exit the BioLogic application and restart the Controller. 2. If the problem persists, contact Bio-Rad Technical Service.

12.2 TROUBLESHOOTING THE BIOLOGIC HR GRADIENT PUMP

Problem	Possible Cause	Solution
Pump is not delivering the correct flow rate.	There may be problems with the fittings or tubing sizes. It is important to use the wide bore 1/8" OD tubing for the pump inlets and from all pre-pump buffer selection valves to the buffer containers.	- Check pumps A & B and pre-pump valves. - Make sure the inlet lines into pumps A and B are tight. - If pre-pump valves are used, eg. for buffer selection, make sure all tubing connections are secure and any non-used ports are plugged.
	The pump may not be receiving buffer.	- Check that there is buffer flowing to the pump. - Ensure that the pump inlet lines are immersed in buffer. - Ensure that the buffer bottles are positioned at or above the level of the Workstation.
	There may be air bubbles trapped in the pumpheads, causing erratic liquid delivery. Note: Always degas buffers before use.	- Re-prime the pumps and purge the lines. - Re-prime the pumps and ensure that the priming port is shut tight. - Make sure the Inject Valve is set to Purge position and purge all lines. Refer to the Maintenance section on how to prime the gradient pump and remove trapped air bubbles. To minimize the problem of air bubbles, degas buffers by stirring vigorously under vacuum for approximately 20 minutes. Use a heavy-wall side-arm Erlenmeyer flask, as standard flasks may implode under vacuum.
	Check valves are fouled and should be replaced. The check valves should be replaced on a regular basis, depending on the usage and type of solution or sample pumped through the pumphead. Refer to the Maintenance section for more information on how to replace a check valve.	- Replace (or clean) the check valves. - If spare check valves are not available, you can sonicate the existing check valves and continue using them until they can be replaced. Note: Sonication is only a short-term solution. - To replace check valves, use the pump hex keys to remove the pumpheads from the Workstation and remove the check valves. Recalibrate the pumps after changing either check valves or seals by selecting Calibrate Gradient Pump from the Utilities drop-down menu.
Liquid is continuously leaking from the pump-head washout drain trough.	The pump's piston seals are worn or damaged.	- Replace the piston seals on both pumpheads. - Recalibrate the pumps after changing seals by selecting Calibrate Gradient Pump from the Utilities drop-down menu.

Problem	Possible Cause	Solution
Erratic or reduced flow rate when pumping onto columns which produce low back-pressures. (eg. Econo-Pac® cartridges or small, agarose gel affinity columns).	Columns packed with soft gels or large particles typically produce very low back-pressures when run at low flow rates.	1. Insert the 40 psi back-pressure regulator between the pump and the injection valve. CAUTION: These types of columns may burst if the 40-psi backpressure device is placed after the column.
Delivery of liquid is erratic and the pressure readout on the Controller status bar is fluctuating more than 20%.	Severe fluctuation in the back-pressure readout indicates an air bubble trapped in one or both of the pumpheads. Note: Always degas buffers before use.	1. Put the AV7-3 valve into the purge position and purge the system by pressing the Purge buttons on the front of the Workstation panel. 2. Stop or Pause the pumps and re-prime. Loosen the top fitting on the pumphead and force any trapped air out of the pumphead by squirting buffer through the priming port using a syringe. Always use degassed buffers. 3. To minimize this problem, use degassed buffers and solutions. Degas buffers by stirring vigorously under vacuum for approximately 20 minutes. Use a heavy-wall side-arm Erlenmeyer flask, as standard flasks may implode under vacuum.
	The pump's check valves may need replacement.	4. Replace check valves, if necessary. Refer to the Maintenance section for information on how to do this. Recalibrate the pumps after changing either check valves or seals by selecting Calibrate Gradient Pump from the Utilities drop-down menu.

Problem	Possible Cause	Solution
The back-pressure displayed on the Controller status bar drops to zero or is much lower than expected.	Flow rate may have been changed.	1. Confirm that the flow rate setting has not been changed.
	There may be a tubing or fitting leak.	2. Re-inspect all tubing connections, especially at the pump inlet.
	Pump seals may need replacement.	3. Replace pump seals. A build-up of crystallized buffer salts on the rear of the pumphead is a clear sign for seal replacement.
	Check valves may need cleaning or replacement.	4. Clean or replace check valves. This is particularly important if crude samples have been loaded directly through the BioLogic pump.
No pressure (psi) reading on Controller status bar or psi value always reads zero.	Pump may not be running, or inlet lines are not primed, or there is a problem in the buffer flow.	1. Check that the pumps are running, and that the inlet lines are primed and not pulled out of the buffer reservoirs. Check for leaking fittings. 2. Check that the buffer flow is going to the column and not to waste via the injection valve purge position.
	The “zero pump pressure” calibration routine may have been used while the system was under pressure.	3. If this last cause is suspected, stop the pump and undo the line from the pump to the valve such that the pressure transducer is definitely at zero psi. Select Calibrate Gradient Pump from the Utilities drop-down menu and perform a Zero psi routine. 4. The pressure transducer may require replacement. Contact Bio-Rad Technical Services.
Incorrect back-pressure readings for a given column.	Flow rate may have been changed.	1. Check that the pump is delivering the correct flow rate.
	Column may need cleaning along with a frit replacement.	2. Consult the column’s instruction manual for cleaning procedure. Alternatively, the system tubing may have an obstruction, so inspect the tubing path.
	The pressure transducer calibration may be incorrect.	3. Select Calibrate Gradient Pump from the Utilities drop-down menu and perform a “ Zero psi ” routine. If this does not solve the problem, contact Bio-Rad Technical Services.

Problem	Possible Cause	Solution
High back-pressure is shutting down the pumps until the pressure drops.	The BioLogic pumps are capable of pumping against a back-pressure of 1000 psi. Pressure increases are due to a build-up of particulate matter in the system, causing a resistance to flow. An increase in flow rate also increases the pressure, so be sure to check this setting.	1. Remove the column and run the pumps to determine if the high back-pressure is due to the column or not. If the column is suspect, the column may need cleaning along with a frit replacement. Consult the column's instruction manual for cleaning procedure. 2. Alternatively, the system tubing may have an obstruction, so inspect the tubing path. To isolate the blockage, start by loosening connection fittings at the detector and work backwards towards the column and pumphead. 3. Prevent or minimize high back-pressure problems by filtering buffers and samples and by changing the pump seals before they deteriorate completely.

12.3 TROUBLE SHOOTING THE UV OPTICS MODULE AND UV TRACE

Problem	Possible Cause	Solution
UV baseline is unstable or noisy. UV baseline shows a reproducible “zig-zag” or “saw-tooth” trace.	Air bubbles trapped in the analytical 5-mm Z- cell are the most common cause of noisy UV baselines. Note: Always degas buffers before use.	1. Degas buffers by stirring vigorously under vacuum for approximately 20 minutes. Use a heavy-wall side-arm Erlenmeyer flask, as standard flasks may implode under vacuum. 2. Plumb the Z-cell with the column outlet connected to the bottom of the flow cell. This will force air bubbles to rise to the top of the flow cell and dissipate. 3. Use of the 40-psi back-pressure regulator supplied with the BioLogic HR system will remove most air bubbles. Place it directly after the conductivity flow cell and run buffer for several minutes. If the column being used cannot be run with the back-pressure regulator and air bubbles are a continuing problem (for example, during some affinity chromatography steps), use the 2mm path length cuvette cell. This cell is approximately 2.5x less sensitive than the Z-cell and is appropriate in those cases where the concentration of protein sample is high enough to be seen on the chromatogram.
Excessive pump pulsations exhibited as “regular noise” on the baseline.	Air bubbles trapped in the pump heads can produce exaggerated pulsations which appear as a noisy baseline.	4. Purge the pump heads to remove the bubble. Refer to section 11.3.1, Priming the Gradient Pump and Removing Trapped Air Bubbles.
Baseline noise continues when pumps are turned off.	Noise spikes can also be caused by external environmental influences.	1. If the spikes occur at regular intervals (for example, once every 20-30 sec), check for the presence of heating baths, drying ovens, or other heating devices on the same electrical circuit or in close proximity to the BioLogic HR system. Turn off these devices to see if the problem goes away.
UV baseline is “spiky,” particularly at low sensitivities.	Stray light, such as strong sunlight shining through a nearby window, is shining directly on the optics module.	1. Move the Optics module away from the window, or re-plumb the connections to the Optics module with opaque PEEK tubing (0.020” ID).

Problem	Possible Cause	Solution
UV trace will not zero.	If no air bubbles are present in the flow cell, check that the cell itself is clean both internally and externally.	1. Clean the interior of the cell by passing 1M NaOH, water, methanol and water through the cell. Dry the interior of the cell with a stream of high purity nitrogen gas (never use "house" compressed air as it contains oil droplets). 2. Check for leakage of liquid from the top of the optics unit onto the outside quartz windows of the flow cell. Clean the windows with a soft, non-abrasive damp cloth. 3. Condensation may occur on the exterior windows of the flow cell when moving the optics unit from the lab to a cold room and vice versa. Always allow an equilibration period to compensate for such temperature effects.
Drifting baseline.	Non-homogeneous eluant perhaps due to poor mixing or flow rate variation.	1. Check flow rates of both pumps; recalibrate if needed by selecting Calibrate Gradient Pump from the Utilities drop-down menu.
	Slow column equilibration.	2. Certain ion-exchangers are slow to re-equilibrate especially if just sanitized. Allow a longer equilibration period.
	If drift is due to a higher absorbance, suspect that UV absorbing material may be leaching from the column.	3. To confirm this, remove the column and run DI water or an aqueous non-absorbing buffer. If the baseline is stable, then the column is suspect. Consult the column's user manual for cleaning procedures.
Negative peaks.	The sample is applied to the column in a different buffer from that used to equilibrate and elute the column. Refractive index changes may be responsible for the negative peaks.	1. Apply the sample in the same buffer used to equilibrate the column.
	The elution buffer may have a higher UV absorbance than the sample components.	2. Check the elution buffer's UV absorbance.

Problem	Possible Cause	Solution
False or “ghost” peaks.	The injection valve, injection port and loop were not rinsed out between sample injections. Strongly-retained sample components may still be eluting from a previous run.	1. It is a good idea to allow a larger volume of injection buffer to pass through the inject valve and loop. For example, if the sample loop contains 100 µl, then program a load/injection volume of 200 µl to completely flush the valve and loop. 2. Always clean large volume dynamic loops, such as the Bio-Rad DynaLoop, between use to avoid cross contamination. 3. Use stringent elution conditions or consult the column’s user manual for an appropriate wash step to remove such components between runs.

12.4 TROUBLE SHOOTING THE CONDUCTIVITY FLOW CELL AND TRACE

Problem	Possible Cause	Solution
No values for buffer conductivity are displayed on the Controller status bar.	There may be a loose or incorrect cable connection.	1. Make sure the cable from the conductivity flow cell is plugged into the correct connector on the rear of the Workstation and that the pins are not bent.
	Conductivity Monitor may need calibration.	2. Use the Conductivity Flow Cell Constant Calibration function from the Utilities pull-down menu. Either input the constant value which is shown on the tag attached to the flow cell or calibrate the flow cell using 0.5M NaCl. Note that the relationship between the conductivity reading and salt concentration is not linear. The curve will flatten out at higher salt concentrations.
Conductivity trace does not follow exactly the theoretical gradient trace %B.		1. The Conductivity flow cell is typically placed after the UV flow cell. The Conductivity trace will be offset from the theoretical gradient based upon the column and system volume.

12.5 TROUBLE SHOOTING THE MODEL 2128 FRACTION COLLECTOR

Problem	Possible Cause	Solution
The drops are missing the tubes.	The rack being used does not correspond to the rack selected by the BioLogic HR software.	1. Ensure that the correct Rack # (ie., #1 to 5) has been entered in the Method Protocol and matches the actual rack being used.
	The fraction collector may need a drop-head alignment.	2. Model 2128 Fraction Collectors with firmware v. 1.03 or later have a drop-head alignment feature. To align the drop-head: - When you first turn on the fraction collector, note the firmware version as it is briefly displayed. The firmware must be v. 1.03 or greater. - From the startup display, press the Mode softkey. - Simultaneously press the "0" and ." hardkeys. - From the menu, press the Calib softkey. - From the calibration screen, press the D-Arm softkey. - Respond to the message, "Initialize Drop Arm?" by pressing the Yes softkey. - Respond to the message, "Select Item to be Repositioned," by pressing the Rack softkey. - To align the drop-head over a specific tube, press the Locate softkey, followed by the Adjust softkey. - Adjust the position of the drop-head over the tube using the four "arrow" hardkeys. It is helpful to have liquid flowing during this procedure. - It is best to check the alignment at two positions. For example, for rack #1, check the drophead alignment over tubes 1 and 8. Use the Adv softkey to select the next tube. - When you are done, press the Done softkey, followed by the Exit softkey and the Yes softkey to save the adjustment. Press the Exit softkey twice to return to the startup screen. The fraction collector must be in Std mode before it can be returned to BioLogic system control. 3. Make sure that the tubing is not being snagged and restricting the movement of the drophead.

APPENDIX A. SPECIFICATIONS

System Power Requirements: BioLogic HR Controller

- 115 V ~ 5.3 A
- 230 V ~ 3.5 A
- 50 - 60 Hz

System Power Requirements: BioLogic HR Workstation

- 115 V ~ 5.3 A
- 230 V ~ 3.5 A
- 50 - 60 Hz

BioLogic HR Controller

- Internal computer, at least Pentium®/133 MHz, 1.2 GB hard drive
- 3.5" floppy drive
- 14" Super VGA color monitor
- Water-resistant mini-keyboard plus mouse

Gradient pump

- Dual pumphead , dual piston, positive displacement solvent delivery module
- 1000 psi from 0.1-10 ml/min
- High pressure mixing; mixer volume 380 µl
- Built-in pressure transducer; 34 µl volume
- Flow accuracy: 0.5% @ 1.0 ml/min; 2% @ 5.0 ml/min
- Flow reproducibility: 0.003 rsd
- Gradient increments: 1%
- Composition accuracy: ±0.5% of 100% (>1.0 ml/min)
- System fittings: Flangeless 1/4-28 flat-bottom fittings, fingertight nuts and ferrules
- System tubing: 1/16" Tefzel (1/16" OD x 0.020" ID)
- Gradient Pump inlet tubing: 1/8" OD PTFE (1/8" OD x 0.062" ID)
- Wetted materials: 100% bio-compatible; PEEK, Tefzel, ceramic
- Solvent compatibility: 1.0 M NaOH, HCl and organic acids, 1.0 M Sulfuric acid, 7M Urea and Guanidine HCl, 100% Ethanol, IPA and Acetonitrile, 0.1% TFA, 1% detergents (incl. SDS, Triton X-100)

Detection

- Fixed wavelength, mercury lamp with 280/254 nm filter, auto zero; 0.001-2.0 AUFS detection range
- Analytical (5 mm pathlength, 8 µl total volume, 6 µl illum. volume) flow cell
- Optional 2 mm pathlength (30 µl volume) preparative flow cell
- Optional 214 nm detection kit (includes zinc lamp, housing, and 214 nm filter)
- Optional wavelength filters include: 214, 313, 365, 405, 436, and 546 nm
- Conductivity detection (500 mS/cm; 8 µl flow cell; accuracy $\pm$ 2% full scale)

Sample Loading

- Fixed sample loops, 25 µl to 5.0 ml; manual or automated injection
- Large volume loading through the Gradient Pump. Optional inlet select valves (SV5-4, AV9-8) for sanitation and multiple sample use.
- Variable volume sample loading through the DynaLoop sliding-piston sample loop.
- Software control of an auxiliary pump for multiple sample loop fills and large volume dynamic loop filling.

Valve Control

- Up to three low-pressure, automated solenoid valves (SV5-4 and SV3-2) for stream selection.
- Up to three high-pressure, automated valves (AV7-3 and AV9-8) for sample inject, column switching and stream selection.
- Load/Inject position-sensing for manual inject valve V7-3.

Fraction Collection

- Model 2128 Fraction Collector (collection by Collect All, Threshold, Collection Windows, Collection Window + Threshold)
- Model 2110 Fraction Collector (collection by Collect All; when used with optional SV3-2 Diverter valve, it also offers collection by Threshold, Collection Windows, Collection Window + Threshold)

Chart Recorder Control

- Model 1327 Dual pen recorder. Paper feed Start/Stop, Pen Up/Down

Signal Import Module

- Up to two Signal Import Modules for importing analog signals from peripheral chromatography detectors

General

- Dimensions (without collector and recorder): ~68 x 41 x 81 cm (W x D x H), depending upon configuration.
- Weight (Controller, Workstation, Monitor, Rack, Injection valve and Detection modules): 34.5 kg (76 lbs.)

APPENDIX B. PRESSURE CONVERSION TABLE

Use this table to convert pressure units for the pump.

Table B-1
Pressure Conversion

Multiply the units in the left column by the conversion factors listed.

Pressure	PSI	atm	Kg/cm ²	Torr	kPa	Bar	Inches Hg
PSI	1	0.068	0.0703	51.713	6.8948	0.06895	2.0359
atm	14.696	1	1.0332	760	101.32	1.0133	29.921
Kg/cm²	14.223	0.9678	1	735.56	98.06	0.9806	28.958
Torr	0.0193	0.00132	0.00136	1	0.1330	0.00133	0.0394
kPa	0.1450	0.00987	0.0102	7.52	1	0.0100	0.2962
Bar	14.5038	0.9869	1.0197	751.88	100	1	29.5300
Inches Hg	0.4912	0.0334	0.0345	25.400	3.376	0.03376	1

Example 1: Convert 521 psi to Kg/cm²
(521 x 0.0703) or (521 psi = 36.63 Kg/cm²)

Example 2: Convert 60 Bar to psi
(60 x 14.5038) or (60 Bar = 870 psi)

Example 3: Convert 42.70 Bar to Kg/cm²
(42.70 x 1.0197) = (42.70 Bar = 43.54 Kg/cm²)

APPENDIX C. WARRANTY STATEMENT

Bio-Rad Laboratories warrants that every instrument it sells will be free from defects in materials and workmanship when it leaves the factory and that, if such defects appear within the first year following purchase, the defective part(s) will be replaced or the entire unit will be replaced, at Bio-Rad's option, free of any charges to the buyer other than expenses incurred in returning the unit to the factory. Bio-Rad's obligation under this warranty is specifically limited to the aforementioned replacement or repairs. However, the following defects are specifically excluded:

1. Defects caused by improper operation.
2. Repair or modification done by anyone other than Bio-Rad Laboratories or their authorized agent.
3. Use with fittings or other spare parts not specified by Bio-Rad Laboratories.
4. Damage caused by deliberate or accidental misuse.
5. Damage caused by disaster.
6. Damage due to use of improper solvent or sample.
7. Tubing, fittings, lamps, and other consumables.

"The foregoing obligations are in lieu of all other obligations and liabilities including negligence and all warranties, of merchantability, fitness for a particular purpose or otherwise, expressed or implied in fact or by law, and state Bio-Rad's entire and exclusive liability and Buyer's exclusive remedy for any claims or damages in connection with the furnishing of goods or parts, their design, suitability for use, installation or operation, Bio-Rad will in no event be liable for any special, incidental or consequential damages whatsoever, and Bio-Rad's liability under no circumstances will exceed the contract price for the goods for which liability is claimed."

Under this warranty Bio-Rad's obligation with respect to transportation expenses is limited to the cost of shipping the repaired or replacement instrument to the buyer, provided that such repair or replacement comes within the terms of this warranty.

For additional help, contact your local Bio-Rad representative. In the United States, call Technical Service at 1-800-4BIORAD.

APPENDIX D ORDERING INFORMATION

Gradient Pump

750-0164	Gradient Pump Maintenance Kit, services one pump. Kit includes 2 piston seals, 2 check valves, 2 piston wipers.
750-0161	Piston Seals, 4. Two piston seals required to service one pump.
750-0162	Check Valve, 1. Two check valves required to service one pump.
760-0151	Priming Ports, 2

Detector Components

750-0214	214 nm Conversion Kit (Zn lamp, housing, 214 nm filter) only for Model OM-II Optics Modules beginning with serial number 362 BR.
750-0215	214 nm Conversion Kit (Zn lamp, housing, 214 nm filter) only for Model OM-IO Optics Modules, beginning with serial number 345 BR.
750-0210	Flow Cell, 2 mm
750-0212	Flow Cell, 5 mm
750-0216	Hg Lamp, replacement
750-0217	Zn Lamp (214 nm), replacement
750-0220	Filter, 280 + 254 nm
750-0221	Filter, 214 nm
750-0222	Filter, 313 nm
750-0223	Filter, 365 nm
750-0224	Filter, 405 nm
750-0225	Filter, 436 nm
750-0226	Filter, 546 nm
750-0230	Backpressure Regulator
750-0240	Conductivity Flow Cell, replacement

750-0251 BioLogic Rack

750-0268	Rack expansion kit (2 trays, 2 vert. bars, 16 sleeves)
750-0260	Column clamp set
750-0261	Rack tray, 1 (+8 sleeves, 1 drain/plug)
750-0262	Vertical bar, long, 2
750-0263	Vertical bar, short, 2
750-0264	Horizontal bar kit (2 tie bars, 4 bar clamps)
750-0265	Bar clamps, 5
750-0266	Cable Manager clips, 4
750-0269	System wrench set

Peripherals

167-1100	Model AS-100 Automatic Sampling System, 100/120V
167-1101	Model AS-100 Automatic Sampling System, 200/240V
731-8122	Model 2110 Fraction Collector, 110 V
731-8120	Model 2110 Fraction Collector, 220 V
731-8123	Model 2128 Fraction Collector, 110 V
731-8124	Model 2128 Fraction Collector, 220 V
731-8238	Model 2128 Diverter Valve
750-0310	Model 2128 BioLogic Interface Kit (Required for units manufactured in 1993 and 1994. Consult with local Bio-Rad representative prior to purchase.)

Peripherals (continued)

750-0505	BioLogic HR pH monitor (1 SIM-HR, 1 pH probe, 1 flow cell)
750-0502	Signal Import Module-HR
750-0503	pH electrode for BioLogic HR
750-0504	pH flow cell for BioLogic HR
731-8250	Model 1327 Econo Recorder, with USA, Canada, Japan, Mexico, Taiwan, and Latin America power adaptor
731-8253	Model 1327 Econo Recorder, with UK, Commonwealth power adaptor
731-8254	Model 1327 Econo Recorder, with Australia, New Zealand power adaptor
731-8255	Model 1327 Econo Recorder, with European power adaptor
750-0111	EZ Logic Integration Software Package

BioLogic Valves

750-0401	Manual Injection Valve, V7-3
750-0406	Automated Injection Valve, AV7-3
750-0403	Manual Stream Select Valve, V9-8
750-0408	Automated Stream Select Valve, AV9-8
750-0410	Diverter Valve
750-0415	Select Valve, SV5-4

Sample Injection Loops

750-0480	Small volume inject loop kit
750-0481	Large volume Inject loop kit
750-0482	25 µl injection loop
750-0483	50 µl injection loop
750-0484	100 µl injection loop
750-0485	250 µl injection loop
750-0486	500 µl injection loop
750-0487	1 ml injection loop
750-0488	2 ml injection loop
750-0489	5 ml injection loop

BioLogic System Fittings

750-0550	BioLogic System Fittings Kit
750-0555	1/16" nut, ferrule, lock ring, 10 pk
750-0556	1/16" ferrule, lock ring, 10 pk
750-0557	1/8" nut, ferrule, 5 pk
750-0558	1/8" ferrule, 5 pk
750-0559	Caps, 5 pk
750-0561	1/4-28 to M6 (FPLC column adaptor), 2 pk
750-0562	1/4-28 to 1/4-28 union, 5 pk
750-0563	1/4-28 plug, 5 pk
750-0471	Inject port, 1
125-0224	Inject Needle, 22 gauge blunt, 1
750-0471	Econo-Column® to BioLogic System Fittings Kit, 1 set
732-0113	Econo-Pac® Cartridge to BioLogic System Fittings Kit, 1 set
750-0564	HPLC Column to BioLogic System Fittings Kit, 1 set
750-0703	Inline Filter Kit, 1 inline filter, 2 replacement Ultra High Molecular Weight PolyEthylene (UHMWPE) filter frits
750-0704	Replacement UHMWPE filter frits, 5
750-0566	Bottle Cap Kit, 2 bottle caps, 2 cap plugs

APPENDIX E COLUMN INFORMATION

The following contain information about the different columns available from Bio-Rad for use with the BioLogic HR system.

APPENDIX F DECLARATION OF CONFORMITY ISO 9000 CERTIFICATION

The following pages show the Declaration of Conformity and ISO 9001 Certificate of Registration for the BioLogic HR system.



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Denmark Ph. 45 44 52-1000, Fx. 45 4452 1001 **Finland** Ph. 358 (0)9 804 2200, Fx. 358 (0)9 804 1100 **France** Ph. 01 47 95 69 65, Fx. 01 47 41 9133
Germany Ph. 089 318 84-177, Fx. 089 318 84-123 **Hong Kong** Ph. 852-2789-3300, Fx. 852-2789-1257 **India** Ph. (91-124)-6398112/113/114, Fx. (91-124)-6398115
Israel Ph. 03 951 4124, Fx. 03 951 4129 **Italy** Ph. 34 91 590 5200, Fx. 34 91 590 5211 **Japan** Ph. 03-5811-6270, Fx. 03-5811-6272
Korea Ph. 82-2-3473-4460, Fx. 82-2-3472-7003 **Latin America** Ph. 305-894-5950, Fx. 305-894-5960 **Mexico** Ph. 52 5 534 2552 to 54, Fx. 52 5 524 5971
The Netherlands Ph. 0318-540666, Fx. 0318-542216 **New Zealand** Ph. 64-9-4152280, Fx. 64-9-443 3097 **Norway** Ph. 47-23-38-41-30, Fx. 47-23-38-41-39
Russia Ph. 7 095 979 98 00, Fx. 7 095 979 98 56 **Singapore** Ph. 65-2729877, Fx. 65-2734835 **Spain** Ph. 34-91-590-5200, Fx. 34-91-590-5211
Sweden Ph. 46 (0)8-55 51 27 00, Fx. 46 (0)8-55 51 27 80 **Switzerland** Ph. 061-717-9555, Fx. 061-717-9550 **United Kingdom** Ph. 0800-181134, Fx. 01442-259118